EGYPTIAN BACCALAUREATE • 2026

MATHEMATICS

CLASSROOM WORKING EDITION

Chapter 05

Trigonometric Functions

1 Concepts | 82 questions

B5 Landscape | one question per teaching page

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Chapter 05 | Trigonometric Functions

CLASSROOM ROUTE

Concept 1 | Defining Angles and Basic Trigonometric Functions

82 questions

Official lessons: 5-1, 5-2, 5-3, 5-4

The Formula Bank changes at each Concept boundary.

Concept 1 | Defining Angles and Basic Trigonometric Functions OFFICIAL LESSONS 5-1, 5-2, 5-3, 5-4

Defining Angles and Basic Trigonometric Functions

Equation-first recall | use the same rail beside every question in this Concept

Angle family

$$\theta = \alpha + 360^\circ \times n$$

Degree / radian

$$\theta_{rad} = \theta_{deg} \times \frac{\pi}{180}$$

Sector measures

$$L = r\theta \qquad S = \frac{1}{2}r^2\theta$$

Unit circle

$$\sin\theta = y \qquad \cos\theta = x \qquad \tan\theta = \frac{y}{x}$$

Start with the family cue, write the first move, then use the working grid.

Concept 1 | Defining Angles and Basic Trigonometric Functions

F01 | V01

Q001 | WRITTEN C05-K01-5-1-WU1

Find the measures of θ in the two figures: (a) one full counter-clockwise revolution plus 75°; (b) two full clockwise revolutions and a further 135°.

WORKING AREA

FORMULA BANK

Angle family

$$\theta = \alpha + 360° \times n$$

Degree / radian

$$\theta_{rad} = \theta_{deg} \times \frac{\pi}{180}$$

Sector measures

$$L = r\theta \qquad S = \frac{1}{2}r^2\theta$$

Unit circle

$$\sin\theta = y \qquad \cos\theta = x \qquad \tan\theta = \frac{y}{x}$$

Concept 1 | Defining Angles and Basic Trigonometric Functions

F01 | V02

Q002 | WRITTEN

C05-K01-5-1-WU2

Draw the terminal side for 300° and −450°.

WORKING AREA

FORMULA BANK

Angle family

$$\theta = \alpha + 360^\circ \times n$$

Degree / radian

$$\theta_{rad} = \theta_{deg} \times \frac{\pi}{180}$$

Sector measures

$$L = r\theta \qquad S = \frac{1}{2}r^2\theta$$

Unit circle

$$\sin\theta = y \qquad \cos\theta = x \qquad \tan\theta = \frac{y}{x}$$

Concept 1 | Defining Angles and Basic Trigonometric Functions

F01 | V03

Q003 | WRITTEN

C05-K01-5-1-WU3

Express $420°$ and $-210°$ as $\alpha + 360° \times n$, where n is an integer and $0° \leq \alpha < 360°$.

WORKING AREA

FORMULA BANK

Angle family

$$\theta = \alpha + 360° \times n$$

Degree / radian

$$\theta_{rad} = \theta_{deg} \times \frac{\pi}{180}$$

Sector measures

$$L = r\theta \qquad S = \frac{1}{2}r^2\theta$$

Unit circle

$$\sin\theta = y \quad \cos\theta = x \quad \tan\theta = \frac{y}{x}$$

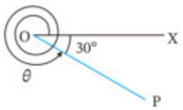
Concept 1 | Defining Angles and Basic Trigonometric Functions

F01 | V01

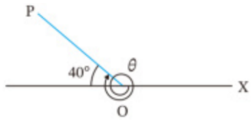
Q004 | WRITTEN · C05-K01-5-1-TR1

Find the measures of θ in the two figures: (a) a clockwise turn of 30° from the positive x-axis; (b) a terminal side 40° above the negative x-axis, measured counter-clockwise from the positive x-axis).

(a)



(b)



WORKING AREA

FORMULA BANK

Angle family

$$\theta = \alpha + 360^\circ \times n$$

Degree / radian

$$\theta_{rad} = \theta_{deg} \times \frac{\pi}{180}$$

Sector measures

$$L = r\theta \qquad S = \frac{1}{2}r^2\theta$$

Unit circle

$$\sin\theta = y \qquad \cos\theta = x \qquad \tan\theta = \frac{y}{x}$$

Concept 1 | Defining Angles and Basic Trigonometric Functions

F01 | V02

Q005 | WRITTEN

C05-K01-5-1-TR2

Draw the terminal side for $500°$ and $-120°$.

WORKING AREA

FORMULA BANK

Angle family

$$\theta = \alpha + 360° \times n$$

Degree / radian

$$\theta_{rad} = \theta_{deg} \times \frac{\pi}{180}$$

Sector measures

$$L = r\theta \qquad S = \frac{1}{2}r^2\theta$$

Unit circle

$\sin\theta = y \quad \cos\theta = x \quad \tan\theta = \frac{y}{x}$

Concept 1 | Defining Angles and Basic Trigonometric Functions

F01 | V03

Q006 | WRITTEN

C05-K01-5-1-TR3

Express 520° and −140° as $\alpha + 360° \times n$.

WORKING AREA

FORMULA BANK

Angle family

$$\theta = \alpha + 360° \times n$$

Degree / radian

$$\theta_{rad} = \theta_{deg} \times \frac{\pi}{180}$$

Sector measures

$$L = r\theta \qquad S = \frac{1}{2}r^2\theta$$

Unit circle

$$\sin\theta = y \qquad \cos\theta = x \qquad \tan\theta = \frac{y}{x}$$

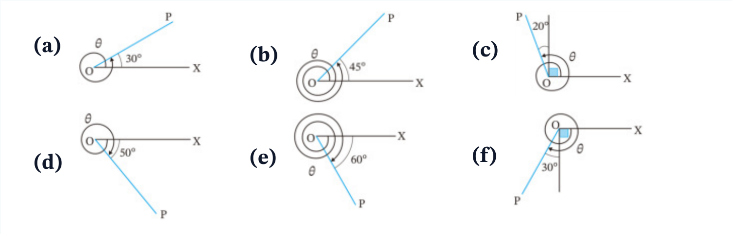
Concept 1 | Defining Angles and Basic Trigonometric Functions

F01 | V01

Q007 | WRITTEN

C05-K01-5-1-EX1

Find the six diagram measures θ shown in the given figure (a)–(f).



WORKING AREA

FORMULA BANK

Angle family

$$\theta = \alpha + 360° \times n$$

Degree / radian

$$\theta_{rad} = \theta_{deg} \times \frac{\pi}{180}$$

Sector measures

$$L = r\theta \qquad S = \frac{1}{2}r^2\theta$$

Unit circle

$$\sin\theta = y \qquad \cos\theta = x \qquad \tan\theta = \frac{y}{x}$$

Concept 1 | Defining Angles and Basic Trigonometric Functions

F01 | V02

Q008 | WRITTEN

C05-K01-5-1-EX2

Draw the terminal sides for $470°,\ 780°,\ 225°,\ -380°,\ -990°,\ -200°$.

WORKING AREA

FORMULA BANK

Angle family

$$\theta = \alpha + 360° \times n$$

Degree / radian

$$\theta_{rad} = \theta_{deg} \times \frac{\pi}{180}$$

Sector measures

$$L = r\theta \qquad S = \frac{1}{2}r^2\theta$$

Unit circle

$$\sin\theta = y \quad \cos\theta = x \quad \tan\theta = \frac{y}{x}$$

Concept 1 | Defining Angles and Basic Trigonometric Functions

F01 | V03

Q009 | WRITTEN

C05-K01-5-1-EX3

Express $610°,\ 920°,\ 840°,\ -430°,\ -960°,\ -20°$ as $\alpha + 360° \times n$.

WORKING AREA

FORMULA BANK

Angle family

$$\theta = \alpha + 360° \times n$$

Degree / radian

$$\theta_{rad} = \theta_{deg} \times \frac{\pi}{180}$$

Sector measures

$$L = r\theta \qquad S = \frac{1}{2}r^2\theta$$

Unit circle

$$\sin\theta = y \qquad \cos\theta = x \qquad \tan\theta = \frac{y}{x}$$

Concept 1 | Defining Angles and Basic Trigonometric Functions

F01 | V03

Q010 | WRITTEN

C05-K01-5-1-CH1

Express 129720° and −2592100° as $\alpha + 360° \times n$, with $0° \leq \alpha < 360°$.

WORKING AREA

FORMULA BANK

Angle family

$$\theta = \alpha + 360° \times n$$

Degree / radian

$$\theta_{rad} = \theta_{deg} \times \frac{\pi}{180}$$

Sector measures

$$L = r\theta \qquad S = \frac{1}{2}r^2\theta$$

Unit circle

$$\sin\theta = y \quad \cos\theta = x \quad \tan\theta = \frac{y}{x}$$

Concept 1 | Defining Angles and Basic Trigonometric Functions

F01 | V03

Q011 | MCQ

C05-K01-5-1-GM01

The principal angle coterminal with 765° is:

A 45°	**B** 135°
C 225°	**D** 315°

WORKING AREA

FORMULA BANK

Angle family

$$\theta = \alpha + 360^\circ \times n$$

Degree / radian

$$\theta_{rad} = \theta_{deg} \times \frac{\pi}{180}$$

Sector measures

$$L = r\theta \qquad S = \frac{1}{2} r^2 \theta$$

Unit circle

$\sin\theta = y \quad \cos\theta = x \quad \tan\theta = \frac{y}{x}$

Concept 1 | Defining Angles and Basic Trigonometric Functions

F01 | V03

Q012 | MCQ

C05-K01-5-1-GM02

The principal angle coterminal with $-410°$ is:

A $-50°$

B $50°$

C $310°$

D $410°$

WORKING AREA

FORMULA BANK

Angle family

$$\theta = \alpha + 360° \times n$$

Degree / radian

$$\theta_{rad} = \theta_{deg} \times \frac{\pi}{180}$$

Sector measures

$$L = r\theta \qquad S = \frac{1}{2}r^2\theta$$

Unit circle

$$\sin\theta = y \qquad \cos\theta = x \qquad \tan\theta = \frac{y}{x}$$

Concept 1 | Defining Angles and Basic Trigonometric Functions

F01 | V02

Q013 | MCQ

C05-K01-5-1-GM03

The terminal side of 540° lies on the:

A positive x-axis

B negative x-axis

C positive y-axis

D negative y-axis

WORKING AREA

FORMULA BANK

Angle family

$$\theta = \alpha + 360° \times n$$

Degree / radian

$$\theta_{rad} = \theta_{deg} \times \frac{\pi}{180}$$

Sector measures

$$L = r\theta \qquad S = \frac{1}{2}r^2\theta$$

Unit circle

$$\sin\theta = y \quad \cos\theta = x \quad \tan\theta = \frac{y}{x}$$

Concept 1 | Defining Angles and Basic Trigonometric Functions

F01 | V02

Q014 | MCQ

C05-K01-5-1-GM04

A positive angle of 315° terminates in:

A QI

B QII

C QIII

D QIV

WORKING AREA

FORMULA BANK

Angle family

$$\theta = \alpha + 360° \times n$$

Degree / radian

$$\theta_{rad} = \theta_{deg} \times \frac{\pi}{180}$$

Sector measures

$$L = r\theta \qquad S = \frac{1}{2}r^2\theta$$

Unit circle

$$\sin\theta = y \qquad \cos\theta = x \qquad \tan\theta = \frac{y}{x}$$

Concept 1 | Defining Angles and Basic Trigonometric Functions

F01 | V03

Q015 | MCQ

C05-K01-5-1-GM05

Which is the general-angle form for $-75°$?:

A $75° + 360°n$

B $285° + 360°n$

C $-285° + 360°n$

D $360°n - 75°$

WORKING AREA

FORMULA BANK

Angle family

$$\theta = \alpha + 360° \times n$$

Degree / radian

$$\theta_{rad} = \theta_{deg} \times \frac{\pi}{180}$$

Sector measures

$$L = r\theta \qquad S = \frac{1}{2}r^2\theta$$

Unit circle

$$\sin\theta = y \qquad \cos\theta = x \qquad \tan\theta = \frac{y}{x}$$

Concept 1 | Defining Angles and Basic Trigonometric Functions

F01 | V01

Q016 | MCQ

C05-K01-5-1-GM06

The direction of a negative angle is:

A counter-clockwise

B clockwise

C radial only

D undefined

WORKING AREA

FORMULA BANK

Angle family

$$\theta = \alpha + 360^\circ \times n$$

Degree / radian

$$\theta_{rad} = \theta_{deg} \times \frac{\pi}{180}$$

Sector measures

$$L = r\theta \qquad S = \frac{1}{2}r^2\theta$$

Unit circle

$$\sin\theta = y \qquad \cos\theta = x \qquad \tan\theta = \frac{y}{x}$$

Concept 1 | Defining Angles and Basic Trigonometric Functions

F01 | V02

Q017 | MCQ — C05-K01-5-1-GM07

The terminal side of 1080° is the:

A positive x-axis

B negative x-axis

C positive y-axis

D negative y-axis

WORKING AREA

FORMULA BANK

Angle family

$$\theta = \alpha + 360^\circ \times n$$

Degree / radian

$$\theta_{rad} = \theta_{deg} \times \frac{\pi}{180}$$

Sector measures

$$L = r\theta \qquad S = \frac{1}{2}r^2\theta$$

Unit circle

$$\sin\theta = y \qquad \cos\theta = x \qquad \tan\theta = \frac{y}{x}$$

Concept 1 | Defining Angles and Basic Trigonometric Functions

F01 | V03

Q018 | MCQ

C05-K01-5-1-GM08

For $\alpha = 0°$, the general-angle family is:

A $360°n$

B $0°n$

C $180° + 360°n$

D $90° + 360°n$

WORKING AREA

FORMULA BANK

Angle family

$$\theta = \alpha + 360° \times n$$

Degree / radian

$$\theta_{rad} = \theta_{deg} \times \frac{\pi}{180}$$

Sector measures

$$L = r\theta \qquad S = \frac{1}{2}r^2\theta$$

Unit circle

$$\sin\theta = y \qquad \cos\theta = x \qquad \tan\theta = \frac{y}{x}$$

Concept 1 | Defining Angles and Basic Trigonometric Functions

F01 | V03

Q019 | MCQ

C05-K01-5-1-GM09

The principal angle for $-990°$ is:

A $90°$

B $180°$

C $270°$

D $-270°$

WORKING AREA

FORMULA BANK

Angle family

$$\theta = \alpha + 360° \times n$$

Degree / radian

$$\theta_{rad} = \theta_{deg} \times \frac{\pi}{180}$$

Sector measures

$$L = r\theta \qquad S = \frac{1}{2}r^2\theta$$

Unit circle

$$\sin\theta = y \qquad \cos\theta = x \qquad \tan\theta = \frac{y}{x}$$

Concept 1 | Defining Angles and Basic Trigonometric Functions

F01 | V01

Q020 | MCQ

C05-K01-5-1-GM10

A ray (P) is 25° above the negative x-axis. Its standard-position angle is:

A 25°

B 155°

C 205°

D 335°

WORKING AREA

FORMULA BANK

Angle family

$$\theta = \alpha + 360^\circ \times n$$

Degree / radian

$$\theta_{rad} = \theta_{deg} \times \frac{\pi}{180}$$

Sector measures

$$L = r\theta \qquad S = \frac{1}{2}r^2\theta$$

Unit circle

$\sin\theta = y$ $\cos\theta = x$ $\tan\theta = \frac{y}{x}$

Concept 1 | Defining Angles and Basic Trigonometric Functions

F02 | V01

Q021 | WRITTEN   C05-K01-5-2-WU

(1) Convert 630° to radians. (2) Convert $-\frac{7\pi}{4}$ to degrees. (3) For a sector with $r = 3$ and $\theta = \frac{4\pi}{3}$, find arc length L and area S.

WORKING AREA

FORMULA BANK

Angle family

$$\theta = \alpha + 360° \times n$$

Degree / radian

$$\theta_{rad} = \theta_{deg} \times \frac{\pi}{180}$$

Sector measures

$$L = r\theta \qquad S = \frac{1}{2}r^2\theta$$

Unit circle

$$\sin\theta = y \qquad \cos\theta = x \qquad \tan\theta = \frac{y}{x}$$

Concept 1 | Defining Angles and Basic Trigonometric Functions

F02 | V01

Q022 | WRITTEN

C05-K01-5-2-TR

Complete the standard-angle degree/radian table, then: convert $288°$, $-400°$ to radians; convert $\frac{11\pi}{6}$, $-\frac{3\pi}{5}$ to degrees; and find L, S for $r = 6$, $\theta = \frac{3\pi}{4}$.

WORKING AREA

FORMULA BANK

Angle family

$$\theta = \alpha + 360° \times n$$

Degree / radian

$$\theta_{rad} = \theta_{deg} \times \frac{\pi}{180}$$

Sector measures

$$L = r\theta \qquad S = \frac{1}{2}r^2\theta$$

Unit circle

$$\sin\theta = y \quad \cos\theta = x \quad \tan\theta = \frac{y}{x}$$

Concept 1 | Defining Angles and Basic Trigonometric Functions

F02 | V01

Q023 | WRITTEN

C05-K01-5-2-EX1

Convert $108°,\ 36°,\ -240°,\ -225°,\ 420°,\ 400°,\ -570°,\ -2700°$ to radians.

WORKING AREA

FORMULA BANK

Angle family

$$\theta = \alpha + 360° \times n$$

Degree / radian

$$\theta_{rad} = \theta_{deg} \times \frac{\pi}{180}$$

Sector measures

$$L = r\theta \qquad S = \frac{1}{2}r^2\theta$$

Unit circle

$$\sin\theta = y \quad \cos\theta = x \quad \tan\theta = \frac{y}{x}$$

Concept 1 | Defining Angles and Basic Trigonometric Functions

F02 | V02

Q024 | WRITTEN

C05-K01-5-2-EX2

Convert $\frac{7\pi}{6}, \frac{4\pi}{5}, \frac{2\pi}{3}, \frac{4\pi}{3}, -\frac{5\pi}{4}, -\frac{\pi}{2}, -\frac{11\pi}{2}, -\frac{7\pi}{18}$ to degrees.

WORKING AREA

FORMULA BANK

Angle family

$$\theta = \alpha + 360^\circ \times n$$

Degree / radian

$$\theta_{rad} = \theta_{deg} \times \frac{\pi}{180}$$

Sector measures

$$L = r\theta \qquad S = \frac{1}{2}r^2\theta$$

Unit circle

$$\sin\theta = y \quad \cos\theta = x \quad \tan\theta = \frac{y}{x}$$

Concept 1 | Defining Angles and Basic Trigonometric Functions

F02 | V03

Q025 | WRITTEN

C05-K01-5-2-EX3

Find L and S for: (a) $r = 6,\ \theta = \frac{7\pi}{6}$; (b) $r = 4,\ \theta = \frac{2\pi}{5}$; (c) $r = 6,\ \theta = \frac{5\pi}{4}$.

WORKING AREA

FORMULA BANK

Angle family

$$\theta = \alpha + 360^\circ \times n$$

Degree / radian

$$\theta_{rad} = \theta_{deg} \times \frac{\pi}{180}$$

Sector measures

$$L = r\theta \qquad S = \frac{1}{2}r^2\theta$$

Unit circle

$\sin\theta = y \quad \cos\theta = x \quad \tan\theta = \frac{y}{x}$

Concept 1 | Defining Angles and Basic Trigonometric Functions

F02 | V04

Q026 | WRITTEN

C05-K01-5-2-EX4

Draw terminal sides for $\frac{\pi}{4}, \frac{\pi}{3}, \frac{5\pi}{6}, \frac{3\pi}{4}, \frac{4\pi}{3}, \frac{7\pi}{6}, -\frac{\pi}{3}, -\frac{\pi}{4}$.

WORKING AREA

FORMULA BANK

Angle family

$$\theta = \alpha + 360° \times n$$

Degree / radian

$$\theta_{rad} = \theta_{deg} \times \frac{\pi}{180}$$

Sector measures

$$L = r\theta \qquad S = \frac{1}{2}r^2\theta$$

Unit circle

$$\sin\theta = y \quad \cos\theta = x \quad \tan\theta = \frac{y}{x}$$

Concept 1 | Defining Angles and Basic Trigonometric Functions

F02 | V04

Q027 | WRITTEN

C05-K01-5-2-CH1

A sector of radius 9 cm and central angle $4\pi/3$ is joined to form a cone. Determine the exact vertical height and total volume.

WORKING AREA

FORMULA BANK

Angle family

$$\theta = \alpha + 360° \times n$$

Degree / radian

$$\theta_{rad} = \theta_{deg} \times \frac{\pi}{180}$$

Sector measures

$$L = r\theta \qquad S = \frac{1}{2}r^2\theta$$

Unit circle

$$\sin\theta = y \quad \cos\theta = x \quad \tan\theta = \frac{y}{x}$$

Concept 1 | Defining Angles and Basic Trigonometric Functions

F02 | V01

Q028 | MCQ

C05-K01-5-2-GM01

Convert 150° to radians:

A $\frac{5\pi}{6}$

B $\frac{3\pi}{5}$

C $\frac{5\pi}{3}$

D $\frac{\pi}{6}$

WORKING AREA

FORMULA BANK

Angle family

$$\theta = \alpha + 360^\circ \times n$$

Degree / radian

$$\theta_{rad} = \theta_{deg} \times \frac{\pi}{180}$$

Sector measures

$$L = r\theta \qquad S = \frac{1}{2}r^2\theta$$

Unit circle

$\sin\theta = y \quad \cos\theta = x \quad \tan\theta = \frac{y}{x}$

Concept 1 | Defining Angles and Basic Trigonometric Functions

F02 | V02

Q029 | MCQ

C05-K01-5-2-GM02

Convert $-\frac{5\pi}{6}$ to degrees:

A $-120°$

B $-150°$

C $150°$

D $-300°$

WORKING AREA

FORMULA BANK

Angle family

$$\theta = \alpha + 360° \times n$$

Degree / radian

$$\theta_{rad} = \theta_{deg} \times \frac{\pi}{180}$$

Sector measures

$$L = r\theta \qquad S = \frac{1}{2}r^2\theta$$

Unit circle

$$\sin\theta = y \qquad \cos\theta = x \qquad \tan\theta = \frac{y}{x}$$

Concept 1 | Defining Angles and Basic Trigonometric Functions

F02 | V03

Q030 | MCQ C05-K01-5-2-GM03

For $r = 5,\ \theta = \dfrac{2\pi}{3}$, the arc length is:

A $\dfrac{5\pi}{3}$

B $\dfrac{10\pi}{3}$

C $\dfrac{25\pi}{3}$

D 10π

WORKING AREA

FORMULA BANK

Angle family

$$\theta = \alpha + 360^\circ \times n$$

Degree / radian

$$\theta_{rad} = \theta_{deg} \times \frac{\pi}{180}$$

Sector measures

$$L = r\theta \qquad S = \frac{1}{2}r^2\theta$$

Unit circle

$\sin\theta = y \quad \cos\theta = x \quad \tan\theta = \dfrac{y}{x}$

Concept 1 | Defining Angles and Basic Trigonometric Functions

F02 | V03

Q031 | MCQ

C05-K01-5-2-GM04

For $r = 4,\ \theta = \frac{\pi}{2}$, the sector area is:

A π

B 2π

C 4π

D 8π

WORKING AREA

FORMULA BANK

Angle family

$$\theta = \alpha + 360° \times n$$

Degree / radian

$$\theta_{rad} = \theta_{deg} \times \frac{\pi}{180}$$

Sector measures

$$L = r\theta \qquad S = \frac{1}{2}r^2\theta$$

Unit circle

$\sin\theta = y \quad \cos\theta = x \quad \tan\theta = \frac{y}{x}$

Concept 1 | Defining Angles and Basic Trigonometric Functions

F02 | V01

Q032 | MCQ

C05-K01-5-2-GM05

The radian measure of a full turn is:

A π

B 2π

C 180

D 360

WORKING AREA

FORMULA BANK

Angle family

$$\theta = \alpha + 360^\circ \times n$$

Degree / radian

$$\theta_{rad} = \theta_{deg} \times \frac{\pi}{180}$$

Sector measures

$$L = r\theta \qquad S = \frac{1}{2} r^2 \theta$$

Unit circle

$\sin\theta = y \quad \cos\theta = x \quad \tan\theta = \frac{y}{x}$

Concept 1 | Defining Angles and Basic Trigonometric Functions

F02 | V02

Q033 | MCQ

C05-K01-5-2-GM06

The degree measure of $\frac{7\pi}{4}$ is:

A 210°	**B** 270°
C 315°	**D** 360°

WORKING AREA

FORMULA BANK

Angle family

$$\theta = \alpha + 360^\circ \times n$$

Degree / radian

$$\theta_{rad} = \theta_{deg} \times \frac{\pi}{180}$$

Sector measures

$$L = r\theta \qquad S = \frac{1}{2}r^2\theta$$

Unit circle

$\sin\theta = y \quad \cos\theta = x \quad \tan\theta = \frac{y}{x}$

Concept 1 | Defining Angles and Basic Trigonometric Functions

F02 | V04

Q034 | MCQ C05-K01-5-2-GM07

The principal angle for $-\frac{11\pi}{6}$ is:

A $\frac{\pi}{6}$

B $-\frac{\pi}{6}$

C $\frac{5\pi}{6}$

D $\frac{11\pi}{6}$

WORKING AREA

FORMULA BANK

Angle family

$$\theta = \alpha + 360^\circ \times n$$

Degree / radian

$$\theta_{rad} = \theta_{deg} \times \frac{\pi}{180}$$

Sector measures

$$L = r\theta \qquad S = \frac{1}{2}r^2\theta$$

Unit circle

$$\sin\theta = y \quad \cos\theta = x \quad \tan\theta = \frac{y}{x}$$

Concept 1 | Defining Angles and Basic Trigonometric Functions

F02 | V03

Q035 | MCQ

C05-K01-5-2-GM08

For $r = 3,\ \theta = \pi$, the sector area is:

A $\frac{3\pi}{2}$

B 3π

C 9π

D $\frac{9\pi}{2}$

WORKING AREA

FORMULA BANK

Angle family

$$\theta = \alpha + 360^\circ \times n$$

Degree / radian

$$\theta_{rad} = \theta_{deg} \times \frac{\pi}{180}$$

Sector measures

$$L = r\theta \qquad S = \frac{1}{2}r^2\theta$$

Unit circle

$\sin\theta = y \quad \cos\theta = x \quad \tan\theta = \frac{y}{x}$

Concept 1 | Defining Angles and Basic Trigonometric Functions

F02 | V04

Q036 | MCQ

C05-K01-5-2-GM09

The angle $\frac{5\pi}{3}$ lies in:

A QI

B QII

C QIII

D QIV

WORKING AREA

FORMULA BANK

Angle family

$$\theta = \alpha + 360° \times n$$

Degree / radian

$$\theta_{rad} = \theta_{deg} \times \frac{\pi}{180}$$

Sector measures

$$L = r\theta \qquad S = \frac{1}{2}r^2\theta$$

Unit circle

$$\sin\theta = y \quad \cos\theta = x \quad \tan\theta = \frac{y}{x}$$

Concept 1 | Defining Angles and Basic Trigonometric Functions

F02 | V01

Q037 | WRITTEN

C05-K01-5-2-GW01

Convert 252° to radians.

WORKING AREA

FORMULA BANK

Angle family

$$\theta = \alpha + 360^\circ \times n$$

Degree / radian

$$\theta_{rad} = \theta_{deg} \times \frac{\pi}{180}$$

Sector measures

$$L = r\theta \qquad S = \frac{1}{2}r^2\theta$$

Unit circle

$$\sin\theta = y \qquad \cos\theta = x \qquad \tan\theta = \frac{y}{x}$$

Concept 1 | Defining Angles and Basic Trigonometric Functions

F02 | V03

Q038 | WRITTEN

C05-K01-5-2-GW02

A sector has $r = 8$ and $\theta = \frac{3\pi}{4}$. Find L and S.

WORKING AREA

FORMULA BANK

Angle family

$$\theta = \alpha + 360^\circ \times n$$

Degree / radian

$$\theta_{rad} = \theta_{deg} \times \frac{\pi}{180}$$

Sector measures

$$L = r\theta \qquad S = \frac{1}{2}r^2\theta$$

Unit circle

$$\sin\theta = y \qquad \cos\theta = x \qquad \tan\theta = \frac{y}{x}$$

Concept 1 | Defining Angles and Basic Trigonometric Functions

F03 | V01

Q039 | WRITTEN

C05-K01-5-3-WU

If $\theta = \frac{4\pi}{3}$, find $\sin\theta,\ \cos\theta,\ \tan\theta$.

WORKING AREA

FORMULA BANK

Angle family

$$\theta = \alpha + 360^\circ \times n$$

Degree / radian

$$\theta_{rad} = \theta_{deg} \times \frac{\pi}{180}$$

Sector measures

$$L = r\theta \qquad S = \frac{1}{2}r^2\theta$$

Unit circle

$$\sin\theta = y \quad \cos\theta = x \quad \tan\theta = \frac{y}{x}$$

Concept 1 | Defining Angles and Basic Trigonometric Functions

F03 | V01

Q040 | WRITTEN

C05-K01-5-3-TR

Find the exact values (write undefined where appropriate): $\sin\frac{7\pi}{4}$, $\cos\frac{\pi}{3}$, $\tan\frac{\pi}{6}$, $\sin(-\frac{2\pi}{3})$, $\cos(-\pi)$, $\tan(-\frac{\pi}{2})$.

WORKING AREA

FORMULA BANK

Angle family

$$\theta = \alpha + 360° \times n$$

Degree / radian

$$\theta_{rad} = \theta_{deg} \times \frac{\pi}{180}$$

Sector measures

$$L = r\theta \qquad S = \frac{1}{2}r^2\theta$$

Unit circle

$$\sin\theta = y \qquad \cos\theta = x \qquad \tan\theta = \frac{y}{x}$$

Concept 1 | Defining Angles and Basic Trigonometric Functions

F03 | V01

Q041 | WRITTEN | C05-K01-5-3-EX

Find the exact values of the eighteen trigonometric functions listed in the given exercise (a)–(r), including negative and undefined cases.

Find the values of the following trigonometric functions. If the value is undefined, write "undefined".

(a) $\sin\frac{\pi}{3}$	**(b)** $\sin\frac{7}{6}\pi$	**(c)** $\sin\pi$
(d) $\cos\frac{5}{4}\pi$	**(e)** $\cos\frac{5}{3}\pi$	**(f)** $\cos\frac{7}{6}\pi$
(g) $\tan\frac{2}{3}\pi$	**(h)** $\tan\frac{5}{4}\pi$	**(i)** $\tan\pi$
(j) $\sin\left(-\frac{\pi}{6}\right)$	**(k)** $\sin\left(-\frac{5}{4}\pi\right)$	**(l)** $\sin\left(-\frac{5}{3}\pi\right)$
(m) $\cos\left(-\frac{\pi}{4}\right)$	**(n)** $\cos\left(-\frac{5}{6}\pi\right)$	**(o)** $\cos\left(-\frac{4}{3}\pi\right)$
(p) $\tan\left(-\frac{4}{3}\pi\right)$	**(q)** $\tan\left(-\frac{11}{6}\pi\right)$	**(r)** $\tan\left(-\frac{3}{2}\pi\right)$

FORMULA BANK

Angle family

$$\theta = \alpha + 360^\circ \times n$$

Degree / radian

$$\theta_{rad} = \theta_{deg} \times \frac{\pi}{180}$$

Sector measures

$$L = r\theta \qquad S = \frac{1}{2}r^2\theta$$

Unit circle

$$\sin\theta = y \qquad \cos\theta = x \qquad \tan\theta = \frac{y}{x}$$

WORKING AREA

Concept 1 | Defining Angles and Basic Trigonometric Functions

F03 | V02

Q042 | WRITTEN

C05-K01-5-3-CH1

The panel height is $h(t) = 3\sin(\frac{2\pi}{3}t - \frac{\pi}{4}) + 1.5$. Find the exact height at $t = 0$.

WORKING AREA

FORMULA BANK

Angle family

$$\theta = \alpha + 360° \times n$$

Degree / radian

$$\theta_{rad} = \theta_{deg} \times \frac{\pi}{180}$$

Sector measures

$$L = r\theta \qquad S = \frac{1}{2}r^2\theta$$

Unit circle

$\sin\theta = y$ $\cos\theta = x$ $\tan\theta = \frac{y}{x}$

Concept 1 | Defining Angles and Basic Trigonometric Functions

F03 | V01

Q043 | MCQ

C05-K01-5-3-GM01

$\sin\frac{\pi}{6}$ equals:

A 0

B $\frac{1}{2}$

C $\frac{\sqrt{3}}{2}$

D 1

WORKING AREA

FORMULA BANK

Angle family

$$\theta = \alpha + 360^\circ \times n$$

Degree / radian

$$\theta_{rad} = \theta_{deg} \times \frac{\pi}{180}$$

Sector measures

$$L = r\theta \qquad S = \frac{1}{2}r^2\theta$$

Unit circle

$$\sin\theta = y \qquad \cos\theta = x \qquad \tan\theta = \frac{y}{x}$$

Concept 1 | Defining Angles and Basic Trigonometric Functions

F03 | V01

Q044 | MCQ C05-K01-5-3-GM02

$\cos \frac{2\pi}{3}$ equals:

A -1

B $-\frac{1}{2}$

C $\frac{1}{2}$

D $\frac{\sqrt{3}}{2}$

WORKING AREA

FORMULA BANK

Angle family

$$\theta = \alpha + 360° \times n$$

Degree / radian

$$\theta_{rad} = \theta_{deg} \times \frac{\pi}{180}$$

Sector measures

$$L = r\theta \qquad S = \frac{1}{2}r^2\theta$$

Unit circle

$$\sin\theta = y \qquad \cos\theta = x \qquad \tan\theta = \frac{y}{x}$$

Concept 1 | Defining Angles and Basic Trigonometric Functions

F03 | V01

Q045 | MCQ

C05-K01-5-3-GM03

$\tan \frac{3\pi}{4}$ equals:

A -1

B 0

C 1

D undefined

WORKING AREA

FORMULA BANK

Angle family

$$\theta = \alpha + 360^\circ \times n$$

Degree / radian

$$\theta_{rad} = \theta_{deg} \times \frac{\pi}{180}$$

Sector measures

$$L = r\theta \qquad S = \frac{1}{2}r^2\theta$$

Unit circle

$$\sin\theta = y \qquad \cos\theta = x \qquad \tan\theta = \frac{y}{x}$$

Concept 1 | Defining Angles and Basic Trigonometric Functions

F03 | V02

Q046 | MCQ

C05-K01-5-3-GM04

$\sin(-\frac{\pi}{2})$ equals:

A -1

B 0

C 1

D undefined

WORKING AREA

FORMULA BANK

Angle family

$$\theta = \alpha + 360^\circ \times n$$

Degree / radian

$$\theta_{rad} = \theta_{deg} \times \frac{\pi}{180}$$

Sector measures

$$L = r\theta \qquad S = \frac{1}{2}r^2\theta$$

Unit circle

$$\sin\theta = y \qquad \cos\theta = x \qquad \tan\theta = \frac{y}{x}$$

Concept 1 | Defining Angles and Basic Trigonometric Functions

F03 | V01

Q047 | MCQ

C05-K01-5-3-GM05

$\tan \frac{\pi}{2}$ is:

A 0

B 1

C -1

D undefined

WORKING AREA

FORMULA BANK

Angle family

$$\theta = \alpha + 360° \times n$$

Degree / radian

$$\theta_{rad} = \theta_{deg} \times \frac{\pi}{180}$$

Sector measures

$$L = r\theta \qquad S = \frac{1}{2}r^2\theta$$

Unit circle

$$\sin\theta = y \qquad \cos\theta = x \qquad \tan\theta = \frac{y}{x}$$

Concept 1 | Defining Angles and Basic Trigonometric Functions

F03 | V01

Q048 | MCQ

C05-K01-5-3-GM06

$\cos \frac{5\pi}{3}$ equals:

A $-\frac{1}{2}$

B $\frac{1}{2}$

C $-\frac{\sqrt{3}}{2}$

D $\frac{\sqrt{3}}{2}$

WORKING AREA

FORMULA BANK

Angle family

$$\theta = \alpha + 360° \times n$$

Degree / radian

$$\theta_{rad} = \theta_{deg} \times \frac{\pi}{180}$$

Sector measures

$$L = r\theta \qquad S = \frac{1}{2} r^2 \theta$$

Unit circle

$$\sin\theta = y \quad \cos\theta = x \quad \tan\theta = \frac{y}{x}$$

Concept 1 | Defining Angles and Basic Trigonometric Functions

F03 | V01

Q049 | MCQ

C05-K01-5-3-GM07

If $P(x,\ y)$ lies on the unit circle, then $\sin\theta$ is:

A x

B y

C x/y

D 1/x

WORKING AREA

FORMULA BANK

Angle family

$$\theta = \alpha + 360° \times n$$

Degree / radian

$$\theta_{rad} = \theta_{deg} \times \frac{\pi}{180}$$

Sector measures

$$L = r\theta \qquad S = \frac{1}{2}r^2\theta$$

Unit circle

$\sin\theta = y$ $\quad \cos\theta = x$ $\quad \tan\theta = \frac{y}{x}$

Concept 1 | Defining Angles and Basic Trigonometric Functions

F03 | V02

Q050 | MCQ

C05-K01-5-3-GM08

$\tan(-\frac{3\pi}{4})$ equals:

A -1

B 0

C 1

D undefined

WORKING AREA

FORMULA BANK

Angle family

$$\theta = \alpha + 360° \times n$$

Degree / radian

$$\theta_{rad} = \theta_{deg} \times \frac{\pi}{180}$$

Sector measures

$$L = r\theta \qquad S = \frac{1}{2}r^2\theta$$

Unit circle

$$\sin\theta = y \qquad \cos\theta = x \qquad \tan\theta = \frac{y}{x}$$

Concept 1 | Defining Angles and Basic Trigonometric Functions

F03 | V01

Q051 | MCQ

C05-K01-5-3-GM09

$\sin \frac{11\pi}{6}$ equals:

A $-\frac{1}{2}$

B $\frac{1}{2}$

C $-\frac{\sqrt{3}}{2}$

D $\frac{\sqrt{3}}{2}$

WORKING AREA

FORMULA BANK

Angle family

$$\theta = \alpha + 360^\circ \times n$$

Degree / radian

$$\theta_{rad} = \theta_{deg} \times \frac{\pi}{180}$$

Sector measures

$$L = r\theta \qquad S = \frac{1}{2}r^2\theta$$

Unit circle

$$\sin\theta = y \qquad \cos\theta = x \qquad \tan\theta = \frac{y}{x}$$

Concept 1 | Defining Angles and Basic Trigonometric Functions

F03 | V01

Q052 | WRITTEN

C05-K01-5-3-GW01

Find $\sin\frac{5\pi}{6}$, $\cos\frac{4\pi}{3}$, $\tan\frac{7\pi}{4}$.

WORKING AREA

FORMULA BANK

Angle family

$$\theta = \alpha + 360^\circ \times n$$

Degree / radian

$$\theta_{rad} = \theta_{deg} \times \frac{\pi}{180}$$

Sector measures

$$L = r\theta \qquad S = \frac{1}{2}r^2\theta$$

Unit circle

$$\sin\theta = y \quad \cos\theta = x \quad \tan\theta = \frac{y}{x}$$

Concept 1 | Defining Angles and Basic Trigonometric Functions

F03 | V02

Q053 | WRITTEN

C05-K01-5-3-GW02

Find $\sin(-\frac{5\pi}{6}),\ \cos(-\frac{7\pi}{6}),\ \tan(-\frac{\pi}{3})$.

WORKING AREA

FORMULA BANK

Angle family

$$\theta = \alpha + 360^\circ \times n$$

Degree / radian

$$\theta_{rad} = \theta_{deg} \times \frac{\pi}{180}$$

Sector measures

$$L = r\theta \qquad S = \frac{1}{2}r^2\theta$$

Unit circle

$$\sin\theta = y \qquad \cos\theta = x \qquad \tan\theta = \frac{y}{x}$$

Concept 1 | Defining Angles and Basic Trigonometric Functions

F03 | V01

Q054 | WRITTEN

C05-K01-5-3-GW03

For $P = \left(-\frac{1}{2}, \frac{\sqrt{3}}{2}\right)$ on the unit circle, find the corresponding $\sin\theta,\ \cos\theta,\ \tan\theta$.

WORKING AREA

FORMULA BANK

Angle family

$$\theta = \alpha + 360° \times n$$

Degree / radian

$$\theta_{rad} = \theta_{deg} \times \frac{\pi}{180}$$

Sector measures

$$L = r\theta \qquad S = \frac{1}{2}r^2\theta$$

Unit circle

$$\sin\theta = y \quad \cos\theta = x \quad \tan\theta = \frac{y}{x}$$

Concept 1 | Defining Angles and Basic Trigonometric Functions

F03 | V01

Q055 | WRITTEN

C05-K01-5-3-GW04

Find the exact value of $3\sin\frac{3\pi}{2} - 2\cos\pi$.

WORKING AREA

FORMULA BANK

Angle family

$$\theta = \alpha + 360^\circ \times n$$

Degree / radian

$$\theta_{rad} = \theta_{deg} \times \frac{\pi}{180}$$

Sector measures

$$L = r\theta \qquad S = \frac{1}{2}r^2\theta$$

Unit circle

$$\sin\theta = y \qquad \cos\theta = x \qquad \tan\theta = \frac{y}{x}$$

Concept 1 | Defining Angles and Basic Trigonometric Functions

F03 | V01

Q056 | WRITTEN

C05-K01-5-3-GW05

State the two standard angles where $\tan\theta$ is undefined in $[0,\ 2\pi)$.

WORKING AREA

FORMULA BANK

Angle family

$$\theta = \alpha + 360^\circ \times n$$

Degree / radian

$$\theta_{rad} = \theta_{deg} \times \frac{\pi}{180}$$

Sector measures

$$L = r\theta \qquad S = \frac{1}{2}r^2\theta$$

Unit circle

$$\sin\theta = y \quad \cos\theta = x \quad \tan\theta = \frac{y}{x}$$

Concept 1 | Defining Angles and Basic Trigonometric Functions

F04 | V01

Q057 | WRITTEN

C05-K01-5-4-WU

Solve for θ, $0 \le \theta < 2\pi$: (a) $\sin\theta = -\frac{1}{2}$; (b) $\cos\theta = \frac{1}{\sqrt{2}}$; (c) $\tan\theta = -\sqrt{3}$.

WORKING AREA

FORMULA BANK

Angle family

$$\theta = \alpha + 360^\circ \times n$$

Degree / radian

$$\theta_{rad} = \theta_{deg} \times \frac{\pi}{180}$$

Sector measures

$$L = r\theta \qquad S = \frac{1}{2}r^2\theta$$

Unit circle

$\sin\theta = y$ $\quad \cos\theta = x$ $\quad \tan\theta = \frac{y}{x}$

Concept 1 | Defining Angles and Basic Trigonometric Functions

F04 | V01

Q058 | WRITTEN C05-K01-5-4-TR

Solve the six trigonometric equations in the given Try set for $0 \leq \theta < 2\pi$: $\sin\theta = 1/\sqrt{2}$, $\cos\theta = -\sqrt{3}/2$, $\tan\theta = 1/\sqrt{3}$, $\sin\theta = 0$, $\cos\theta = 1/2$, $\tan\theta = -1$.

WORKING AREA

FORMULA BANK

Angle family

$$\theta = \alpha + 360^\circ \times n$$

Degree / radian

$$\theta_{rad} = \theta_{deg} \times \frac{\pi}{180}$$

Sector measures

$$L = r\theta \qquad S = \frac{1}{2}r^2\theta$$

Unit circle

$$\sin\theta = y \qquad \cos\theta = x \qquad \tan\theta = \frac{y}{x}$$

Concept 1 | Defining Angles and Basic Trigonometric Functions

F04 | V01

Q059 | WRITTEN C05-K01-5-4-EX

Solve the sixteen equations in the given Exercise set (a)–(p) for $0 \le \theta < 2\pi$.

WORKING AREA

FORMULA BANK

Angle family

$$\theta = \alpha + 360^\circ \times n$$

Degree / radian

$$\theta_{rad} = \theta_{deg} \times \frac{\pi}{180}$$

Sector measures

$$L = r\theta \qquad S = \frac{1}{2}r^2\theta$$

Unit circle

$$\sin\theta = y \qquad \cos\theta = x \qquad \tan\theta = \frac{y}{x}$$

Concept 1 | Defining Angles and Basic Trigonometric Functions

F04 | V03

Q060 | WRITTEN

C05-K01-5-4-CH1

Solve $\tan^2\theta + (1-\sqrt{3})\tan\theta = \sqrt{3}$ for $0 \le \theta < 360°$.

WORKING AREA

FORMULA BANK

Angle family

$$\theta = \alpha + 360° \times n$$

Degree / radian

$$\theta_{rad} = \theta_{deg} \times \frac{\pi}{180}$$

Sector measures

$$L = r\theta \qquad S = \frac{1}{2}r^2\theta$$

Unit circle

$\sin\theta = y$ $\cos\theta = x$ $\tan\theta = \frac{y}{x}$

Concept 1 | Defining Angles and Basic Trigonometric Functions

F04 | V01

Q061 | MCQ

C05-K01-5-4-GM01

For $0 \le \theta < 2\pi$, $\sin\theta = \frac{1}{2}$ gives:

A $\frac{\pi}{6}, \frac{5\pi}{6}$

B $\frac{\pi}{3}, \frac{2\pi}{3}$

C $\frac{\pi}{6}, \frac{7\pi}{6}$

D $\frac{5\pi}{6}, \frac{11\pi}{6}$

WORKING AREA

FORMULA BANK

Angle family

$$\theta = \alpha + 360^\circ \times n$$

Degree / radian

$$\theta_{rad} = \theta_{deg} \times \frac{\pi}{180}$$

Sector measures

$$L = r\theta \qquad S = \frac{1}{2}r^2\theta$$

Unit circle

$\sin\theta = y \quad \cos\theta = x \quad \tan\theta = \frac{y}{x}$

Concept 1 | Defining Angles and Basic Trigonometric Functions

F04 | V01

Q062 | MCQ C05-K01-5-4-GM02

For $0 \le \theta < 2\pi$, $\cos\theta = -\frac{1}{2}$ gives:

A $\frac{\pi}{3}, \frac{5\pi}{3}$

B $\frac{2\pi}{3}, \frac{4\pi}{3}$

C $\frac{\pi}{6}, \frac{5\pi}{6}$

D $\frac{4\pi}{3}, \frac{5\pi}{3}$

WORKING AREA

FORMULA BANK

Angle family

$$\theta = \alpha + 360^\circ \times n$$

Degree / radian

$$\theta_{rad} = \theta_{deg} \times \frac{\pi}{180}$$

Sector measures

$$L = r\theta \qquad S = \frac{1}{2}r^2\theta$$

Unit circle

$\sin\theta = y$ $\cos\theta = x$ $\tan\theta = \frac{y}{x}$

Concept 1 | Defining Angles and Basic Trigonometric Functions

F04 | V01

Q063 | MCQ

C05-K01-5-4-GM03

For $0 \leq \theta < 2\pi$, $\tan\theta = 1$ gives:

A $\frac{\pi}{4}, \frac{5\pi}{4}$

B $\frac{3\pi}{4}, \frac{7\pi}{4}$

C $\frac{\pi}{4}, \frac{7\pi}{4}$

D $\frac{3\pi}{4}, \frac{5\pi}{4}$

WORKING AREA

FORMULA BANK

Angle family

$$\theta = \alpha + 360^\circ \times n$$

Degree / radian

$$\theta_{rad} = \theta_{deg} \times \frac{\pi}{180}$$

Sector measures

$$L = r\theta \qquad S = \frac{1}{2}r^2\theta$$

Unit circle

$\sin\theta = y$ $\cos\theta = x$ $\tan\theta = \frac{y}{x}$

Concept 1 | Defining Angles and Basic Trigonometric Functions

F04 | V01

Q064 | MCQ

C05-K01-5-4-GM04

The solutions of $\sin\theta = 0$ in $[0,\ 2\pi)$ are:

A $0,\ \pi$

B $\frac{\pi}{2},\ \frac{3\pi}{2}$

C $0,\ 2\pi$

D π

WORKING AREA

FORMULA BANK

Angle family

$$\theta = \alpha + 360° \times n$$

Degree / radian

$$\theta_{rad} = \theta_{deg} \times \frac{\pi}{180}$$

Sector measures

$$L = r\theta \qquad S = \frac{1}{2}r^2\theta$$

Unit circle

$\sin\theta = y \quad \cos\theta = x \quad \tan\theta = \frac{y}{x}$

Concept 1 | Defining Angles and Basic Trigonometric Functions

F04 | V01

Q065 | MCQ

C05-K01-5-4-GM05

If $\cos\theta = 1$, then in $[0,\ 2\pi)$, θ is:

A 0

B $\frac{\pi}{2}$

C π

D $\frac{3\pi}{2}$

WORKING AREA

FORMULA BANK

Angle family

$$\theta = \alpha + 360° \times n$$

Degree / radian

$$\theta_{rad} = \theta_{deg} \times \frac{\pi}{180}$$

Sector measures

$$L = r\theta \qquad S = \frac{1}{2}r^2\theta$$

Unit circle

$\sin\theta = y \quad \cos\theta = x \quad \tan\theta = \frac{y}{x}$

Concept 1 | Defining Angles and Basic Trigonometric Functions

F04 | V01

Q066 | MCQ C05-K01-5-4-GM06

For $0 \le \theta < 2\pi$, $\tan\theta = -\sqrt{3}$ gives:

A $\frac{\pi}{3}, \frac{4\pi}{3}$

B $\frac{2\pi}{3}, \frac{5\pi}{3}$

C $\frac{\pi}{6}, \frac{7\pi}{6}$

D $\frac{5\pi}{6}, \frac{11\pi}{6}$

WORKING AREA

FORMULA BANK

Angle family

$$\theta = \alpha + 360^\circ \times n$$

Degree / radian

$$\theta_{rad} = \theta_{deg} \times \frac{\pi}{180}$$

Sector measures

$$L = r\theta \qquad S = \frac{1}{2}r^2\theta$$

Unit circle

$\sin\theta = y$ $\cos\theta = x$ $\tan\theta = \frac{y}{x}$

Concept 1 | Defining Angles and Basic Trigonometric Functions

F04 | V01

Q067 | MCQ

C05-K01-5-4-GM07

For $0 \le \theta < 2\pi$, $\sin\theta = -1$ gives:

A $\frac{\pi}{2}$

B $\frac{3\pi}{2}$

C π

D 0

WORKING AREA

FORMULA BANK

Angle family

$$\theta = \alpha + 360° \times n$$

Degree / radian

$$\theta_{rad} = \theta_{deg} \times \frac{\pi}{180}$$

Sector measures

$$L = r\theta \qquad S = \frac{1}{2}r^2\theta$$

Unit circle

$$\sin\theta = y \qquad \cos\theta = x \qquad \tan\theta = \frac{y}{x}$$

Concept 1 | Defining Angles and Basic Trigonometric Functions

F04 | V01

Q068 | MCQ

C05-K01-5-4-GM08

For $0 \leq \theta < 2\pi$, $\cos\theta = 0$ gives:

A $0, \pi$

B $\frac{\pi}{2}, \frac{3\pi}{2}$

C π

D $\frac{\pi}{4}, \frac{5\pi}{4}$

WORKING AREA

FORMULA BANK

Angle family

$$\theta = \alpha + 360^\circ \times n$$

Degree / radian

$$\theta_{rad} = \theta_{deg} \times \frac{\pi}{180}$$

Sector measures

$$L = r\theta \qquad S = \frac{1}{2}r^2\theta$$

Unit circle

$\sin\theta = y$ $\cos\theta = x$ $\tan\theta = \frac{y}{x}$

Concept 1 | Defining Angles and Basic Trigonometric Functions

F04 | V01

Q069 | MCQ

C05-K01-5-4-GM09

The equation $\tan\theta = 0$ has solutions in $[0,\ 2\pi)$:

A $0,\ \pi$

B $\frac{\pi}{2},\ \frac{3\pi}{2}$

C $\frac{\pi}{4},\ \frac{5\pi}{4}$

D $\pi,\ 2\pi$

WORKING AREA

FORMULA BANK

Angle family

$$\theta = \alpha + 360° \times n$$

Degree / radian

$$\theta_{rad} = \theta_{deg} \times \frac{\pi}{180}$$

Sector measures

$$L = r\theta \qquad S = \frac{1}{2}r^2\theta$$

Unit circle

$\sin\theta = y$ $\quad\cos\theta = x$ $\quad\tan\theta = \frac{y}{x}$

Concept 1 | Defining Angles and Basic Trigonometric Functions

F04 | V01

Q070 | WRITTEN

C05-K01-5-4-GW01

Solve $\cos\theta = \frac{\sqrt{3}}{2}$ for $0 \le \theta < 2\pi$.

WORKING AREA

FORMULA BANK

Angle family

$$\theta = \alpha + 360° \times n$$

Degree / radian

$$\theta_{rad} = \theta_{deg} \times \frac{\pi}{180}$$

Sector measures

$$L = r\theta \qquad S = \frac{1}{2}r^2\theta$$

Unit circle

$\sin\theta = y$ $\cos\theta = x$ $\tan\theta = \frac{y}{x}$

Concept 1 | Defining Angles and Basic Trigonometric Functions

F04 | V01

Q071 | WRITTEN

C05-K01-5-4-GW02

Solve $\sin\theta = -\frac{\sqrt{3}}{2}$ for $0 \le \theta < 2\pi$.

WORKING AREA

FORMULA BANK

Angle family

$$\theta = \alpha + 360^\circ \times n$$

Degree / radian

$$\theta_{rad} = \theta_{deg} \times \frac{\pi}{180}$$

Sector measures

$$L = r\theta \qquad S = \frac{1}{2}r^2\theta$$

Unit circle

$$\sin\theta = y \qquad \cos\theta = x \qquad \tan\theta = \frac{y}{x}$$

Concept 1 | Defining Angles and Basic Trigonometric Functions

F04 | V01

Q072 | WRITTEN

C05-K01-5-4-GW03

Solve $\tan\theta = \frac{1}{\sqrt{3}}$ for $0 \le \theta < 2\pi$.

WORKING AREA

FORMULA BANK

Angle family

$$\theta = \alpha + 360^\circ \times n$$

Degree / radian

$$\theta_{rad} = \theta_{deg} \times \frac{\pi}{180}$$

Sector measures

$$L = r\theta \qquad S = \frac{1}{2}r^2\theta$$

Unit circle

$$\sin\theta = y \qquad \cos\theta = x \qquad \tan\theta = \frac{y}{x}$$

Concept 1 | Defining Angles and Basic Trigonometric Functions

F04 | V01

Q073 | WRITTEN

C05-K01-5-4-GW04

Solve $2\cos\theta - 1 = 0$ for $0 \le \theta < 2\pi$.

WORKING AREA

FORMULA BANK

Angle family

$$\theta = \alpha + 360^\circ \times n$$

Degree / radian

$$\theta_{rad} = \theta_{deg} \times \frac{\pi}{180}$$

Sector measures

$$L = r\theta \qquad S = \frac{1}{2}r^2\theta$$

Unit circle

$$\sin\theta = y \qquad \cos\theta = x \qquad \tan\theta = \frac{y}{x}$$

Concept 1 | Defining Angles and Basic Trigonometric Functions

F04 | V01

Q074 | WRITTEN

C05-K01-5-4-GW05

Solve $2\sin\theta + \sqrt{2} = 0$ for $0 \le \theta < 2\pi$.

WORKING AREA

FORMULA BANK

Angle family

$$\theta = \alpha + 360° \times n$$

Degree / radian

$$\theta_{rad} = \theta_{deg} \times \frac{\pi}{180}$$

Sector measures

$$L = r\theta \qquad S = \frac{1}{2}r^2\theta$$

Unit circle

$$\sin\theta = y \quad \cos\theta = x \quad \tan\theta = \frac{y}{x}$$

Concept 1 | Defining Angles and Basic Trigonometric Functions

F02 | V01

Quiz M1 | QUIZ MCQ

C05-K01-Q-M01

The radian measure of 240° is:

A $\frac{2\pi}{3}$

B $\frac{4\pi}{3}$

C $\frac{3\pi}{4}$

D $\frac{5\pi}{3}$

WORKING AREA

FORMULA BANK

Angle family

$$\theta = \alpha + 360^\circ \times n$$

Degree / radian

$$\theta_{rad} = \theta_{deg} \times \frac{\pi}{180}$$

Sector measures

$$L = r\theta \qquad S = \frac{1}{2}r^2\theta$$

Unit circle

$\sin\theta = y$ $\cos\theta = x$ $\tan\theta = \frac{y}{x}$

Concept 1 | Defining Angles and Basic Trigonometric Functions

F04 | V01

Quiz M2 | QUIZ MCQ

C05-K01-Q-M02

If $\sin\theta = -\frac{\sqrt{3}}{2}$, $0 \leq \theta < 2\pi$, then θ is:

A $\frac{\pi}{3}, \frac{2\pi}{3}$

B $\frac{4\pi}{3}, \frac{5\pi}{3}$

C $\frac{2\pi}{3}, \frac{4\pi}{3}$

D $\frac{5\pi}{6}, \frac{11\pi}{6}$

WORKING AREA

FORMULA BANK

Angle family

$$\theta = \alpha + 360° \times n$$

Degree / radian

$$\theta_{rad} = \theta_{deg} \times \frac{\pi}{180}$$

Sector measures

$$L = r\theta \qquad S = \frac{1}{2}r^2\theta$$

Unit circle

$\sin\theta = y$ $\quad\cos\theta = x$ $\quad\tan\theta = \frac{y}{x}$

Concept 1 | Defining Angles and Basic Trigonometric Functions

F02 | V03

Quiz M3 | QUIZ MCQ

C05-K01-Q-M03

For a sector with $r = 6$ and $\theta = \frac{\pi}{3}$, the area is:

A 3π

B 6π

C 12π

D 18π

WORKING AREA

FORMULA BANK

Angle family

$$\theta = \alpha + 360^\circ \times n$$

Degree / radian

$$\theta_{rad} = \theta_{deg} \times \frac{\pi}{180}$$

Sector measures

$$L = r\theta \qquad S = \frac{1}{2}r^2\theta$$

Unit circle

$$\sin\theta = y \qquad \cos\theta = x \qquad \tan\theta = \frac{y}{x}$$

Concept 1 | Defining Angles and Basic Trigonometric Functions

F01 | V03

Quiz M4 | QUIZ MCQ

C05-K01-Q-M04

The general-angle family for $-300°$ is:

A $60° + 360° n$

B $300° + 360° n$

C $-60° + 360° n$

D $120° + 360° n$

WORKING AREA

FORMULA BANK

Angle family

$$\theta = \alpha + 360° \times n$$

Degree / radian

$$\theta_{rad} = \theta_{deg} \times \frac{\pi}{180}$$

Sector measures

$$L = r\theta \qquad S = \frac{1}{2} r^2 \theta$$

Unit circle

$$\sin\theta = y \qquad \cos\theta = x \qquad \tan\theta = \frac{y}{x}$$

Concept 1 | Defining Angles and Basic Trigonometric Functions

F04 | V01

Quiz W1 | QUIZ WRITTEN

C05-K01-Q-W01

Solve $\cos\theta = -\frac{\sqrt{2}}{2}$ for $0 \le \theta < 2\pi$.

WORKING AREA

FORMULA BANK

Angle family

$$\theta = \alpha + 360^\circ \times n$$

Degree / radian

$$\theta_{rad} = \theta_{deg} \times \frac{\pi}{180}$$

Sector measures

$$L = r\theta \qquad S = \frac{1}{2}r^2\theta$$

Unit circle

$$\sin\theta = y \qquad \cos\theta = x \qquad \tan\theta = \frac{y}{x}$$

Concept 1 | Defining Angles and Basic Trigonometric Functions

F02 | V02

Quiz W2 | QUIZ WRITTEN C05-K01-Q-W02

Convert $-\frac{13\pi}{6}$ to degrees and state its principal angle.

WORKING AREA

FORMULA BANK

Angle family

$$\theta = \alpha + 360° \times n$$

Degree / radian

$$\theta_{rad} = \theta_{deg} \times \frac{\pi}{180}$$

Sector measures

$$L = r\theta \qquad S = \frac{1}{2}r^2\theta$$

Unit circle

$$\sin\theta = y \qquad \cos\theta = x \qquad \tan\theta = \frac{y}{x}$$

Concept 1 | Defining Angles and Basic Trigonometric Functions

F02 | V03

Quiz W3 | QUIZ WRITTEN

C05-K01-Q-W03

Find L and S for a sector with $r = 10$ and $\theta = \frac{2\pi}{5}$.

WORKING AREA

FORMULA BANK

Angle family

$$\theta = \alpha + 360^\circ \times n$$

Degree / radian

$$\theta_{rad} = \theta_{deg} \times \frac{\pi}{180}$$

Sector measures

$$L = r\theta \qquad S = \frac{1}{2}r^2\theta$$

Unit circle

$$\sin\theta = y \qquad \cos\theta = x \qquad \tan\theta = \frac{y}{x}$$

Concept 1 | Defining Angles and Basic Trigonometric Functions

F03 | V01

Quiz W4 | QUIZ WRITTEN

C05-K01-Q-W04

Evaluate $3\sin\frac{7\pi}{6} - 2\cos\frac{4\pi}{3}$.

WORKING AREA

FORMULA BANK

Angle family

$$\theta = \alpha + 360° \times n$$

Degree / radian

$$\theta_{rad} = \theta_{deg} \times \frac{\pi}{180}$$

Sector measures

$$L = r\theta \qquad S = \frac{1}{2}r^2\theta$$

Unit circle

$$\sin\theta = y \qquad \cos\theta = x \qquad \tan\theta = \frac{y}{x}$$

EGYPTIAN BACCALAUREATE • 2026

MATHEMATICS

CLASSROOM WORKING EDITION

Chapter 05

Trigonometric Functions

1 Concept | 62 questions

B5 Landscape | one question per teaching page

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Chapter 05 | Trigonometric Functions

CLASSROOM ROUTE

Concept 2 | Trigonometric Identities and Expression Evaluation

62 questions

Official lessons: 5-5, 5-6, 5-7

Formula Bank changes at each Concept boundary.

Concept 2 | Trigonometric Identities and Expression Evaluation **OFFICIAL LESSONS 5-5, 5-6, 5-7**

Trigonometric Identities and Expression Evaluation

Equation-first recall | use the same rail beside every question in this Concept

Pythagorean

$$\sin^2\theta + \cos^2\theta = 1$$

Quotient

$$\tan\theta = \frac{\sin\theta}{\cos\theta} \qquad \cot\theta = \frac{\cos\theta}{\sin\theta}$$

Tangent identity

$$1 + \tan^2\theta = \frac{1}{\cos^2\theta}$$

Cubes

$$a^3 + b^3 = (a+b)(a^2 - ab + b^2) \quad a^3 - b^3 = (a-b)(a^2 + ab + b^2)$$

Proof target

$$\text{LHS} = \text{RHS}$$

Start with the family cue, write the first move, then use the working grid.

Concept 2 | Trigonometric Identities and Expression Evaluation

F01 | V01

Q001 | WRITTEN C05-K02-5-5-WU1

Given (a) $\sin\theta = -\frac{1}{5}$ in QIII and (b) $\tan\theta = -\sqrt{6}$ with $\frac{3\pi}{2} < \theta < 2\pi$, find the other two trigonometric values in each case.

WORKING AREA

FORMULA BANK

Pythagorean

$$\sin^2\theta + \cos^2\theta = 1$$

Quotient

$$\tan\theta = \frac{\sin\theta}{\cos\theta} \quad \cot\theta = \frac{\cos\theta}{\sin\theta}$$

Tangent identity

$$1 + \tan^2\theta = \frac{1}{\cos^2\theta}$$

Cubes

$$a^3 + b^3 = (a+b)(a^2 - ab + b^2) \quad a^3 - b^3 = (a-b)(a^2 + ab + b^2)$$

Proof target

$$\text{LHS} = \text{RHS}$$

Concept 2 | Trigonometric Identities and Expression Evaluation

F01 | V01

Q002 | WRITTEN

C05-K02-5-5-TR1

Given one value and the quadrant, find the other two: (a) $\cos\theta = -1/\sqrt{3}$, QIII; (b) $\tan\theta = -2$, QIV; (c) $\sin\theta = 3/5$, QII; (d) $\tan\theta = \sqrt{3}/2$, $\pi < \theta < 3\pi/2$.

WORKING AREA

FORMULA BANK

Pythagorean

$$\sin^2\theta + \cos^2\theta = 1$$

Quotient

$$\tan\theta = \frac{\sin\theta}{\cos\theta} \qquad \cot\theta = \frac{\cos\theta}{\sin\theta}$$

Tangent identity

$$1 + \tan^2\theta = \frac{1}{\cos^2\theta}$$

Cubes

$$a^3 + b^3 = (a+b)(a^2 - ab + b^2) \qquad a^3 - b^3 = (a-b)(a^2 + ab + b^2)$$

Proof target

$$\text{LHS} = \text{RHS}$$

Concept 2 | Trigonometric Identities and Expression Evaluation

F01 | V02

Q003 | WRITTEN

C05-K02-5-5-EX1

For each of the twelve numbered cases in the given Exercise, one of $\sin\theta, \cos\theta, \tan\theta$ and a quadrant/interval is given. Find the other two values.

WORKING AREA

FORMULA BANK

Pythagorean

$$\sin^2\theta + \cos^2\theta = 1$$

Quotient

$$\tan\theta = \frac{\sin\theta}{\cos\theta} \qquad \cot\theta = \frac{\cos\theta}{\sin\theta}$$

Tangent identity

$$1 + \tan^2\theta = \frac{1}{\cos^2\theta}$$

Cubes

$$a^3 + b^3 = (a+b)(a^2 - ab + b^2) \qquad a^3 - b^3 = (a-b)(a^2 + ab + b^2)$$

Proof target

$$\text{LHS} = \text{RHS}$$

Concept 2 | Trigonometric Identities and Expression Evaluation

F01 | V03

Q004 | WRITTEN C05-K02-5-5-CH1

Using trigonometric identities, find the other two values given $\tan\theta + \cot\theta = 12$ and $\pi < \theta < 2\pi$.

WORKING AREA

FORMULA BANK

Pythagorean

$$\sin^2\theta + \cos^2\theta = 1$$

Quotient

$$\tan\theta = \frac{\sin\theta}{\cos\theta} \qquad \cot\theta = \frac{\cos\theta}{\sin\theta}$$

Tangent identity

$$1 + \tan^2\theta = \frac{1}{\cos^2\theta}$$

Cubes

$$a^3 + b^3 = (a+b)(a^2 - ab + b^2) \qquad a^3 - b^3 = (a-b)(a^2 + ab + b^2)$$

Proof target

$$\mathrm{LHS} = \mathrm{RHS}$$

Concept 2 | Trigonometric Identities and Expression Evaluation

F01 | V01

Q005 | WRITTEN

C05-K02-5-5-GW01

Given $\sin\theta = \frac{5}{13}$ in QII, find $\cos\theta$ and $\tan\theta$.

WORKING AREA

FORMULA BANK

Pythagorean

$$\sin^2\theta + \cos^2\theta = 1$$

Quotient

$$\tan\theta = \frac{\sin\theta}{\cos\theta} \qquad \cot\theta = \frac{\cos\theta}{\sin\theta}$$

Tangent identity

$$1 + \tan^2\theta = \frac{1}{\cos^2\theta}$$

Cubes

$$a^3 + b^3 = (a+b)(a^2 - ab + b^2) \quad a^3 - b^3 = (a-b)(a^2 + ab + b^2)$$

Proof target

$$\text{LHS} = \text{RHS}$$

Concept 2 | Trigonometric Identities and Expression Evaluation

F01 | V01

Q006 | WRITTEN

C05-K02-5-5-GW02

Given $\cos\theta = -\frac{3}{5}$ in QIII, find $\sin\theta$ and $\tan\theta$.

WORKING AREA

FORMULA BANK

Pythagorean

$$\sin^2\theta + \cos^2\theta = 1$$

Quotient

$$\tan\theta = \frac{\sin\theta}{\cos\theta} \qquad \cot\theta = \frac{\cos\theta}{\sin\theta}$$

Tangent identity

$$1 + \tan^2\theta = \frac{1}{\cos^2\theta}$$

Cubes

$$a^3 + b^3 = (a+b)(a^2 - ab + b^2) \quad a^3 - b^3 = (a-b)(a^2 + ab + b^2)$$

Proof target

$$\text{LHS} = \text{RHS}$$

Concept 2 | Trigonometric Identities and Expression Evaluation

F01 | V01

Q007 | WRITTEN

C05-K02-5-5-GW03

Given $\tan\theta = \frac{3}{4}$ in QI, find $\sin\theta$ and $\cos\theta$.

WORKING AREA

FORMULA BANK

Pythagorean

$$\sin^2\theta + \cos^2\theta = 1$$

Quotient

$$\tan\theta = \frac{\sin\theta}{\cos\theta} \qquad \cot\theta = \frac{\cos\theta}{\sin\theta}$$

Tangent identity

$$1 + \tan^2\theta = \frac{1}{\cos^2\theta}$$

Cubes

$$a^3 + b^3 = (a+b)(a^2 - ab + b^2) \quad a^3 - b^3 = (a-b)(a^2 + ab + b^2)$$

Proof target

$$\text{LHS} = \text{RHS}$$

Concept 2 | Trigonometric Identities and Expression Evaluation

F01 | V01

Q008 | WRITTEN

C05-K02-5-5-GW04

Given $\tan\theta = -\frac{5}{12}$ in QIV, find $\sin\theta$ and $\cos\theta$.

WORKING AREA

FORMULA BANK

Pythagorean

$$\sin^2\theta + \cos^2\theta = 1$$

Quotient

$$\tan\theta = \frac{\sin\theta}{\cos\theta} \qquad \cot\theta = \frac{\cos\theta}{\sin\theta}$$

Tangent identity

$$1 + \tan^2\theta = \frac{1}{\cos^2\theta}$$

Cubes

$$a^3 + b^3 = (a+b)(a^2 - ab + b^2) \quad a^3 - b^3 = (a-b)(a^2 + ab + b^2)$$

Proof target

$$\text{LHS} = \text{RHS}$$

Concept 2 | Trigonometric Identities and Expression Evaluation

F01 | V01

Q009 | WRITTEN

C05-K02-5-5-GW05

Given $\sin\theta = -\frac{\sqrt{2}}{2}$ in QIII, find $\cos\theta$ and $\tan\theta$.

WORKING AREA

FORMULA BANK

Pythagorean

$$\sin^2\theta + \cos^2\theta = 1$$

Quotient

$$\tan\theta = \frac{\sin\theta}{\cos\theta} \qquad \cot\theta = \frac{\cos\theta}{\sin\theta}$$

Tangent identity

$$1 + \tan^2\theta = \frac{1}{\cos^2\theta}$$

Cubes

$$a^3 + b^3 = (a+b)(a^2 - ab + b^2) \quad a^3 - b^3 = (a-b)(a^2 + ab + b^2)$$

Proof target

$$\text{LHS} = \text{RHS}$$

Concept 2 | Trigonometric Identities and Expression Evaluation

F01 | V01

Q010 | MCQ

C05-K02-5-5-GM01

If $\sin\theta = -\frac{3}{5}$ in QIII, $\cos\theta$ is:

A $\frac{4}{5}$

B $-\frac{4}{5}$

C $\frac{3}{5}$

D $-\frac{3}{5}$

WORKING AREA

FORMULA BANK

Pythagorean

$$\sin^2\theta + \cos^2\theta = 1$$

Quotient

$$\tan\theta = \frac{\sin\theta}{\cos\theta} \qquad \cot\theta = \frac{\cos\theta}{\sin\theta}$$

Tangent identity

$$1 + \tan^2\theta = \frac{1}{\cos^2\theta}$$

Cubes

$$a^3 + b^3 = (a+b)(a^2 - ab + b^2) \qquad a^3 - b^3 = (a-b)(a^2 + ab + b^2)$$

Proof target

$$\text{LHS} = \text{RHS}$$

Concept 2 | Trigonometric Identities and Expression Evaluation

F01 | V01

Q011 | MCQ — C05-K02-5-5-GM02

If $\tan\theta = 2$ in QI, $\sin\theta$ is:

A $\frac{2\sqrt{5}}{5}$

B $\frac{\sqrt{5}}{5}$

C $-\frac{2\sqrt{5}}{5}$

D $\frac{2}{5}$

WORKING AREA

FORMULA BANK

Pythagorean

$$\sin^2\theta + \cos^2\theta = 1$$

Quotient

$$\tan\theta = \frac{\sin\theta}{\cos\theta} \qquad \cot\theta = \frac{\cos\theta}{\sin\theta}$$

Tangent identity

$$1 + \tan^2\theta = \frac{1}{\cos^2\theta}$$

Cubes

$$a^3 + b^3 = (a+b)(a^2 - ab + b^2) \qquad a^3 - b^3 = (a-b)(a^2 + ab + b^2)$$

Proof target

$$\text{LHS} = \text{RHS}$$

Concept 2 | Trigonometric Identities and Expression Evaluation

F01 | V01

Q012 | MCQ

C05-K02-5-5-GM03

If $\cos\theta = \frac{1}{\sqrt{2}}$ in QIV, $\tan\theta$ is:

A 1

B -1

C $\sqrt{2}$

D $-\sqrt{2}$

WORKING AREA

FORMULA BANK

Pythagorean

$$\sin^2\theta + \cos^2\theta = 1$$

Quotient

$$\tan\theta = \frac{\sin\theta}{\cos\theta} \qquad \cot\theta = \frac{\cos\theta}{\sin\theta}$$

Tangent identity

$$1 + \tan^2\theta = \frac{1}{\cos^2\theta}$$

Cubes

$$a^3 + b^3 = (a+b)(a^2 - ab + b^2) \quad a^3 - b^3 = (a-b)(a^2 + ab + b^2)$$

Proof target

$$\text{LHS} = \text{RHS}$$

Concept 2 | Trigonometric Identities and Expression Evaluation

F01 | V02

Q013 | MCQ C05-K02-5-5-GM04

For $\tan\theta = -\sqrt{3}$ in QII, $\sin\theta$ is:

A $\frac{1}{2}$

B $-\frac{1}{2}$

C $\frac{\sqrt{3}}{2}$

D $-\frac{\sqrt{3}}{2}$

WORKING AREA

FORMULA BANK

Pythagorean

$$\sin^2\theta + \cos^2\theta = 1$$

Quotient

$$\tan\theta = \frac{\sin\theta}{\cos\theta} \quad \cot\theta = \frac{\cos\theta}{\sin\theta}$$

Tangent identity

$$1 + \tan^2\theta = \frac{1}{\cos^2\theta}$$

Cubes

$$a^3 + b^3 = (a+b)(a^2 - ab + b^2) \quad a^3 - b^3 = (a-b)(a^2 + ab + b^2)$$

Proof target

$$\text{LHS} = \text{RHS}$$

Concept 2 | Trigonometric Identities and Expression Evaluation

F01 | V01

Q014 | MCQ

C05-K02-5-5-GM05

If $\sin\theta = \dfrac{7}{25}$ in QI, $\tan\theta$ is:

A $\dfrac{7}{24}$

B $\dfrac{24}{7}$

C $-\dfrac{7}{24}$

D $\dfrac{25}{7}$

WORKING AREA

FORMULA BANK

Pythagorean

$$\sin^2\theta + \cos^2\theta = 1$$

Quotient

$$\tan\theta = \frac{\sin\theta}{\cos\theta} \qquad \cot\theta = \frac{\cos\theta}{\sin\theta}$$

Tangent identity

$$1 + \tan^2\theta = \frac{1}{\cos^2\theta}$$

Cubes

$$a^3 + b^3 = (a+b)(a^2 - ab + b^2) \qquad a^3 - b^3 = (a-b)(a^2 + ab + b^2)$$

Proof target

$$\text{LHS} = \text{RHS}$$

Concept 2 | Trigonometric Identities and Expression Evaluation

F02 | V01

Q015 | WRITTEN

C05-K02-5-6-WU

Given $\sin\theta + \cos\theta = \frac{1}{2}$, with θ in QII, evaluate (a) $\sin\theta\cos\theta$, (b) $\sin^3\theta + \cos^3\theta$, (c) $\sin\theta - \cos\theta$.

WORKING AREA

FORMULA BANK

Pythagorean

$$\sin^2\theta + \cos^2\theta = 1$$

Quotient

$$\tan\theta = \frac{\sin\theta}{\cos\theta} \qquad \cot\theta = \frac{\cos\theta}{\sin\theta}$$

Tangent identity

$$1 + \tan^2\theta = \frac{1}{\cos^2\theta}$$

Cubes

$$a^3 + b^3 = (a+b)(a^2 - ab + b^2) \qquad a^3 - b^3 = (a-b)(a^2 + ab + b^2)$$

Proof target

$$\text{LHS} = \text{RHS}$$

Concept 2 | Trigonometric Identities and Expression Evaluation

F02 | V01

Q016 | WRITTEN

C05-K02-5-6-TR

Given $\sin\theta + \cos\theta = \frac{1}{3}$, with θ in QII, evaluate (a) $\sin\theta\cos\theta$, (b) $\sin^3\theta + \cos^3\theta$, (c) $\sin\theta - \cos\theta$.

WORKING AREA

FORMULA BANK

Pythagorean

$$\sin^2\theta + \cos^2\theta = 1$$

Quotient

$$\tan\theta = \frac{\sin\theta}{\cos\theta} \quad \cot\theta = \frac{\cos\theta}{\sin\theta}$$

Tangent identity

$$1 + \tan^2\theta = \frac{1}{\cos^2\theta}$$

Cubes

$$a^3 + b^3 = (a+b)(a^2 - ab + b^2) \quad a^3 - b^3 = (a-b)(a^2 + ab + b^2)$$

Proof target

$$\text{LHS} = \text{RHS}$$

Concept 2 | Trigonometric Identities and Expression Evaluation

F02 | V02

Q017 | WRITTEN

C05-K02-5-6-EX1

Solve the three numbered Exercise problems: (1) given $\sin\theta - \cos\theta = \sqrt{3}/2$ in QI; (2) given $\sin\theta + \cos\theta = \sqrt{3}/2$ in QIV; (3) given $\sin\theta - \cos\theta = 1/2$ in QIII. Evaluate the three expressions requested in each case.

WORKING AREA

FORMULA BANK

Pythagorean

$$\sin^2\theta + \cos^2\theta = 1$$

Quotient

$$\tan\theta = \frac{\sin\theta}{\cos\theta} \qquad \cot\theta = \frac{\cos\theta}{\sin\theta}$$

Tangent identity

$$1 + \tan^2\theta = \frac{1}{\cos^2\theta}$$

Cubes

$$a^3 + b^3 = (a+b)(a^2 - ab + b^2) \quad a^3 - b^3 = (a-b)(a^2 + ab + b^2)$$

Proof target

$$\text{LHS} = \text{RHS}$$

Concept 2 | Trigonometric Identities and Expression Evaluation

F02 | V03

Q018 | WRITTEN

C05-K02-5-6-CH1

Given $\sin\theta - \cos\theta = 0.5$ and θ in QIII, evaluate the exact value of $\sin^6\theta - \cos^6\theta$.

WORKING AREA

FORMULA BANK

Pythagorean

$$\sin^2\theta + \cos^2\theta = 1$$

Quotient

$$\tan\theta = \frac{\sin\theta}{\cos\theta} \qquad \cot\theta = \frac{\cos\theta}{\sin\theta}$$

Tangent identity

$$1 + \tan^2\theta = \frac{1}{\cos^2\theta}$$

Cubes

$$a^3 + b^3 = (a+b)(a^2 - ab + b^2) \quad a^3 - b^3 = (a-b)(a^2 + ab + b^2)$$

Proof target

$$\text{LHS} = \text{RHS}$$

Concept 2 | Trigonometric Identities and Expression Evaluation

F02 | V01

Q019 | WRITTEN C05-K02-5-6-GW01

Given $\sin\theta + \cos\theta = \frac{2}{3}$ in QII, find $\sin\theta\cos\theta$ and $\sin\theta - \cos\theta$.

WORKING AREA

FORMULA BANK

Pythagorean

$$\sin^2\theta + \cos^2\theta = 1$$

Quotient

$$\tan\theta = \frac{\sin\theta}{\cos\theta} \qquad \cot\theta = \frac{\cos\theta}{\sin\theta}$$

Tangent identity

$$1 + \tan^2\theta = \frac{1}{\cos^2\theta}$$

Cubes

$$a^3 + b^3 = (a+b)(a^2 - ab + b^2) \quad a^3 - b^3 = (a-b)(a^2 + ab + b^2)$$

Proof target

$$\text{LHS} = \text{RHS}$$

Concept 2 | Trigonometric Identities and Expression Evaluation

F02 | V02

Q020 | WRITTEN

C05-K02-5-6-GW02

Given $\sin\theta - \cos\theta = \frac{\sqrt{5}}{2}$ in QI, find $\sin\theta\cos\theta$ and $\sin^3\theta - \cos^3\theta$.

WORKING AREA

FORMULA BANK

Pythagorean

$$\sin^2\theta + \cos^2\theta = 1$$

Quotient

$$\tan\theta = \frac{\sin\theta}{\cos\theta} \qquad \cot\theta = \frac{\cos\theta}{\sin\theta}$$

Tangent identity

$$1 + \tan^2\theta = \frac{1}{\cos^2\theta}$$

Cubes

$$a^3 + b^3 = (a+b)(a^2 - ab + b^2) \qquad a^3 - b^3 = (a-b)(a^2 + ab + b^2)$$

Proof target

$$\text{LHS} = \text{RHS}$$

Concept 2 | Trigonometric Identities and Expression Evaluation

F02 | V01

Q021 | WRITTEN

C05-K02-5-6-GW03

Given $\sin\theta + \cos\theta = -\frac{1}{2}$ in QIII, find $\sin^3\theta + \cos^3\theta$.

WORKING AREA

FORMULA BANK

Pythagorean

$$\sin^2\theta + \cos^2\theta = 1$$

Quotient

$$\tan\theta = \frac{\sin\theta}{\cos\theta} \qquad \cot\theta = \frac{\cos\theta}{\sin\theta}$$

Tangent identity

$$1 + \tan^2\theta = \frac{1}{\cos^2\theta}$$

Cubes

$$a^3 + b^3 = (a+b)(a^2 - ab + b^2) \qquad a^3 - b^3 = (a-b)(a^2 + ab + b^2)$$

Proof target

$$\text{LHS} = \text{RHS}$$

Concept 2 | Trigonometric Identities and Expression Evaluation

F02 | V01

Q022 | WRITTEN

C05-K02-5-6-GW04

Given $\sin\theta - \cos\theta = \frac{1}{3}$ in QIV, find $\sin\theta + \cos\theta$.

WORKING AREA

FORMULA BANK

Pythagorean

$$\sin^2\theta + \cos^2\theta = 1$$

Quotient

$$\tan\theta = \frac{\sin\theta}{\cos\theta} \qquad \cot\theta = \frac{\cos\theta}{\sin\theta}$$

Tangent identity

$$1 + \tan^2\theta = \frac{1}{\cos^2\theta}$$

Cubes

$$a^3 + b^3 = (a+b)(a^2 - ab + b^2) \quad a^3 - b^3 = (a-b)(a^2 + ab + b^2)$$

Proof target

$$\text{LHS} = \text{RHS}$$

Concept 2 | Trigonometric Identities and Expression Evaluation

F02 | V02

Q023 | WRITTEN

C05-K02-5-6-GW05

Given $\sin\theta - \cos\theta = 1$ in QIII, find $\sin^3\theta - \cos^3\theta$.

WORKING AREA

FORMULA BANK

Pythagorean

$$\sin^2\theta + \cos^2\theta = 1$$

Quotient

$$\tan\theta = \frac{\sin\theta}{\cos\theta} \qquad \cot\theta = \frac{\cos\theta}{\sin\theta}$$

Tangent identity

$$1 + \tan^2\theta = \frac{1}{\cos^2\theta}$$

Cubes

$$a^3 + b^3 = (a+b)(a^2 - ab + b^2) \quad a^3 - b^3 = (a-b)(a^2 + ab + b^2)$$

Proof target

$$\text{LHS} = \text{RHS}$$

Concept 2 | Trigonometric Identities and Expression Evaluation

F02 | V01

Q024 | MCQ

C05-K02-5-6-GM01

If $s + c = \frac{1}{2}$, then sc equals:

A $-\frac{3}{8}$

B $\frac{3}{8}$

C $-\frac{1}{4}$

D $\frac{1}{4}$

WORKING AREA

FORMULA BANK

Pythagorean

$$\sin^2\theta + \cos^2\theta = 1$$

Quotient

$$\tan\theta = \frac{\sin\theta}{\cos\theta} \qquad \cot\theta = \frac{\cos\theta}{\sin\theta}$$

Tangent identity

$$1 + \tan^2\theta = \frac{1}{\cos^2\theta}$$

Cubes

$$a^3 + b^3 = (a+b)(a^2 - ab + b^2) \qquad a^3 - b^3 = (a-b)(a^2 + ab + b^2)$$

Proof target

$$\text{LHS} = \text{RHS}$$

Concept 2 | Trigonometric Identities and Expression Evaluation

F02 | V01

Q025 | MCQ C05-K02-5-6-GM02

If $s + c = \frac{1}{2}$, then $s^3 + c^3$ equals:

A $\frac{5}{16}$

B $\frac{11}{16}$

C $-\frac{11}{16}$

D $\frac{3}{16}$

WORKING AREA

FORMULA BANK

Pythagorean

$$\sin^2\theta + \cos^2\theta = 1$$

Quotient

$$\tan\theta = \frac{\sin\theta}{\cos\theta} \qquad \cot\theta = \frac{\cos\theta}{\sin\theta}$$

Tangent identity

$$1 + \tan^2\theta = \frac{1}{\cos^2\theta}$$

Cubes

$$a^3 + b^3 = (a+b)(a^2 - ab + b^2) \quad a^3 - b^3 = (a-b)(a^2 + ab + b^2)$$

Proof target

$$\text{LHS} = \text{RHS}$$

Concept 2 | Trigonometric Identities and Expression Evaluation

F02 | V02

Q026 | MCQ

C05-K02-5-6-GM03

If $s - c = \frac{1}{2}$ in QIII, then $s + c$ equals:

A $\frac{\sqrt{7}}{2}$

B $-\frac{\sqrt{7}}{2}$

C $\frac{1}{2}$

D $-\frac{1}{2}$

WORKING AREA

FORMULA BANK

Pythagorean

$$\sin^2\theta + \cos^2\theta = 1$$

Quotient

$$\tan\theta = \frac{\sin\theta}{\cos\theta} \qquad \cot\theta = \frac{\cos\theta}{\sin\theta}$$

Tangent identity

$$1 + \tan^2\theta = \frac{1}{\cos^2\theta}$$

Cubes

$$a^3 + b^3 = (a+b)(a^2 - ab + b^2) \qquad a^3 - b^3 = (a-b)(a^2 + ab + b^2)$$

Proof target

$$\text{LHS} = \text{RHS}$$

Concept 2 | Trigonometric Identities and Expression Evaluation

F02 | V02

Q027 | MCQ

C05-K02-5-6-GM04

If $s - c = \frac{\sqrt{3}}{2}$ in QI, then sc equals:

A $\frac{1}{8}$

B $-\frac{1}{8}$

C $\frac{3}{8}$

D $-\frac{3}{8}$

WORKING AREA

FORMULA BANK

Pythagorean

$$\sin^2\theta + \cos^2\theta = 1$$

Quotient

$$\tan\theta = \frac{\sin\theta}{\cos\theta} \qquad \cot\theta = \frac{\cos\theta}{\sin\theta}$$

Tangent identity

$$1 + \tan^2\theta = \frac{1}{\cos^2\theta}$$

Cubes

$$a^3 + b^3 = (a+b)(a^2 - ab + b^2) \quad a^3 - b^3 = (a-b)(a^2 + ab + b^2)$$

Proof target

$$\text{LHS} = \text{RHS}$$

Concept 2 | Trigonometric Identities and Expression Evaluation

F02 | V03

Q028 | MCQ

C05-K02-5-6-GM05

If $s - c = 0.5$ in QIII, then $s^6 - c^6$ equals:

A $-\frac{55\sqrt{7}}{256}$

B $\frac{55\sqrt{7}}{256}$

C $-\frac{11\sqrt{7}}{16}$

D $\frac{11}{16}$

WORKING AREA

FORMULA BANK

Pythagorean

$$\sin^2\theta + \cos^2\theta = 1$$

Quotient

$$\tan\theta = \frac{\sin\theta}{\cos\theta} \qquad \cot\theta = \frac{\cos\theta}{\sin\theta}$$

Tangent identity

$$1 + \tan^2\theta = \frac{1}{\cos^2\theta}$$

Cubes

$$a^3 + b^3 = (a+b)(a^2 - ab + b^2) \qquad a^3 - b^3 = (a-b)(a^2 + ab + b^2)$$

Proof target

$$\text{LHS} = \text{RHS}$$

Concept 2 | Trigonometric Identities and Expression Evaluation

F03 | V01

Q029 | WRITTEN

C05-K02-5-7-WU

Prove $\frac{\sin\theta}{1+\cos\theta}+\frac{1}{\tan\theta}=\frac{1}{\sin\theta}$.

WORKING AREA

FORMULA BANK

Pythagorean

$$\sin^2\theta+\cos^2\theta=1$$

Quotient

$$\tan\theta=\frac{\sin\theta}{\cos\theta}\qquad\cot\theta=\frac{\cos\theta}{\sin\theta}$$

Tangent identity

$$1+\tan^2\theta=\frac{1}{\cos^2\theta}$$

Cubes

$$a^3+b^3=(a+b)(a^2-ab+b^2)\quad a^3-b^3=(a-b)(a^2+ab+b^2)$$

Proof target

$$\mathrm{LHS}=\mathrm{RHS}$$

Concept 2 | Trigonometric Identities and Expression Evaluation

F03 | V01

Q030 | WRITTEN

C05-K02-5-7-TR

Prove (a) $\frac{1-\tan\theta}{1+\tan\theta} = \frac{\cos\theta-\sin\theta}{\cos\theta+\sin\theta}$; (b) $\frac{\cos\theta}{1-\sin\theta} - \tan\theta = \frac{1}{\cos\theta}$.

WORKING AREA

FORMULA BANK

Pythagorean

$$\sin^2\theta + \cos^2\theta = 1$$

Quotient

$$\tan\theta = \frac{\sin\theta}{\cos\theta} \qquad \cot\theta = \frac{\cos\theta}{\sin\theta}$$

Tangent identity

$$1 + \tan^2\theta = \frac{1}{\cos^2\theta}$$

Cubes

$$a^3 + b^3 = (a+b)(a^2 - ab + b^2) \quad a^3 - b^3 = (a-b)(a^2 + ab + b^2)$$

Proof target

$$\text{LHS} = \text{RHS}$$

Concept 2 | Trigonometric Identities and Expression Evaluation

F03 | V02

Q031 | WRITTEN C05-K02-5-7-EX1

Prove the six identities in the given Exercise: (a) $\tan^2\theta - \sin^2\theta = \tan^2\theta\sin^2\theta$; (b) $\tan\theta + 1/\tan\theta = 1/(\sin\theta\cos\theta)$; (c) $\cos\theta/(1-\sin\theta) - 1/\cos\theta = \tan\theta$; (d) $\tan\theta/\sin\theta - \sin\theta/\tan\theta = \sin\theta\tan\theta$; (e) $(1+\sin\theta)/\cos\theta + \cos\theta/(1+\sin\theta) = 2/\cos\theta$; (f) $(\tan^2\theta - \sin^2\theta)/\sin^2\theta = \tan^2\theta$.

WORKING AREA

FORMULA BANK

Pythagorean

$$\sin^2\theta + \cos^2\theta = 1$$

Quotient

$$\tan\theta = \frac{\sin\theta}{\cos\theta} \qquad \cot\theta = \frac{\cos\theta}{\sin\theta}$$

Tangent identity

$$1 + \tan^2\theta = \frac{1}{\cos^2\theta}$$

Cubes

$$a^3 + b^3 = (a+b)(a^2 - ab + b^2) \quad a^3 - b^3 = (a-b)(a^2 + ab + b^2)$$

Proof target

$$\mathrm{LHS} = \mathrm{RHS}$$

Concept 2 | Trigonometric Identities and Expression Evaluation

F03 | V03

Q032 | WRITTEN

C05-K02-5-7-CH1

Prove that, for all defined values of x, $\frac{1+\sin x-\cos x}{1+\sin x+\cos x}+\frac{1+\sin x+\cos x}{1+\sin x-\cos x}=2\csc x$.

WORKING AREA

FORMULA BANK

Pythagorean

$$\sin^2\theta+\cos^2\theta=1$$

Quotient

$$\tan\theta=\frac{\sin\theta}{\cos\theta}\qquad \cot\theta=\frac{\cos\theta}{\sin\theta}$$

Tangent identity

$$1+\tan^2\theta=\frac{1}{\cos^2\theta}$$

Cubes

$$a^3+b^3=(a+b)(a^2-ab+b^2)\quad a^3-b^3=(a-b)(a^2+ab+b^2)$$

Proof target

$$\text{LHS}=\text{RHS}$$

Concept 2 | Trigonometric Identities and Expression Evaluation

F03 | V01

Q033 | WRITTEN

C05-K02-5-7-GW01

Prove $\frac{1+\sin\theta}{\cos\theta} = \frac{\cos\theta}{1-\sin\theta}$.

WORKING AREA

FORMULA BANK

Pythagorean

$$\sin^2\theta + \cos^2\theta = 1$$

Quotient

$$\tan\theta = \frac{\sin\theta}{\cos\theta} \qquad \cot\theta = \frac{\cos\theta}{\sin\theta}$$

Tangent identity

$$1 + \tan^2\theta = \frac{1}{\cos^2\theta}$$

Cubes

$$a^3 + b^3 = (a+b)(a^2 - ab + b^2) \quad a^3 - b^3 = (a-b)(a^2 + ab + b^2)$$

Proof target

$$\text{LHS} = \text{RHS}$$

Concept 2 | Trigonometric Identities and Expression Evaluation

F03 | V01

Q034 | WRITTEN

C05-K02-5-7-GW02

Prove $\frac{1}{1+\cos\theta} + \frac{1}{1-\cos\theta} = 2\sec^2\theta$.

WORKING AREA

FORMULA BANK

Pythagorean

$$\sin^2\theta + \cos^2\theta = 1$$

Quotient

$$\tan\theta = \frac{\sin\theta}{\cos\theta} \qquad \cot\theta = \frac{\cos\theta}{\sin\theta}$$

Tangent identity

$$1 + \tan^2\theta = \frac{1}{\cos^2\theta}$$

Cubes

$$a^3 + b^3 = (a+b)(a^2 - ab + b^2) \qquad a^3 - b^3 = (a-b)(a^2 + ab + b^2)$$

Proof target

$$\text{LHS} = \text{RHS}$$

Concept 2 | Trigonometric Identities and Expression Evaluation

F03 | V01

Q035 | WRITTEN

C05-K02-5-7-GW03

Prove $\frac{\tan\theta}{1+\tan\theta} = \frac{\sin\theta}{\sin\theta+\cos\theta}$.

WORKING AREA

FORMULA BANK

Pythagorean

$$\sin^2\theta + \cos^2\theta = 1$$

Quotient

$$\tan\theta = \frac{\sin\theta}{\cos\theta} \quad \cot\theta = \frac{\cos\theta}{\sin\theta}$$

Tangent identity

$$1 + \tan^2\theta = \frac{1}{\cos^2\theta}$$

Cubes

$$a^3 + b^3 = (a+b)(a^2 - ab + b^2) \quad a^3 - b^3 = (a-b)(a^2 + ab + b^2)$$

Proof target

$$\text{LHS} = \text{RHS}$$

Concept 2 | Trigonometric Identities and Expression Evaluation

F03 | V02

Q036 | WRITTEN

C05-K02-5-7-GW04

Prove $\frac{\sin\theta}{1-\cos\theta} = \frac{1+\cos\theta}{\sin\theta}$.

WORKING AREA

FORMULA BANK

Pythagorean

$$\sin^2\theta + \cos^2\theta = 1$$

Quotient

$$\tan\theta = \frac{\sin\theta}{\cos\theta} \qquad \cot\theta = \frac{\cos\theta}{\sin\theta}$$

Tangent identity

$$1 + \tan^2\theta = \frac{1}{\cos^2\theta}$$

Cubes

$$a^3 + b^3 = (a+b)(a^2 - ab + b^2) \quad a^3 - b^3 = (a-b)(a^2 + ab + b^2)$$

Proof target

$$\text{LHS} = \text{RHS}$$

Concept 2 | Trigonometric Identities and Expression Evaluation

F03 | V02

Q037 | WRITTEN

C05-K02-5-7-GW05

Prove $\frac{\cos\theta}{1+\sin\theta} = \frac{1-\sin\theta}{\cos\theta}$.

WORKING AREA

FORMULA BANK

Pythagorean

$$\sin^2\theta + \cos^2\theta = 1$$

Quotient

$$\tan\theta = \frac{\sin\theta}{\cos\theta} \qquad \cot\theta = \frac{\cos\theta}{\sin\theta}$$

Tangent identity

$$1 + \tan^2\theta = \frac{1}{\cos^2\theta}$$

Cubes

$$a^3 + b^3 = (a+b)(a^2 - ab + b^2) \quad a^3 - b^3 = (a-b)(a^2 + ab + b^2)$$

Proof target

$$\text{LHS} = \text{RHS}$$

Concept 2 | Trigonometric Identities and Expression Evaluation

F03 | V01

Q038 | MCQ

C05-K02-5-7-GM01

The first move in proving a trig identity is usually to:

A Change both sides at once

B Simplify one side

C Substitute random values

D Take a square root

WORKING AREA

FORMULA BANK

Pythagorean

$$\sin^2\theta + \cos^2\theta = 1$$

Quotient

$$\tan\theta = \frac{\sin\theta}{\cos\theta} \qquad \cot\theta = \frac{\cos\theta}{\sin\theta}$$

Tangent identity

$$1 + \tan^2\theta = \frac{1}{\cos^2\theta}$$

Cubes

$$a^3 + b^3 = (a+b)(a^2 - ab + b^2) \quad a^3 - b^3 = (a-b)(a^2 + ab + b^2)$$

Proof target

$$\text{LHS} = \text{RHS}$$

Concept 2 | Trigonometric Identities and Expression Evaluation

F03 | V01

Q039 | MCQ

C05-K02-5-7-GM02

The identity used to replace $1 - \sin^2\theta$ is:

A $\cos^2\theta$

B $\tan^2\theta$

C $\sec^2\theta$

D $\sin^2\theta$

WORKING AREA

FORMULA BANK

Pythagorean

$$\sin^2\theta + \cos^2\theta = 1$$

Quotient

$$\tan\theta = \frac{\sin\theta}{\cos\theta} \qquad \cot\theta = \frac{\cos\theta}{\sin\theta}$$

Tangent identity

$$1 + \tan^2\theta = \frac{1}{\cos^2\theta}$$

Cubes

$$a^3 + b^3 = (a+b)(a^2 - ab + b^2) \quad a^3 - b^3 = (a-b)(a^2 + ab + b^2)$$

Proof target

$$\text{LHS} = \text{RHS}$$

Concept 2 | Trigonometric Identities and Expression Evaluation

F03 | V01

Q040 | MCQ

C05-K02-5-7-GM03

$\tan\theta$ should be rewritten as:

A $\sin\theta\cos\theta$

B $\sin\theta/\cos\theta$

C $\cos\theta/\sin\theta$

D $1/\sin\theta$

WORKING AREA

FORMULA BANK

Pythagorean

$$\sin^2\theta + \cos^2\theta = 1$$

Quotient

$$\tan\theta = \frac{\sin\theta}{\cos\theta} \quad \cot\theta = \frac{\cos\theta}{\sin\theta}$$

Tangent identity

$$1 + \tan^2\theta = \frac{1}{\cos^2\theta}$$

Cubes

$$a^3 + b^3 = (a+b)(a^2 - ab + b^2) \quad a^3 - b^3 = (a-b)(a^2 + ab + b^2)$$

Proof target

$$\text{LHS} = \text{RHS}$$

Concept 2 | Trigonometric Identities and Expression Evaluation

F03 | V02

Q041 | MCQ

C05-K02-5-7-GM04

To add two trig fractions, first find a:

A common numerator

B common denominator

C new quadrant

D reference angle

WORKING AREA

FORMULA BANK

Pythagorean

$$\sin^2\theta + \cos^2\theta = 1$$

Quotient

$$\tan\theta = \frac{\sin\theta}{\cos\theta} \quad \cot\theta = \frac{\cos\theta}{\sin\theta}$$

Tangent identity

$$1 + \tan^2\theta = \frac{1}{\cos^2\theta}$$

Cubes

$$a^3 + b^3 = (a+b)(a^2 - ab + b^2) \quad a^3 - b^3 = (a-b)(a^2 + ab + b^2)$$

Proof target

$$\text{LHS} = \text{RHS}$$

Concept 2 | Trigonometric Identities and Expression Evaluation

F03 | V01

Q042 | MCQ

C05-K02-5-7-GM05

The result of a valid identity proof is:

A LHS=0

B LHS=RHS

C RHS=1

D undefined

WORKING AREA

FORMULA BANK

Pythagorean

$$\sin^2\theta + \cos^2\theta = 1$$

Quotient

$$\tan\theta = \frac{\sin\theta}{\cos\theta} \qquad \cot\theta = \frac{\cos\theta}{\sin\theta}$$

Tangent identity

$$1 + \tan^2\theta = \frac{1}{\cos^2\theta}$$

Cubes

$$a^3 + b^3 = (a+b)(a^2 - ab + b^2) \quad a^3 - b^3 = (a-b)(a^2 + ab + b^2)$$

Proof target

$$\text{LHS} = \text{RHS}$$

Concept 2 | Trigonometric Identities and Expression Evaluation

F01 | V01

Q043 | MCQ   C05-K02-5-5-GM06

If $\cos\theta = -\frac{12}{13}$ in QII, $\sin\theta$ is:

A $\frac{5}{13}$

B $-\frac{5}{13}$

C $\frac{12}{13}$

D $-\frac{12}{5}$

WORKING AREA

FORMULA BANK

Pythagorean

$$\sin^2\theta + \cos^2\theta = 1$$

Quotient

$$\tan\theta = \frac{\sin\theta}{\cos\theta} \qquad \cot\theta = \frac{\cos\theta}{\sin\theta}$$

Tangent identity

$$1 + \tan^2\theta = \frac{1}{\cos^2\theta}$$

Cubes

$$a^3 + b^3 = (a+b)(a^2 - ab + b^2) \qquad a^3 - b^3 = (a-b)(a^2 + ab + b^2)$$

Proof target

$$\text{LHS} = \text{RHS}$$

Concept 2 | Trigonometric Identities and Expression Evaluation

F01 | V02

Q044 | MCQ   C05-K02-5-5-GM07

If $\tan\theta = -\frac{1}{2}$ in QII, $\cos\theta$ is:

A $\frac{2\sqrt{5}}{5}$

B $-\frac{2\sqrt{5}}{5}$

C $\frac{\sqrt{5}}{5}$

D $-\frac{\sqrt{5}}{5}$

WORKING AREA

FORMULA BANK

Pythagorean

$$\sin^2\theta + \cos^2\theta = 1$$

Quotient

$$\tan\theta = \frac{\sin\theta}{\cos\theta} \qquad \cot\theta = \frac{\cos\theta}{\sin\theta}$$

Tangent identity

$$1 + \tan^2\theta = \frac{1}{\cos^2\theta}$$

Cubes

$$a^3 + b^3 = (a+b)(a^2 - ab + b^2) \qquad a^3 - b^3 = (a-b)(a^2 + ab + b^2)$$

Proof target

$$\text{LHS} = \text{RHS}$$

Concept 2 | Trigonometric Identities and Expression Evaluation

F01 | V01

Q045 | MCQ

C05-K02-5-5-GM08

If $\sin\theta = \frac{8}{17}$ in QIV, $\tan\theta$ is:

A $\frac{8}{15}$

B $-\frac{8}{15}$

C $\frac{15}{8}$

D $-\frac{15}{8}$

WORKING AREA

FORMULA BANK

Pythagorean

$$\sin^2\theta + \cos^2\theta = 1$$

Quotient

$$\tan\theta = \frac{\sin\theta}{\cos\theta} \qquad \cot\theta = \frac{\cos\theta}{\sin\theta}$$

Tangent identity

$$1 + \tan^2\theta = \frac{1}{\cos^2\theta}$$

Cubes

$$a^3 + b^3 = (a+b)(a^2 - ab + b^2) \qquad a^3 - b^3 = (a-b)(a^2 + ab + b^2)$$

Proof target

$$\text{LHS} = \text{RHS}$$

Concept 2 | Trigonometric Identities and Expression Evaluation

F01 | V03

Q046 | MCQ C05-K02-5-5-GM09

If $\tan\theta = \sqrt{3}$ and θ is in QIII, $\sin\theta$ is:

A $\frac{1}{2}$

B $-\frac{1}{2}$

C $\frac{\sqrt{3}}{2}$

D $-\frac{\sqrt{3}}{2}$

WORKING AREA

FORMULA BANK

Pythagorean

$$\sin^2\theta + \cos^2\theta = 1$$

Quotient

$$\tan\theta = \frac{\sin\theta}{\cos\theta} \qquad \cot\theta = \frac{\cos\theta}{\sin\theta}$$

Tangent identity

$$1 + \tan^2\theta = \frac{1}{\cos^2\theta}$$

Cubes

$$a^3 + b^3 = (a+b)(a^2 - ab + b^2) \quad a^3 - b^3 = (a-b)(a^2 + ab + b^2)$$

Proof target

$$\text{LHS} = \text{RHS}$$

Concept 2 | Trigonometric Identities and Expression Evaluation

F02 | V01

Q047 | MCQ

C05-K02-5-6-GM06

If $s + c = 1$, then sc equals:

A 0

B $\frac{1}{2}$

C $-\frac{1}{2}$

D 1

WORKING AREA

FORMULA BANK

Pythagorean

$$\sin^2\theta + \cos^2\theta = 1$$

Quotient

$$\tan\theta = \frac{\sin\theta}{\cos\theta} \qquad \cot\theta = \frac{\cos\theta}{\sin\theta}$$

Tangent identity

$$1 + \tan^2\theta = \frac{1}{\cos^2\theta}$$

Cubes

$$a^3 + b^3 = (a+b)(a^2 - ab + b^2) \qquad a^3 - b^3 = (a-b)(a^2 + ab + b^2)$$

Proof target

$$\text{LHS} = \text{RHS}$$

Concept 2 | Trigonometric Identities and Expression Evaluation

F02 | V02

Q048 | MCQ

C05-K02-5-6-GM07

If $s - c = \sqrt{2}$, then sc equals:

A $-\frac{1}{2}$

B $\frac{1}{2}$

C -1

D 1

WORKING AREA

FORMULA BANK

Pythagorean

$$\sin^2\theta + \cos^2\theta = 1$$

Quotient

$$\tan\theta = \frac{\sin\theta}{\cos\theta} \qquad \cot\theta = \frac{\cos\theta}{\sin\theta}$$

Tangent identity

$$1 + \tan^2\theta = \frac{1}{\cos^2\theta}$$

Cubes

$$a^3 + b^3 = (a+b)(a^2 - ab + b^2) \quad a^3 - b^3 = (a-b)(a^2 + ab + b^2)$$

Proof target

$$\text{LHS} = \text{RHS}$$

Concept 2 | Trigonometric Identities and Expression Evaluation

F02 | V01

Q049 | MCQ

C05-K02-5-6-GM08

If $s + c = \frac{2}{3}$, then $s^3 + c^3$ equals:

A $\frac{19}{27}$

B $\frac{5}{27}$

C $-\frac{19}{27}$

D $\frac{11}{27}$

WORKING AREA

FORMULA BANK

Pythagorean

$$\sin^2\theta + \cos^2\theta = 1$$

Quotient

$$\tan\theta = \frac{\sin\theta}{\cos\theta} \qquad \cot\theta = \frac{\cos\theta}{\sin\theta}$$

Tangent identity

$$1 + \tan^2\theta = \frac{1}{\cos^2\theta}$$

Cubes

$$a^3 + b^3 = (a+b)(a^2 - ab + b^2) \quad a^3 - b^3 = (a-b)(a^2 + ab + b^2)$$

Proof target

$$\text{LHS} = \text{RHS}$$

Concept 2 | Trigonometric Identities and Expression Evaluation

F02 | V03

Q050 | MCQ — C05-K02-5-6-GM09

If $s - c = \frac{1}{2}$ in QIII, then $s^3 - c^3$ equals:

A $\frac{11}{16}$

B $-\frac{11}{16}$

C $\frac{7}{16}$

D $-\frac{7}{16}$

WORKING AREA

FORMULA BANK

Pythagorean

$$\sin^2\theta + \cos^2\theta = 1$$

Quotient

$$\tan\theta = \frac{\sin\theta}{\cos\theta} \qquad \cot\theta = \frac{\cos\theta}{\sin\theta}$$

Tangent identity

$$1 + \tan^2\theta = \frac{1}{\cos^2\theta}$$

Cubes

$$a^3 + b^3 = (a+b)(a^2 - ab + b^2) \qquad a^3 - b^3 = (a-b)(a^2 + ab + b^2)$$

Proof target

$$\text{LHS} = \text{RHS}$$

Concept 2 | Trigonometric Identities and Expression Evaluation

F03 | V01

Q051 | MCQ

C05-K02-5-7-GM06

Which identity converts $\frac{1}{\tan\theta}$ into sine and cosine?:

A $\frac{\sin\theta}{\cos\theta}$

B $\frac{\cos\theta}{\sin\theta}$

C $\sin\theta\cos\theta$

D $\sec\theta$

WORKING AREA

FORMULA BANK

Pythagorean

$$\sin^2\theta + \cos^2\theta = 1$$

Quotient

$$\tan\theta = \frac{\sin\theta}{\cos\theta} \qquad \cot\theta = \frac{\cos\theta}{\sin\theta}$$

Tangent identity

$$1 + \tan^2\theta = \frac{1}{\cos^2\theta}$$

Cubes

$$a^3 + b^3 = (a+b)(a^2 - ab + b^2) \qquad a^3 - b^3 = (a-b)(a^2 + ab + b^2)$$

Proof target

$$\text{LHS} = \text{RHS}$$

Concept 2 | Trigonometric Identities and Expression Evaluation

F03 | V02

Q052 | MCQ

C05-K02-5-7-GM07

The identity $1 - \cos^2\theta$ simplifies to:

A $\sin^2\theta$

B $\tan^2\theta$

C $\cos^2\theta$

D 1

WORKING AREA

FORMULA BANK

Pythagorean

$$\sin^2\theta + \cos^2\theta = 1$$

Quotient

$$\tan\theta = \frac{\sin\theta}{\cos\theta} \qquad \cot\theta = \frac{\cos\theta}{\sin\theta}$$

Tangent identity

$$1 + \tan^2\theta = \frac{1}{\cos^2\theta}$$

Cubes

$$a^3 + b^3 = (a+b)(a^2 - ab + b^2) \qquad a^3 - b^3 = (a-b)(a^2 + ab + b^2)$$

Proof target

$$\text{LHS} = \text{RHS}$$

Concept 2 | Trigonometric Identities and Expression Evaluation

F03 | V01

Q053 | MCQ

C05-K02-5-7-GM08

When proving an identity, which side should normally be transformed?:

A Both sides simultaneously

B One side only

C Neither side

D Only the denominator

WORKING AREA

FORMULA BANK

Pythagorean

$$\sin^2\theta + \cos^2\theta = 1$$

Quotient

$$\tan\theta = \frac{\sin\theta}{\cos\theta} \qquad \cot\theta = \frac{\cos\theta}{\sin\theta}$$

Tangent identity

$$1 + \tan^2\theta = \frac{1}{\cos^2\theta}$$

Cubes

$$a^3 + b^3 = (a+b)(a^2 - ab + b^2) \qquad a^3 - b^3 = (a-b)(a^2 + ab + b^2)$$

Proof target

$$\text{LHS} = \text{RHS}$$

Concept 2 | Trigonometric Identities and Expression Evaluation

F03 | V03

Q054 | MCQ C05-K02-5-7-GM09

After combining trig fractions, the next step is usually to use:

A A random angle

B The Pythagorean identity

C A logarithm

D A derivative

WORKING AREA

FORMULA BANK

Pythagorean

$$\sin^2\theta + \cos^2\theta = 1$$

Quotient

$$\tan\theta = \frac{\sin\theta}{\cos\theta} \qquad \cot\theta = \frac{\cos\theta}{\sin\theta}$$

Tangent identity

$$1 + \tan^2\theta = \frac{1}{\cos^2\theta}$$

Cubes

$$a^3 + b^3 = (a+b)(a^2 - ab + b^2) \qquad a^3 - b^3 = (a-b)(a^2 + ab + b^2)$$

Proof target

$$\text{LHS} = \text{RHS}$$

Concept 2 | Trigonometric Identities and Expression Evaluation

F02 | V01

Quiz M1 | QUIZ MCQ

C05-K02-Q-M01

If $\sin\theta + \cos\theta = \frac{1}{2}$, then $\sin\theta\cos\theta$ is:

A $-\frac{3}{8}$

B $\frac{3}{8}$

C $-\frac{1}{4}$

D $\frac{1}{4}$

WORKING AREA

FORMULA BANK

Pythagorean

$$\sin^2\theta + \cos^2\theta = 1$$

Quotient

$$\tan\theta = \frac{\sin\theta}{\cos\theta} \qquad \cot\theta = \frac{\cos\theta}{\sin\theta}$$

Tangent identity

$$1 + \tan^2\theta = \frac{1}{\cos^2\theta}$$

Cubes

$$a^3 + b^3 = (a+b)(a^2 - ab + b^2) \qquad a^3 - b^3 = (a-b)(a^2 + ab + b^2)$$

Proof target

$$\text{LHS} = \text{RHS}$$

Concept 2 | Trigonometric Identities and Expression Evaluation

F01 | V01

Quiz M2 | QUIZ MCQ C05-K02-Q-M02

If $\tan\theta = \frac{3}{4}$ in QI, then $\cos\theta$ is:

A $\frac{3}{5}$

B $\frac{4}{5}$

C $-\frac{4}{5}$

D $\frac{3}{4}$

WORKING AREA

FORMULA BANK

Pythagorean

$$\sin^2\theta + \cos^2\theta = 1$$

Quotient

$$\tan\theta = \frac{\sin\theta}{\cos\theta} \qquad \cot\theta = \frac{\cos\theta}{\sin\theta}$$

Tangent identity

$$1 + \tan^2\theta = \frac{1}{\cos^2\theta}$$

Cubes

$$a^3 + b^3 = (a+b)(a^2 - ab + b^2) \quad a^3 - b^3 = (a-b)(a^2 + ab + b^2)$$

Proof target

$$\text{LHS} = \text{RHS}$$

Concept 2 | Trigonometric Identities and Expression Evaluation

F03 | V01

Quiz M3 | QUIZ MCQ

C05-K02-Q-M03

Which identity is always valid where defined?:

A $\tan\theta = \dfrac{\cos\theta}{\sin\theta}$

B $1 + \tan^2\theta = \dfrac{1}{\cos^2\theta}$

C $\sin\theta + \cos\theta = 1$

D $\sin\theta\cos\theta = 1$

WORKING AREA

FORMULA BANK

Pythagorean

$$\sin^2\theta + \cos^2\theta = 1$$

Quotient

$$\tan\theta = \frac{\sin\theta}{\cos\theta} \qquad \cot\theta = \frac{\cos\theta}{\sin\theta}$$

Tangent identity

$$1 + \tan^2\theta = \frac{1}{\cos^2\theta}$$

Cubes

$$a^3 + b^3 = (a+b)(a^2 - ab + b^2) \qquad a^3 - b^3 = (a-b)(a^2 + ab + b^2)$$

Proof target

$$\text{LHS} = \text{RHS}$$

Concept 2 | Trigonometric Identities and Expression Evaluation

F02 | V02

Quiz M4 | QUIZ MCQ

C05-K02-Q-M04

If $s - c = \frac{1}{2}$ in QIII, then $s + c$ is:

A $\frac{\sqrt{7}}{2}$

B $-\frac{\sqrt{7}}{2}$

C $\frac{1}{2}$

D $-\frac{1}{2}$

WORKING AREA

FORMULA BANK

Pythagorean

$$\sin^2\theta + \cos^2\theta = 1$$

Quotient

$$\tan\theta = \frac{\sin\theta}{\cos\theta} \qquad \cot\theta = \frac{\cos\theta}{\sin\theta}$$

Tangent identity

$$1 + \tan^2\theta = \frac{1}{\cos^2\theta}$$

Cubes

$$a^3 + b^3 = (a+b)(a^2 - ab + b^2) \quad a^3 - b^3 = (a-b)(a^2 + ab + b^2)$$

Proof target

$$\text{LHS} = \text{RHS}$$

Concept 2 | Trigonometric Identities and Expression Evaluation

F01 | V01

Quiz W1 | QUIZ WRITTEN

C05-K02-Q-W01

Given $\cos\theta = -\frac{5}{13}$ in QIII, find $\sin\theta$ and $\tan\theta$.

WORKING AREA

FORMULA BANK

Pythagorean

$$\sin^2\theta + \cos^2\theta = 1$$

Quotient

$$\tan\theta = \frac{\sin\theta}{\cos\theta} \qquad \cot\theta = \frac{\cos\theta}{\sin\theta}$$

Tangent identity

$$1 + \tan^2\theta = \frac{1}{\cos^2\theta}$$

Cubes

$$a^3 + b^3 = (a+b)(a^2 - ab + b^2) \qquad a^3 - b^3 = (a-b)(a^2 + ab + b^2)$$

Proof target

$$\text{LHS} = \text{RHS}$$

Concept 2 | Trigonometric Identities and Expression Evaluation

F02 | V01

Quiz W2 | QUIZ WRITTEN

C05-K02-Q-W02

Given $s + c = \frac{2}{3}$, find sc and $s^3 + c^3$.

WORKING AREA

FORMULA BANK

Pythagorean

$$\sin^2\theta + \cos^2\theta = 1$$

Quotient

$$\tan\theta = \frac{\sin\theta}{\cos\theta} \qquad \cot\theta = \frac{\cos\theta}{\sin\theta}$$

Tangent identity

$$1 + \tan^2\theta = \frac{1}{\cos^2\theta}$$

Cubes

$$a^3 + b^3 = (a+b)(a^2 - ab + b^2) \quad a^3 - b^3 = (a-b)(a^2 + ab + b^2)$$

Proof target

$$\text{LHS} = \text{RHS}$$

Concept 2 | Trigonometric Identities and Expression Evaluation

F03 | V02

Quiz W3 | QUIZ WRITTEN

C05-K02-Q-W03

Prove $\frac{1-\sin\theta}{\cos\theta} = \frac{\cos\theta}{1+\sin\theta}$.

WORKING AREA

FORMULA BANK

Pythagorean

$$\sin^2\theta + \cos^2\theta = 1$$

Quotient

$$\tan\theta = \frac{\sin\theta}{\cos\theta} \qquad \cot\theta = \frac{\cos\theta}{\sin\theta}$$

Tangent identity

$$1 + \tan^2\theta = \frac{1}{\cos^2\theta}$$

Cubes

$$a^3 + b^3 = (a+b)(a^2 - ab + b^2) \quad a^3 - b^3 = (a-b)(a^2 + ab + b^2)$$

Proof target

$$\text{LHS} = \text{RHS}$$

Concept 2 | Trigonometric Identities and Expression Evaluation

F01 | V02

Quiz W4 | QUIZ WRITTEN C05-K02-Q-W04

Given $\tan\theta = -2$ in QIV, find $\sin\theta$ and $\cos\theta$, then verify $1 + \tan^2\theta = 1/\cos^2\theta$.

WORKING AREA

FORMULA BANK

Pythagorean

$$\sin^2\theta + \cos^2\theta = 1$$

Quotient

$$\tan\theta = \frac{\sin\theta}{\cos\theta} \qquad \cot\theta = \frac{\cos\theta}{\sin\theta}$$

Tangent identity

$$1 + \tan^2\theta = \frac{1}{\cos^2\theta}$$

Cubes

$$a^3 + b^3 = (a+b)(a^2 - ab + b^2) \quad a^3 - b^3 = (a-b)(a^2 + ab + b^2)$$

Proof target

$$\text{LHS} = \text{RHS}$$

EGYPTIAN BACCALAUREATE • 2026

MATHEMATICS

CLASSROOM WORKING EDITION

Chapter 05

Trigonometric Functions

1 Concepts | 88 questions

B5 Landscape | one question per teaching page

Eng Eslam Ahmed

Assistant Lecturer • Faculty of Engineering • Cairo University

01120009622

Chapter 05 | Trigonometric Functions

CLASSROOM ROUTE

Concept 3 | Properties and Graphs of Trigonometric Functions

88 questions

Official lessons: 5-8, 5-9, 5-10, 5-11, 5-12

The Formula Bank changes at each Concept boundary.

Concept 3 | Properties and Graphs of Trigonometric Functions OFFICIAL LESSONS 5-8, 5-9, 5-10, 5-11, 5-12

Properties and Graphs of Trigonometric Functions

Equation-first recall | use the same rail beside every question in this Concept

Parent graphs

$$y = \sin\theta,\ y = \cos\theta,\ y = \tan\theta$$

Periods

$$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$$

Range

$$\sin,\cos:\ -1 \le y \le 1 \quad \tan:\ y \in \mathbb{R}$$

Combined form

$$y = af(b\theta - q) = af\left(b\left(\theta - \frac{q}{b}\right)\right)$$

Transform order

shift $\frac{q}{b}$ $\rightarrow$ **horizontal scale** $\frac{1}{|b|}$ $\rightarrow$ **vertical scale** $|a|$

Start with the family cue, write the first move, then use the working grid.

Concept 3 | Properties and Graphs of Trigonometric Functions

F01 | V01

Q001 | WRITTEN

C05-K03-5-8-WU

(1) Express $\sin\frac{9\pi}{5}$ as a trigonometric function of an acute angle. (2) Simplify $\sin\theta+\sin(\theta+\frac{\pi}{2})+\sin(\theta+\pi)+\sin(\theta+\frac{3\pi}{2})$.

WORKING AREA

FORMULA BANK

Parent graphs

$y=\sin\theta,\ y=\cos\theta,\ y=\tan\theta$

Periods

$T_{\sin,\cos}=2\pi \quad T_{\tan}=\pi$

Range

$\sin,\cos:\ -1\le y\le 1 \quad \tan:\ y\in\mathbb{R}$

Combined form

$y=af(b\theta-q)=af\left(b\left(\theta-\frac{q}{b}\right)\right)$

Transform order

shift $\frac{q}{b}$ → horizontal scale $\frac{1}{|b|}$ → vertical scale $|a|$

Concept 3 | Properties and Graphs of Trigonometric Functions

F01 | V02

Q002 | WRITTEN

C05-K03-5-8-TR

(1) Express $\cos(-\frac{\pi}{6})$ as a trigonometric function of an acute angle. (2) Express $\sin\frac{8\pi}{7}$ as a trigonometric function of an acute angle. (3) Simplify $\cos(\frac{\pi}{2}-\theta)\sin(\pi-\theta)-\sin(\frac{3\pi}{2}+\theta)\cos(\pi-\theta)$.

WORKING AREA

FORMULA BANK

Parent graphs
$y=\sin\theta,\ y=\cos\theta,\ y=\tan\theta$

Periods
$T_{\sin,\cos}=2\pi \quad T_{\tan}=\pi$

Range
$\sin,\cos:\ -1\le y\le 1 \quad \tan:\ y\in\mathbb{R}$

Combined form
$y=af(b\theta-q)=af\left(b\left(\theta-\frac{q}{b}\right)\right)$

Transform order
shift $\frac{q}{b}$ → horizontal scale $\frac{1}{|b|}$ → vertical scale $|a|$

Concept 3 | Properties and Graphs of Trigonometric Functions

F01 | V03

Q003 | WRITTEN

C05-K03-5-8-EX

Solve all three given demands: (1) express $\cos\frac{17\pi}{5}, \cos\frac{5\pi}{7}, \sin(-\frac{7\pi}{8}), \sin(-\frac{19\pi}{8})$ using acute angles; (2) find $\tan(-\frac{\pi}{6})$ and $\tan(-\frac{\pi}{4})$; (3) simplify (a) $\sin\theta + \cos(\pi+\theta) - \cos(\pi-\theta) + \sin(\pi+\theta)$, (b) $\sin(\theta+\frac{\pi}{2})\cos(\theta+\pi) + \sin(-\theta)\cos(\frac{\pi}{2}-\theta)$.

WORKING AREA

FORMULA BANK

Parent graphs
$y = \sin\theta,\ y = \cos\theta,\ y = \tan\theta$

Periods
$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$

Range
$\sin, \cos:\ -1 \le y \le 1 \quad \tan:\ y \in \mathbb{R}$

Combined form
$y = af(b\theta - q) = af\left(b\left(\theta - \frac{q}{b}\right)\right)$

Transform order
shift $\frac{q}{b}$ → horizontal scale $\frac{1}{|b|}$ → vertical scale $|a|$

Concept 3 | Properties and Graphs of Trigonometric Functions

F01 | V03

Q004 | WRITTEN

C05-K03-5-8-CH

Simplify $\sin(\frac{3\pi}{2} - \theta)\sec(\pi + \theta) - \tan(\theta - \frac{\pi}{2})\cot(\frac{5\pi}{2} + \theta)$.

WORKING AREA

FORMULA BANK

Parent graphs

$y = \sin\theta,\ y = \cos\theta,\ y = \tan\theta$

Periods

$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$

Range

$\sin, \cos:\ -1 \leq y \leq 1 \quad \tan:\ y \in \mathbb{R}$

Combined form

$y = af(b\theta - q) = af\left(b\left(\theta - \frac{q}{b}\right)\right)$

Transform order

shift $\frac{q}{b}$ → horizontal scale $\frac{1}{|b|}$ → vertical scale $|a|$

Concept 3 | Properties and Graphs of Trigonometric Functions

F02 | V01

Q005 | WRITTEN

C05-K03-5-9-WU

Draw the parent graphs $y = \sin\theta$, $y = \cos\theta$, and $y = \tan\theta$, and state each period and range.

WORKING AREA

FORMULA BANK

Parent graphs

$y = \sin\theta,\ y = \cos\theta,\ y = \tan\theta$

Periods

$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$

Range

$\sin,\cos:\ -1 \le y \le 1 \quad \tan:\ y \in \mathbb{R}$

Combined form

$y = af(b\theta - q) = af\left(b\left(\theta - \frac{q}{b}\right)\right)$

Transform order

shift $\frac{q}{b}$ → horizontal scale $\frac{1}{|b|}$ → vertical scale $|a|$

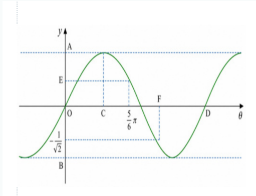
Concept 3 | Properties and Graphs of Trigonometric Functions

F02 | V02

Q006 | WRITTEN

C05-K03-5-9-WU2

For the given graph of $y = \sin\theta$, determine the labelled scales A to F.



WORKING AREA

FORMULA BANK

Parent graphs

$y = \sin\theta,\ y = \cos\theta,\ y = \tan\theta$

Periods

$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$

Range

$\sin, \cos:\ -1 \le y \le 1 \quad \tan:\ y \in \mathbb{R}$

Combined form

$y = af(b\theta - q) = af\left(b\left(\theta - \frac{q}{b}\right)\right)$

Transform order

shift $\frac{q}{b}$ → horizontal scale $\frac{1}{|b|}$ → vertical scale $|a|$

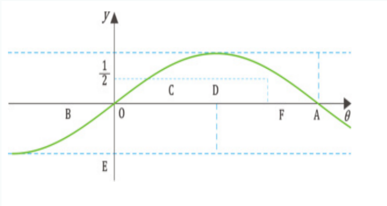
Concept 3 | Properties and Graphs of Trigonometric Functions

F02 | V02

Q007 | WRITTEN

C05-K03-5-9-TRY

(1) Draw the parent graphs $y = \sin\theta$, $y = \cos\theta$, $y = \tan\theta$, and state their periods. (2) For the given graph of $y = \cos\theta$, determine the labelled scales A to F.



WORKING AREA

FORMULA BANK

Parent graphs
$y = \sin\theta,\ y = \cos\theta,\ y = \tan\theta$

Periods
$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$

Range
$\sin, \cos:\ -1 \le y \le 1 \quad \tan:\ y \in \mathbb{R}$

Combined form
$y = af(b\theta - q) = af\left(b\left(\theta - \frac{q}{b}\right)\right)$

Transform order
shift $\frac{q}{b}$ → horizontal scale $\frac{1}{|b|}$ → vertical scale $|a|$

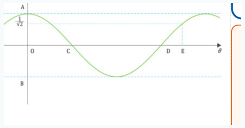
Concept 3 | Properties and Graphs of Trigonometric Functions

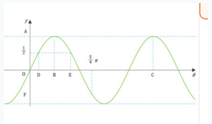
F02 | V03

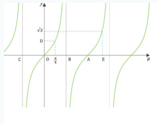
Q008 | WRITTEN

C05-K03-5-9-EX

Complete all four given graph tasks: draw the three parent graphs and find their periods; read the labelled scales on the given cosine graph; read the labelled scales on the given sine graph; and read the labelled scales on the given tangent graph.







FORMULA BANK

Parent graphs
$y = \sin\theta,\ y = \cos\theta,\ y = \tan\theta$

Periods
$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$

Range
$\sin, \cos:\ -1 \le y \le 1 \quad \tan:\ y \in \mathbb{R}$

Combined form
$y = af(b\theta - q) = af\left(b\left(\theta - \frac{q}{b}\right)\right)$

Transform order
shift $\frac{q}{b}$ → horizontal scale $\frac{1}{|b|}$ → vertical scale $|a|$

WORKING AREA

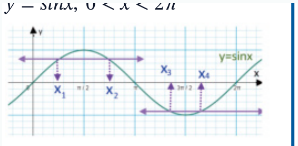
Concept 3 | Properties and Graphs of Trigonometric Functions

F02 | V04

Q009 | WRITTEN

C05-K03-5-9-CH

The given graph represents $y = \sin x$ for $0 < x < 2\pi$. Find $x_1 + x_2 + x_3 + x_4$ from the four marked x-coordinates.



WORKING AREA

FORMULA BANK

Parent graphs

$y = \sin\theta,\ y = \cos\theta,\ y = \tan\theta$

Periods

$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$

Range

$\sin, \cos:\ -1 \le y \le 1 \quad \tan:\ y \in \mathbb{R}$

Combined form

$y = af(b\theta - q) = af\left(b\left(\theta - \frac{q}{b}\right)\right)$

Transform order

shift $\frac{q}{b}$ → horizontal scale $\frac{1}{|b|}$ → vertical scale $|a|$

Concept 3 | Properties and Graphs of Trigonometric Functions

F03 | V01

Q010 | WRITTEN

C05-K03-5-10-WU

Draw the graphs of $y = 2\cos\theta$ and $y = \tan\frac{\theta}{2}$, and find their periods.

WORKING AREA

FORMULA BANK

Parent graphs
$y = \sin\theta,\ y = \cos\theta,\ y = \tan\theta$

Periods
$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$

Range
$\sin, \cos:\ -1 \le y \le 1 \quad \tan:\ y \in \mathbb{R}$

Combined form
$y = af(b\theta - q) = af\left(b\left(\theta - \frac{q}{b}\right)\right)$

Transform order
shift $\frac{q}{b}$ → horizontal scale $\frac{1}{|b|}$ → vertical scale $|a|$

Concept 3 | Properties and Graphs of Trigonometric Functions

F03 | V02

Q011 | WRITTEN

C05-K03-5-10-TRY

Draw the graphs of $y = 2\sin\theta$, $y = \tan 2\theta$, and $y = \cos\frac{\theta}{2}$, and find their periods.

WORKING AREA

FORMULA BANK

Parent graphs
$y = \sin\theta,\ y = \cos\theta,\ y = \tan\theta$

Periods
$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$

Range
$\sin, \cos:\ -1 \le y \le 1 \quad \tan:\ y \in \mathbb{R}$

Combined form
$y = af(b\theta - q) = af\left(b\left(\theta - \frac{q}{b}\right)\right)$

Transform order
shift $\frac{q}{b}$ $\rightarrow$ horizontal scale $\frac{1}{|b|}$ $\rightarrow$ vertical scale $|a|$

Concept 3 | Properties and Graphs of Trigonometric Functions

F03 | V03

Q012 | WRITTEN C05-K03-5-10-EX

Draw the nine given graphs and find each period: (a) $3\sin\theta$, (b) $3\cos\theta$, (c) $\frac{1}{2}\cos\theta$, (d) $\sin 3\theta$, (e) $\cos 2\theta$, (f) $\tan 3\theta$, (g) $\sin\frac{\theta}{2}$, (h) $\cos\frac{\theta}{3}$, (i) $\tan\frac{\theta}{3}$.

WORKING AREA

FORMULA BANK

Parent graphs
$y = \sin\theta,\ y = \cos\theta,\ y = \tan\theta$

Periods
$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$

Range
$\sin,\cos:\ -1 \le y \le 1 \quad \tan:\ y \in \mathbb{R}$

Combined form
$y = af(b\theta - q) = af\left(b\left(\theta - \frac{q}{b}\right)\right)$

Transform order
shift $\frac{q}{b}$ → horizontal scale $\frac{1}{|b|}$ → vertical scale $|a|$

Concept 3 | Properties and Graphs of Trigonometric Functions

F03 | V04

Q013 | WRITTEN

C05-K03-5-10-CH

Draw $y = -2|\sin(3x)|$ and find its period.

WORKING AREA

FORMULA BANK

Parent graphs
$y = \sin\theta,\ y = \cos\theta,\ y = \tan\theta$

Periods
$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$

Range
$\sin,\cos:\ -1 \le y \le 1 \quad \tan:\ y \in \mathbb{R}$

Combined form
$y = af(b\theta - q) = af\left(b\left(\theta - \frac{q}{b}\right)\right)$

Transform order
shift $\frac{q}{b}$ → horizontal scale $\frac{1}{|b|}$ → vertical scale $|a|$

Concept 3 | Properties and Graphs of Trigonometric Functions

F04 | V01

Q014 | WRITTEN

C05-K03-5-11-WU

Draw $y = \sin(\theta - \frac{\pi}{6})$ and find its period.

WORKING AREA

FORMULA BANK

Parent graphs

$y = \sin\theta,\ y = \cos\theta,\ y = \tan\theta$

Periods

$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$

Range

$\sin, \cos: \ -1 \le y \le 1 \quad \tan: \ y \in \mathbb{R}$

Combined form

$y = af(b\theta - q) = af\left(b\left(\theta - \frac{q}{b}\right)\right)$

Transform order

shift $\frac{q}{b}$ → horizontal scale $\frac{1}{|b|}$ → vertical scale $|a|$

Concept 3 | Properties and Graphs of Trigonometric Functions

F04 | V02

Q015 | WRITTEN C05-K03-5-11-TRY

Draw the graphs and find their periods: (1) $y = \sin(\theta - \frac{\pi}{3})$, (2) $y = \cos(\theta + \frac{\pi}{4})$, (3) $y = \tan(\theta - \frac{\pi}{2})$.

WORKING AREA

FORMULA BANK

Parent graphs

$y = \sin\theta,\ y = \cos\theta,\ y = \tan\theta$

Periods

$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$

Range

$\sin,\cos:\ -1 \le y \le 1 \quad \tan:\ y \in \mathbb{R}$

Combined form

$y = af(b\theta - q) = af\left(b\left(\theta - \frac{q}{b}\right)\right)$

Transform order

shift $\frac{q}{b}$ → horizontal scale $\frac{1}{|b|}$ → vertical scale $|a|$

Concept 3 | Properties and Graphs of Trigonometric Functions

F04 | V03

Q016 | WRITTEN
C05-K03-5-11-EX

Draw all given graphs and find their periods: sine group $y = \sin(\theta + \frac{\pi}{4}), \sin(\theta - \frac{\pi}{2}), \sin(\theta + \frac{\pi}{3})$; cosine group $y = \cos(\theta - \frac{\pi}{3}), \cos(\theta - \frac{\pi}{4}), \cos(\theta + \frac{\pi}{6})$; tangent group $y = \tan(\theta + \frac{\pi}{6}), \tan(\theta - \frac{\pi}{3}), \tan(\theta - \frac{\pi}{4})$.

WORKING AREA

FORMULA BANK

Parent graphs
$y = \sin\theta,\ y = \cos\theta,\ y = \tan\theta$

Periods
$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$

Range
$\sin,\cos:\ -1 \le y \le 1 \quad \tan:\ y \in \mathbb{R}$

Combined form
$y = af(b\theta - q) = af\left(b\left(\theta - \frac{q}{b}\right)\right)$

Transform order
shift $\frac{q}{b}$ → horizontal scale $\frac{1}{|b|}$ → vertical scale $|a|$

Concept 3 | Properties and Graphs of Trigonometric Functions

F04 | V04

Q017 | WRITTEN

C05-K03-5-11-CH

Draw $y = 3\cos\left|x + \frac{\pi}{3}\right|$ and find its period.

WORKING AREA

FORMULA BANK

Parent graphs

$y = \sin\theta,\ y = \cos\theta,\ y = \tan\theta$

Periods

$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$

Range

$\sin,\cos:\ -1 \le y \le 1 \quad \tan:\ y \in \mathbb{R}$

Combined form

$y = af(b\theta - q) = af\left(b\left(\theta - \frac{q}{b}\right)\right)$

Transform order

shift $\frac{q}{b}$ $\to$ horizontal scale $\frac{1}{|b|}$ $\to$ vertical scale $|a|$

Concept 3 | Properties and Graphs of Trigonometric Functions

F05 | V01

Q018 | WRITTEN

C05-K03-5-12-WU

Draw the graphs and find their periods: (1) $y = 2\cos(2\theta - \frac{\pi}{2})$, (2) $y = \tan(\frac{\theta}{2} - \frac{\pi}{6})$.

WORKING AREA

FORMULA BANK

Parent graphs
$y = \sin\theta,\ y = \cos\theta,\ y = \tan\theta$

Periods
$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$

Range
$\sin,\cos:\ -1 \le y \le 1 \quad \tan:\ y \in \mathbb{R}$

Combined form
$y = af(b\theta - q) = af\left(b\left(\theta - \frac{q}{b}\right)\right)$

Transform order
shift $\frac{q}{b}$ → horizontal scale $\frac{1}{|b|}$ → vertical scale $|a|$

Concept 3 | Properties and Graphs of Trigonometric Functions

F05 | V02

Q019 | WRITTEN

C05-K03-5-12-TRY

Draw the graphs and find their periods: (a) $y = 3\sin\frac{\theta}{2}$, (b) $y = \tan(2\theta - \frac{\pi}{2})$, (c) $y = 2\cos(2\theta - \frac{\pi}{3})$.

WORKING AREA

FORMULA BANK

Parent graphs
$y = \sin\theta,\ y = \cos\theta,\ y = \tan\theta$

Periods
$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$

Range
$\sin, \cos:\ -1 \le y \le 1 \quad \tan:\ y \in \mathbb{R}$

Combined form
$y = af(b\theta - q) = af\left(b\left(\theta - \frac{q}{b}\right)\right)$

Transform order
shift $\frac{q}{b}$ → horizontal scale $\frac{1}{|b|}$ → vertical scale $|a|$

Concept 3 | Properties and Graphs of Trigonometric Functions

F05 | V03

Q020 | WRITTEN

C05-K03-5-12-EX

Draw all nine given graphs and find their periods: (a) $2\cos\frac{\theta}{2}$, (b) $3\sin 2\theta$, (c) $2\cos 2\theta$, (d) $2\cos(\theta - \frac{\pi}{4})$, (e) $2\sin(\theta - \frac{\pi}{3})$, (f) $\tan(2\theta - \frac{\pi}{3})$, (g) $3\cos(2\theta - \frac{\pi}{2})$, (h) $3\sin(2\theta - \frac{2\pi}{3})$, (i) $2\cos(\frac{\theta}{2} - \frac{\pi}{4})$.

WORKING AREA

FORMULA BANK

Parent graphs
$y = \sin\theta,\ y = \cos\theta,\ y = \tan\theta$

Periods
$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$

Range
$\sin, \cos:\ -1 \le y \le 1 \quad \tan:\ y \in \mathbb{R}$

Combined form
$y = af(b\theta - q) = af\left(b\left(\theta - \frac{q}{b}\right)\right)$

Transform order
shift $\frac{q}{b}$ → horizontal scale $\frac{1}{|b|}$ → vertical scale $|a|$

Concept 3 | Properties and Graphs of Trigonometric Functions

F05 | V04

Q021 | WRITTEN

C05-K03-5-12-CH

Draw $y = 3\sin|2x - \pi|$ and find its period.

WORKING AREA

FORMULA BANK

Parent graphs

$y = \sin\theta,\ y = \cos\theta,\ y = \tan\theta$

Periods

$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$

Range

$\sin,\cos:\ -1 \le y \le 1 \quad \tan:\ y \in \mathbb{R}$

Combined form

$y = af(b\theta - q) = af\left(b\left(\theta - \frac{q}{b}\right)\right)$

Transform order

shift $\frac{q}{b}$ $\to$ horizontal scale $\frac{1}{|b|}$ $\to$ vertical scale $|a|$

Concept 3 | Properties and Graphs of Trigonometric Functions

F01 | V01

Q022 | MCQ

C05-K03-5-8-GM01

$\sin(\theta + \pi)$ equals:

A $\sin\theta$

B $-\sin\theta$

C $\cos\theta$

D $-\cos\theta$

WORKING AREA

FORMULA BANK

Parent graphs
$y = \sin\theta,\ y = \cos\theta,\ y = \tan\theta$

Periods
$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$

Range
$\sin, \cos:\ -1 \le y \le 1 \quad \tan:\ y \in \mathbb{R}$

Combined form
$y = af(b\theta - q) = af\left(b\left(\theta - \frac{q}{b}\right)\right)$

Transform order
shift $\frac{q}{b}$ → horizontal scale $\frac{1}{|b|}$ → vertical scale $|a|$

Concept 3 | Properties and Graphs of Trigonometric Functions

F01 | V01

Q023 | MCQ C05-K03-5-8-GM02

$\cos(\theta + \pi)$ equals:

A $\cos\theta$

B $-\cos\theta$

C $\sin\theta$

D $-\sin\theta$

WORKING AREA

FORMULA BANK

Parent graphs
$y = \sin\theta,\ y = \cos\theta,\ y = \tan\theta$

Periods
$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$

Range
$\sin, \cos:\ -1 \le y \le 1 \quad \tan:\ y \in \mathbb{R}$

Combined form
$y = af(b\theta - q) = af\left(b\left(\theta - \frac{q}{b}\right)\right)$

Transform order
shift $\frac{q}{b}$ $\rightarrow$ horizontal scale $\frac{1}{|b|}$ $\rightarrow$ vertical scale $|a|$

Concept 3 | Properties and Graphs of Trigonometric Functions

F01 | V01

Q024 | MCQ

C05-K03-5-8-GM03

$\sin(\frac{\pi}{2} - \theta)$ equals:

A $\sin\theta$

B $-\sin\theta$

C $\cos\theta$

D $-\cos\theta$

WORKING AREA

FORMULA BANK

Parent graphs
$y = \sin\theta,\ y = \cos\theta,\ y = \tan\theta$

Periods
$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$

Range
$\sin, \cos:\ -1 \le y \le 1 \quad \tan:\ y \in \mathbb{R}$

Combined form
$y = af(b\theta - q) = af\left(b\left(\theta - \frac{q}{b}\right)\right)$

Transform order
shift $\frac{q}{b}$ $\rightarrow$ horizontal scale $\frac{1}{|b|}$ $\rightarrow$ vertical scale $|a|$

Concept 3 | Properties and Graphs of Trigonometric Functions

F01 | V01

Q025 | MCQ

C05-K03-5-8-GM04

$\cos(-\theta)$ equals:

A $\cos\theta$

B $-\cos\theta$

C $\sin\theta$

D $-\sin\theta$

WORKING AREA

FORMULA BANK

Parent graphs
$y = \sin\theta,\ y = \cos\theta,\ y = \tan\theta$

Periods
$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$

Range
$\sin, \cos:\ -1 \le y \le 1 \quad \tan:\ y \in \mathbb{R}$

Combined form
$y = af(b\theta - q) = af\left(b\left(\theta - \frac{q}{b}\right)\right)$

Transform order
shift $\frac{q}{b}$ → horizontal scale $\frac{1}{|b|}$ → vertical scale $|a|$

Concept 3 | Properties and Graphs of Trigonometric Functions

F01 | V01

Q026 | MCQ

C05-K03-5-8-GM05

$\sin(-\theta)$ equals:

A $\sin\theta$

B $-\sin\theta$

C $\cos\theta$

D $-\cos\theta$

WORKING AREA

FORMULA BANK

Parent graphs
$y = \sin\theta,\ y = \cos\theta,\ y = \tan\theta$

Periods
$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$

Range
$\sin,\cos:\ -1 \le y \le 1 \quad \tan:\ y \in \mathbb{R}$

Combined form
$y = af(b\theta - q) = af\left(b\left(\theta - \frac{q}{b}\right)\right)$

Transform order
shift $\frac{q}{b}$ → horizontal scale $\frac{1}{|b|}$ → vertical scale $|a|$

Concept 3 | Properties and Graphs of Trigonometric Functions

F01 | V01

Q027 | MCQ

C05-K03-5-8-GM06

$\cos(\pi - \theta)$ equals:

A $\cos\theta$

B $-\cos\theta$

C $\sin\theta$

D $-\sin\theta$

WORKING AREA

FORMULA BANK

Parent graphs
$y = \sin\theta,\ y = \cos\theta,\ y = \tan\theta$

Periods
$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$

Range
$\sin, \cos:\ -1 \le y \le 1 \quad \tan:\ y \in \mathbb{R}$

Combined form
$y = af(b\theta - q) = af\left(b\left(\theta - \frac{q}{b}\right)\right)$

Transform order
shift $\frac{q}{b}$ $\rightarrow$ horizontal scale $\frac{1}{|b|}$ $\rightarrow$ vertical scale $|a|$

Concept 3 | Properties and Graphs of Trigonometric Functions

F01 | V01

Q028 | MCQ

C05-K03-5-8-GM07

$\sin(\frac{3\pi}{2} + \theta)$ equals:

A $\sin\theta$

B $-\sin\theta$

C $\cos\theta$

D $-\cos\theta$

WORKING AREA

FORMULA BANK

Parent graphs
$y = \sin\theta,\ y = \cos\theta,\ y = \tan\theta$

Periods
$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$

Range
$\sin,\cos:\ -1 \le y \le 1 \quad \tan:\ y \in \mathbb{R}$

Combined form
$y = af(b\theta - q) = af\left(b\left(\theta - \frac{q}{b}\right)\right)$

Transform order
shift $\frac{q}{b}$ $\rightarrow$ horizontal scale $\frac{1}{|b|}$ $\rightarrow$ vertical scale $|a|$

Concept 3 | Properties and Graphs of Trigonometric Functions

F01 | V01

Q029 | MCQ

C05-K03-5-8-GM08

$\tan(\theta - \frac{\pi}{2})$ equals:

A $\tan\theta$

B $-\tan\theta$

C $\cot\theta$

D $-\cot\theta$

WORKING AREA

FORMULA BANK

Parent graphs
$y = \sin\theta,\ y = \cos\theta,\ y = \tan\theta$

Periods
$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$

Range
$\sin,\cos:\ -1 \leq y \leq 1 \quad \tan:\ y \in \mathbb{R}$

Combined form
$y = af(b\theta - q) = af\left(b\left(\theta - \frac{q}{b}\right)\right)$

Transform order
shift $\frac{q}{b}$ → horizontal scale $\frac{1}{|b|}$ → vertical scale $|a|$

Concept 3 | Properties and Graphs of Trigonometric Functions

F02 | V01

Q030 | MCQ

C05-K03-5-9-GM01

The period of $y = \sin\theta$ is:

A $\pi/2$

B π

C 2π

D 4π

WORKING AREA

FORMULA BANK

Parent graphs
$y = \sin\theta,\ y = \cos\theta,\ y = \tan\theta$

Periods
$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$

Range
$\sin,\cos:\ -1 \le y \le 1 \quad \tan:\ y \in \mathbb{R}$

Combined form
$y = af(b\theta - q) = af\left(b\left(\theta - \frac{q}{b}\right)\right)$

Transform order
shift $\frac{q}{b}$ → horizontal scale $\frac{1}{|b|}$ → vertical scale $|a|$

Concept 3 | Properties and Graphs of Trigonometric Functions

F02 | V01

Q031 | MCQ

C05-K03-5-9-GM02

The range of $y = \cos\theta$ is:

A $y \geq 0$

B $-1 \leq y \leq 1$

C $y \in \mathbb{R}$

D y>1

WORKING AREA

FORMULA BANK

Parent graphs
$y = \sin\theta,\ y = \cos\theta,\ y = \tan\theta$

Periods
$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$

Range
$\sin,\cos:\ -1 \leq y \leq 1 \quad \tan:\ y \in \mathbb{R}$

Combined form
$y = af(b\theta - q) = af\left(b\left(\theta - \frac{q}{b}\right)\right)$

Transform order
shift $\frac{q}{b}$ → horizontal scale $\frac{1}{|b|}$ → vertical scale $|a|$

Concept 3 | Properties and Graphs of Trigonometric Functions

F02 | V01

Q032 | MCQ

C05-K03-5-9-GM03

The period of $y = \tan\theta$ is:

A $\pi/2$

B π

C 2π

D 4π

WORKING AREA

FORMULA BANK

Parent graphs
$y = \sin\theta, \ y = \cos\theta, \ y = \tan\theta$

Periods
$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$

Range
$\sin, \cos: \ -1 \le y \le 1 \quad \tan: \ y \in \mathbb{R}$

Combined form
$y = af(b\theta - q) = af\left(b\left(\theta - \frac{q}{b}\right)\right)$

Transform order
shift $\frac{q}{b}$ → horizontal scale $\frac{1}{|b|}$ → vertical scale $|a|$

Concept 3 | Properties and Graphs of Trigonometric Functions

F02 | V01

Q033 | MCQ C05-K03-5-9-GM04

A vertical asymptote of $y = \tan\theta$ occurs at:

A $\theta = k\pi$

B $\theta = \frac{\pi}{2} + k\pi$

C $\theta = 2k\pi$

D $\theta = \frac{\pi}{4} + k\pi$

WORKING AREA

FORMULA BANK

Parent graphs
$y = \sin\theta,\ y = \cos\theta,\ y = \tan\theta$

Periods
$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$

Range
$\sin, \cos:\ -1 \le y \le 1 \quad \tan:\ y \in \mathbb{R}$

Combined form
$y = af(b\theta - q) = af\left(b\left(\theta - \frac{q}{b}\right)\right)$

Transform order
shift $\frac{q}{b}$ → horizontal scale $\frac{1}{|b|}$ → vertical scale $|a|$

Concept 3 | Properties and Graphs of Trigonometric Functions

F02 | V01

Q034 | MCQ

C05-K03-5-9-GM05

The maximum value of $\sin\theta$ is:

A -1

B 0

C 1

D π

WORKING AREA

FORMULA BANK

Parent graphs

$y = \sin\theta,\ y = \cos\theta,\ y = \tan\theta$

Periods

$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$

Range

$\sin,\cos:\ -1 \le y \le 1 \quad \tan:\ y \in \mathbb{R}$

Combined form

$y = af(b\theta - q) = af\left(b\left(\theta - \frac{q}{b}\right)\right)$

Transform order

shift $\frac{q}{b}$ $\rightarrow$ horizontal scale $\frac{1}{|b|}$ $\rightarrow$ vertical scale $|a|$

Concept 3 | Properties and Graphs of Trigonometric Functions

F02 | V01

Q035 | MCQ

C05-K03-5-9-GM06

The x-coordinate of the first positive sine maximum is:

A $\pi/4$

B $\pi/2$

C π

D $3\pi/2$

WORKING AREA

FORMULA BANK

Parent graphs
$y = \sin\theta,\ y = \cos\theta,\ y = \tan\theta$

Periods
$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$

Range
$\sin, \cos:\ -1 \le y \le 1 \quad \tan:\ y \in \mathbb{R}$

Combined form
$y = af(b\theta - q) = af\left(b\left(\theta - \frac{q}{b}\right)\right)$

Transform order
shift $\frac{q}{b}$ → horizontal scale $\frac{1}{|b|}$ → vertical scale $|a|$

Concept 3 | Properties and Graphs of Trigonometric Functions

F02 | V01

Q036 | MCQ

C05-K03-5-9-GM07

At $\theta = 0$, $\cos\theta$ equals:

A -1

B 0

C 1

D undefined

WORKING AREA

FORMULA BANK

Parent graphs
$y = \sin\theta,\ y = \cos\theta,\ y = \tan\theta$

Periods
$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$

Range
$\sin, \cos:\ -1 \le y \le 1 \quad \tan:\ y \in \mathbb{R}$

Combined form
$y = af(b\theta - q) = af\left(b\left(\theta - \frac{q}{b}\right)\right)$

Transform order
shift $\frac{q}{b}$ → horizontal scale $\frac{1}{|b|}$ → vertical scale $|a|$

Concept 3 | Properties and Graphs of Trigonometric Functions

F02 | V01

Q037 | MCQ

C05-K03-5-9-GM08

At $\theta = 0$, $\tan\theta$ equals:

A -1

B 0

C 1

D undefined

WORKING AREA

FORMULA BANK

Parent graphs
$y = \sin\theta,\ y = \cos\theta,\ y = \tan\theta$

Periods
$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$

Range
$\sin,\cos:\ -1 \le y \le 1 \quad \tan:\ y \in \mathbb{R}$

Combined form
$y = af(b\theta - q) = af\left(b\left(\theta - \frac{q}{b}\right)\right)$

Transform order
shift $\frac{q}{b}$ $\to$ horizontal scale $\frac{1}{|b|}$ $\to$ vertical scale $|a|$

Concept 3 | Properties and Graphs of Trigonometric Functions

F03 | V01

Q038 | MCQ

C05-K03-5-10-GM01

The range of $y = 3\sin\theta$ is:

A [-1,1]

B [-3,3]

C [0,3]

D $\mathbb{R}$

WORKING AREA

FORMULA BANK

Parent graphs
$y = \sin\theta,\ y = \cos\theta,\ y = \tan\theta$

Periods
$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$

Range
$\sin,\cos:\ -1 \le y \le 1 \quad \tan:\ y \in \mathbb{R}$

Combined form
$y = af(b\theta - q) = af\left(b\left(\theta - \frac{q}{b}\right)\right)$

Transform order
shift $\frac{q}{b}$ $\rightarrow$ horizontal scale $\frac{1}{|b|}$ $\rightarrow$ vertical scale $|a|$

Concept 3 | Properties and Graphs of Trigonometric Functions

F03 | V01

Q039 | MCQ — C05-K03-5-10-GM02

The period of $y = \sin 4\theta$ is:

A $\pi/4$

B $\pi/2$

C π

D 2π

WORKING AREA

FORMULA BANK

Parent graphs
$y = \sin\theta,\ y = \cos\theta,\ y = \tan\theta$

Periods
$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$

Range
$\sin, \cos:\ -1 \le y \le 1 \quad \tan:\ y \in \mathbb{R}$

Combined form
$y = af(b\theta - q) = af\left(b\left(\theta - \frac{q}{b}\right)\right)$

Transform order
shift $\frac{q}{b}$ → horizontal scale $\frac{1}{|b|}$ → vertical scale $|a|$

Concept 3 | Properties and Graphs of Trigonometric Functions

F03 | V01

Q040 | MCQ

C05-K03-5-10-GM03

The period of $y = \cos(\theta/3)$ is:

A $2\pi/3$

B 2π

C 3π

D 6π

WORKING AREA

FORMULA BANK

Parent graphs
$y = \sin\theta,\ y = \cos\theta,\ y = \tan\theta$

Periods
$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$

Range
$\sin,\cos:\ -1 \le y \le 1 \quad \tan:\ y \in \mathbb{R}$

Combined form
$y = af(b\theta - q) = af\left(b\left(\theta - \frac{q}{b}\right)\right)$

Transform order
shift $\frac{q}{b}$ → horizontal scale $\frac{1}{|b|}$ → vertical scale $|a|$

Concept 3 | Properties and Graphs of Trigonometric Functions

F03 | V01

Q041 | MCQ — C05-K03-5-10-GM04

The period of $y = \tan 5\theta$ is:

A $\pi/5$

B π

C 5π

D $2\pi/5$

WORKING AREA

FORMULA BANK

Parent graphs
$y = \sin\theta,\ y = \cos\theta,\ y = \tan\theta$

Periods
$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$

Range
$\sin,\cos:\ -1 \le y \le 1 \quad \tan:\ y \in \mathbb{R}$

Combined form
$y = af(b\theta - q) = af\left(b\left(\theta - \frac{q}{b}\right)\right)$

Transform order
shift $\frac{q}{b}$ → horizontal scale $\frac{1}{|b|}$ → vertical scale $|a|$

Concept 3 | Properties and Graphs of Trigonometric Functions

F03 | V01

Q042 | MCQ

C05-K03-5-10-GM05

The range of $y = \frac{1}{2}\cos\theta$ is:

A [-2,2]

B [-1,1]

C [-1/2,1/2]

D [0,1/2]

WORKING AREA

FORMULA BANK

Parent graphs
$y = \sin\theta,\ y = \cos\theta,\ y = \tan\theta$

Periods
$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$

Range
$\sin,\cos:\ -1 \le y \le 1 \quad \tan:\ y \in \mathbb{R}$

Combined form
$y = af(b\theta - q) = af\left(b\left(\theta - \frac{q}{b}\right)\right)$

Transform order
shift $\frac{q}{b}$ $\to$ horizontal scale $\frac{1}{|b|}$ $\to$ vertical scale $|a|$

Concept 3 | Properties and Graphs of Trigonometric Functions

F03 | V01

Q043 | MCQ

C05-K03-5-10-GM06

The graph of $y = -2\sin\theta$ is the parent sine graph:

A shifted only

B reflected and stretched vertically

C compressed horizontally only

D undefined

WORKING AREA

FORMULA BANK

Parent graphs
$y = \sin\theta,\ y = \cos\theta,\ y = \tan\theta$

Periods
$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$

Range
$\sin,\cos:\ -1 \le y \le 1 \quad \tan:\ y \in \mathbb{R}$

Combined form
$y = af(b\theta - q) = af\left(b\left(\theta - \frac{q}{b}\right)\right)$

Transform order
shift $\frac{q}{b}$ → horizontal scale $\frac{1}{|b|}$ → vertical scale $|a|$

Concept 3 | Properties and Graphs of Trigonometric Functions

F03 | V01

Q044 | MCQ

C05-K03-5-10-GM07

The period of $y = \tan(\theta/2)$ is:

A $\pi/2$

B π

C 2π

D 4π

WORKING AREA

FORMULA BANK

Parent graphs
$y = \sin\theta,\ y = \cos\theta,\ y = \tan\theta$

Periods
$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$

Range
$\sin,\cos:\ -1 \le y \le 1 \quad \tan:\ y \in \mathbb{R}$

Combined form
$y = af(b\theta - q) = af\left(b\left(\theta - \frac{q}{b}\right)\right)$

Transform order
shift $\frac{q}{b}$ $\rightarrow$ horizontal scale $\frac{1}{|b|}$ $\rightarrow$ vertical scale $|a|$

Concept 3 | Properties and Graphs of Trigonometric Functions

F03 | V01

Q045 | MCQ

C05-K03-5-10-GM08

The period of $y = \sin(3\theta)$ is:

A $\pi/3$

B $2\pi/3$

C 3π

D 6π

WORKING AREA

FORMULA BANK

Parent graphs
$y = \sin\theta, \; y = \cos\theta, \; y = \tan\theta$

Periods
$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$

Range
$\sin,\cos: \; -1 \le y \le 1 \quad \tan: \; y \in \mathbb{R}$

Combined form
$y = af(b\theta - q) = af\left(b\left(\theta - \frac{q}{b}\right)\right)$

Transform order
shift $\frac{q}{b}$ → horizontal scale $\frac{1}{|b|}$ → vertical scale $|a|$

Concept 3 | Properties and Graphs of Trigonometric Functions

F04 | V01

Q046 | MCQ

C05-K03-5-11-GM01

The graph of $y = \sin(\theta - \pi/4)$ is shifted:

A left by $\pi/4$

B right by $\pi/4$

C up by $\pi/4$

D down by $\pi/4$

WORKING AREA

FORMULA BANK

Parent graphs
$y = \sin\theta,\ y = \cos\theta,\ y = \tan\theta$

Periods
$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$

Range
$\sin,\cos:\ -1 \le y \le 1 \quad \tan:\ y \in \mathbb{R}$

Combined form
$y = af(b\theta - q) = af\left(b\left(\theta - \frac{q}{b}\right)\right)$

Transform order
shift $\frac{q}{b}$ → horizontal scale $\frac{1}{|b|}$ → vertical scale $|a|$

Concept 3 | Properties and Graphs of Trigonometric Functions

F04 | V01

Q047 | MCQ

C05-K03-5-11-GM02

The graph of $y = \cos(\theta + \pi/3)$ is shifted:

A right by $\pi/3$

B left by $\pi/3$

C up by $\pi/3$

D down by $\pi/3$

WORKING AREA

FORMULA BANK

Parent graphs
$y = \sin\theta, \; y = \cos\theta, \; y = \tan\theta$

Periods
$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$

Range
$\sin, \cos: \; -1 \le y \le 1 \quad \tan: \; y \in \mathbb{R}$

Combined form
$y = af(b\theta - q) = af\left(b\left(\theta - \frac{q}{b}\right)\right)$

Transform order
shift $\frac{q}{b}$ → horizontal scale $\frac{1}{|b|}$ → vertical scale $|a|$

Concept 3 | Properties and Graphs of Trigonometric Functions

F04 | V01

Q048 | MCQ

C05-K03-5-11-GM03

A horizontal shift changes the sine period from 2π to:

A π

B 2π

C 4π

D 0

WORKING AREA

FORMULA BANK

Parent graphs
$y = \sin\theta,\ y = \cos\theta,\ y = \tan\theta$

Periods
$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$

Range
$\sin,\cos:\ -1 \le y \le 1 \quad \tan:\ y \in \mathbb{R}$

Combined form
$y = af(b\theta - q) = af\left(b\left(\theta - \frac{q}{b}\right)\right)$

Transform order
shift $\frac{q}{b}$ $\rightarrow$ horizontal scale $\frac{1}{|b|}$ $\rightarrow$ vertical scale $|a|$

Concept 3 | Properties and Graphs of Trigonometric Functions

F04 | V01

Q049 | MCQ C05-K03-5-11-GM04

A horizontal shift changes the tangent period from π to:

A π/2

B π

C 2π

D 4π

WORKING AREA

FORMULA BANK

Parent graphs
$y = \sin\theta,\ y = \cos\theta,\ y = \tan\theta$

Periods
$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$

Range
$\sin,\cos:\ -1 \le y \le 1 \quad \tan:\ y \in \mathbb{R}$

Combined form
$y = af(b\theta - q) = af\left(b\left(\theta - \frac{q}{b}\right)\right)$

Transform order
shift $\frac{q}{b}$ → horizontal scale $\frac{1}{|b|}$ → vertical scale $|a|$

Concept 3 | Properties and Graphs of Trigonometric Functions

F04 | V01

Q050 | MCQ

C05-K03-5-11-GM05

The graph of $y = \sin(\theta + \pi/2)$ equals:

A $\sin\theta$

B $-\sin\theta$

C $\cos\theta$

D $-\cos\theta$

WORKING AREA

FORMULA BANK

Parent graphs
$y = \sin\theta,\ y = \cos\theta,\ y = \tan\theta$

Periods
$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$

Range
$\sin, \cos:\ -1 \le y \le 1 \quad \tan:\ y \in \mathbb{R}$

Combined form
$y = af(b\theta - q) = af\left(b\left(\theta - \frac{q}{b}\right)\right)$

Transform order
shift $\frac{q}{b}$ → horizontal scale $\frac{1}{|b|}$ → vertical scale $|a|$

Concept 3 | Properties and Graphs of Trigonometric Functions

F04 | V01

Q051 | MCQ C05-K03-5-11-GM06

The graph of $y = \cos(\theta - \pi)$ equals:

A $\cos\theta$

B $-\cos\theta$

C $\sin\theta$

D $-\sin\theta$

WORKING AREA

FORMULA BANK

Parent graphs
$y = \sin\theta,\ y = \cos\theta,\ y = \tan\theta$

Periods
$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$

Range
$\sin,\cos:\ -1 \le y \le 1 \quad \tan:\ y \in \mathbb{R}$

Combined form
$y = af(b\theta - q) = af\left(b\left(\theta - \frac{q}{b}\right)\right)$

Transform order
shift $\frac{q}{b}$ $\rightarrow$ horizontal scale $\frac{1}{|b|}$ $\rightarrow$ vertical scale $|a|$

Concept 3 | Properties and Graphs of Trigonometric Functions

F04 | V01

Q052 | MCQ C05-K03-5-11-GM07

The graph of $y = \tan(\theta - \pi/2)$ has its central zero at:

A $\theta = 0$

B $\theta = \pi/2$

C $\theta = \pi$

D $\theta = 2\pi$

WORKING AREA

FORMULA BANK

Parent graphs
$y = \sin\theta,\ y = \cos\theta,\ y = \tan\theta$

Periods
$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$

Range
$\sin, \cos:\ -1 \le y \le 1 \quad \tan:\ y \in \mathbb{R}$

Combined form
$y = af(b\theta - q) = af\left(b\left(\theta - \frac{q}{b}\right)\right)$

Transform order
shift $\frac{q}{b}$ → horizontal scale $\frac{1}{|b|}$ → vertical scale $|a|$

Concept 3 | Properties and Graphs of Trigonometric Functions

F04 | V01

Q053 | MCQ

C05-K03-5-11-GM08

Since cosine is even, $\cos|x|$ equals:

A $\cos x$

B $-\cos x$

C $\sin x$

D $|\cos x|$

WORKING AREA

FORMULA BANK

Parent graphs
$y = \sin\theta,\ y = \cos\theta,\ y = \tan\theta$

Periods
$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$

Range
$\sin, \cos:\ -1 \le y \le 1 \quad \tan:\ y \in \mathbb{R}$

Combined form
$y = af(b\theta - q) = af\left(b\left(\theta - \frac{q}{b}\right)\right)$

Transform order
shift $\frac{q}{b}$ → horizontal scale $\frac{1}{|b|}$ → vertical scale $|a|$

Concept 3 | Properties and Graphs of Trigonometric Functions

F05 | V01

Q054 | MCQ C05-K03-5-12-GM01

In $y = a\sin(b\theta - q)$, the horizontal shift is:

A q

B q/b

C b/q

D a/b

WORKING AREA

FORMULA BANK

Parent graphs
$y = \sin\theta,\ y = \cos\theta,\ y = \tan\theta$

Periods
$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$

Range
$\sin,\cos:\ -1 \le y \le 1 \quad \tan:\ y \in \mathbb{R}$

Combined form
$y = af(b\theta - q) = af\left(b\left(\theta - \frac{q}{b}\right)\right)$

Transform order
shift $\frac{q}{b}$ → horizontal scale $\frac{1}{|b|}$ → vertical scale $|a|$

Concept 3 | Properties and Graphs of Trigonometric Functions

F05 | V01

Q055 | MCQ

C05-K03-5-12-GM02

The period of $y = 2\cos(3\theta - \pi)$ is:

A $\pi/3$

B $2\pi/3$

C π

D 2π

WORKING AREA

FORMULA BANK

Parent graphs
$y = \sin\theta,\ y = \cos\theta,\ y = \tan\theta$

Periods
$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$

Range
$\sin, \cos:\ -1 \le y \le 1 \quad \tan:\ y \in \mathbb{R}$

Combined form
$y = af(b\theta - q) = af\left(b\left(\theta - \frac{q}{b}\right)\right)$

Transform order
shift $\frac{q}{b}$ → horizontal scale $\frac{1}{|b|}$ → vertical scale $|a|$

Concept 3 | Properties and Graphs of Trigonometric Functions

F05 | V01

Q056 | MCQ

C05-K03-5-12-GM03

The range of $y = 4\sin(2\theta - 1)$ is:

A [-1,1]

B [-2,2]

C [-4,4]

D $\mathbb{R}$

WORKING AREA

FORMULA BANK

Parent graphs
$y = \sin\theta,\ y = \cos\theta,\ y = \tan\theta$

Periods
$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$

Range
$\sin,\cos:\ -1 \le y \le 1 \quad \tan:\ y \in \mathbb{R}$

Combined form
$y = af(b\theta - q) = af\left(b\left(\theta - \frac{q}{b}\right)\right)$

Transform order
shift $\frac{q}{b}$ $\rightarrow$ horizontal scale $\frac{1}{|b|}$ $\rightarrow$ vertical scale $|a|$

Concept 3 | Properties and Graphs of Trigonometric Functions

F05 | V01

Q057 | MCQ

C05-K03-5-12-GM04

For $y = \tan(2\theta - \pi/3)$, the shift is:

A left by π/3

B right by π/6

C right by π/3

D left by π/6

WORKING AREA

FORMULA BANK

Parent graphs
$y = \sin\theta,\ y = \cos\theta,\ y = \tan\theta$

Periods
$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$

Range
$\sin,\cos:\ -1 \le y \le 1 \quad \tan:\ y \in \mathbb{R}$

Combined form
$y = af(b\theta - q) = af\left(b\left(\theta - \frac{q}{b}\right)\right)$

Transform order
shift $\frac{q}{b}$ → horizontal scale $\frac{1}{|b|}$ → vertical scale $|a|$

Concept 3 | Properties and Graphs of Trigonometric Functions

F05 | V01

Q058 | MCQ

C05-K03-5-12-GM05

The period of $y = 3\sin(\theta/2)$ is:

A π

B 2π

C 3π

D 4π

WORKING AREA

FORMULA BANK

Parent graphs
$y = \sin\theta,\ y = \cos\theta,\ y = \tan\theta$

Periods
$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$

Range
$\sin, \cos:\ -1 \le y \le 1 \quad \tan:\ y \in \mathbb{R}$

Combined form
$y = af(b\theta - q) = af\left(b\left(\theta - \frac{q}{b}\right)\right)$

Transform order
shift $\frac{q}{b}$ → horizontal scale $\frac{1}{|b|}$ → vertical scale $|a|$

Concept 3 | Properties and Graphs of Trigonometric Functions

F05 | V01

Q059 | MCQ

C05-K03-5-12-GM06

The correct transformation order for $y = af(b\theta - q)$ starts with:

A vertical scale

B horizontal shift

C horizontal scale

D reflection only

WORKING AREA

FORMULA BANK

Parent graphs
$y = \sin\theta,\ y = \cos\theta,\ y = \tan\theta$

Periods
$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$

Range
$\sin, \cos:\ -1 \le y \le 1 \quad \tan:\ y \in \mathbb{R}$

Combined form
$y = af(b\theta - q) = af\left(b\left(\theta - \frac{q}{b}\right)\right)$

Transform order
shift $\frac{q}{b}$ → horizontal scale $\frac{1}{|b|}$ → vertical scale $|a|$

Concept 3 | Properties and Graphs of Trigonometric Functions

F05 | V01

Q060 | MCQ

C05-K03-5-12-GM07

The period of $y = \tan(4\theta + q)$ is:

A $\pi/4$

B $\pi/2$

C π

D 2π

WORKING AREA

FORMULA BANK

Parent graphs
$y = \sin\theta,\ y = \cos\theta,\ y = \tan\theta$

Periods
$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$

Range
$\sin, \cos:\ -1 \le y \le 1 \quad \tan:\ y \in \mathbb{R}$

Combined form
$y = af(b\theta - q) = af\left(b\left(\theta - \frac{q}{b}\right)\right)$

Transform order
shift $\frac{q}{b}$ → horizontal scale $\frac{1}{|b|}$ → vertical scale $|a|$

Concept 3 | Properties and Graphs of Trigonometric Functions

F05 | V01

Q061 | MCQ

C05-K03-5-12-GM08

The range of $y = 3\sin|2x - \pi|$ is:

A [-3,3]

B [0,3]

C [-1,1]

D $\mathbb{R}$

WORKING AREA

FORMULA BANK

Parent graphs
$y = \sin\theta,\ y = \cos\theta,\ y = \tan\theta$

Periods
$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$

Range
$\sin,\cos:\ -1 \le y \le 1 \quad \tan:\ y \in \mathbb{R}$

Combined form
$y = af(b\theta - q) = af\left(b\left(\theta - \frac{q}{b}\right)\right)$

Transform order
shift $\frac{q}{b}$ → horizontal scale $\frac{1}{|b|}$ → vertical scale $|a|$

Concept 3 | Properties and Graphs of Trigonometric Functions

F01 | V01

Q062 | WRITTEN

C05-K03-5-8-GW01

Express $\cos \dfrac{11\pi}{6}$ as a function of an acute angle.

WORKING AREA

FORMULA BANK

Parent graphs
$y = \sin\theta,\ y = \cos\theta,\ y = \tan\theta$

Periods
$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$

Range
$\sin,\cos:\ -1 \le y \le 1 \quad \tan:\ y \in \mathbb{R}$

Combined form
$y = af(b\theta - q) = af\left(b\left(\theta - \frac{q}{b}\right)\right)$

Transform order
shift $\frac{q}{b}$ → horizontal scale $\frac{1}{|b|}$ → vertical scale $|a|$

Concept 3 | Properties and Graphs of Trigonometric Functions

F01 | V01

Q063 | WRITTEN

C05-K03-5-8-GW02

Simplify $\sin(\theta+\pi)-\cos(\frac{\pi}{2}-\theta)$.

WORKING AREA

FORMULA BANK

Parent graphs
$y = \sin\theta,\ y = \cos\theta,\ y = \tan\theta$

Periods
$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$

Range
$\sin,\cos:\ -1 \le y \le 1 \quad \tan:\ y \in \mathbb{R}$

Combined form
$y = af(b\theta - q) = af\left(b\left(\theta - \frac{q}{b}\right)\right)$

Transform order
shift $\frac{q}{b}$ → horizontal scale $\frac{1}{|b|}$ → vertical scale $|a|$

Concept 3 | Properties and Graphs of Trigonometric Functions

F01 | V01

Q064 | WRITTEN

C05-K03-5-8-GW03

Simplify $\cos(\theta + \pi) + \cos(\frac{\pi}{2} - \theta)$.

WORKING AREA

FORMULA BANK

Parent graphs
$y = \sin\theta,\ y = \cos\theta,\ y = \tan\theta$

Periods
$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$

Range
$\sin, \cos:\ -1 \le y \le 1 \quad \tan:\ y \in \mathbb{R}$

Combined form
$y = af(b\theta - q) = af\left(b\left(\theta - \frac{q}{b}\right)\right)$

Transform order
shift $\frac{q}{b}$ → horizontal scale $\frac{1}{|b|}$ → vertical scale $|a|$

Concept 3 | Properties and Graphs of Trigonometric Functions

F01 | V01

Q065 | WRITTEN

C05-K03-5-8-GW04

Evaluate $\tan(-\frac{3\pi}{4})$ using an acute reference angle.

WORKING AREA

FORMULA BANK

Parent graphs
$y = \sin\theta,\ y = \cos\theta,\ y = \tan\theta$

Periods
$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$

Range
$\sin, \cos:\ -1 \le y \le 1 \quad \tan:\ y \in \mathbb{R}$

Combined form
$y = af(b\theta - q) = af\left(b\left(\theta - \frac{q}{b}\right)\right)$

Transform order
shift $\frac{q}{b}$ → horizontal scale $\frac{1}{|b|}$ → vertical scale $|a|$

Concept 3 | Properties and Graphs of Trigonometric Functions

F02 | V01

Q066 | WRITTEN C05-K03-5-9-GW01

State the period and range of $y = \cos\theta$, then list its key points on one cycle.

WORKING AREA

FORMULA BANK

Parent graphs
$y = \sin\theta,\ y = \cos\theta,\ y = \tan\theta$

Periods
$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$

Range
$\sin, \cos:\ -1 \le y \le 1 \quad \tan:\ y \in \mathbb{R}$

Combined form
$y = af(b\theta - q) = af\left(b\left(\theta - \frac{q}{b}\right)\right)$

Transform order
shift $\frac{q}{b}$ → horizontal scale $\frac{1}{|b|}$ → vertical scale $|a|$

Concept 3 | Properties and Graphs of Trigonometric Functions

F02 | V01

Q067 | WRITTEN

C05-K03-5-9-GW02

State the period and asymptotes of $y = \tan\theta$ on $[-\pi, \pi]$.

WORKING AREA

FORMULA BANK

Parent graphs
$y = \sin\theta,\ y = \cos\theta,\ y = \tan\theta$

Periods
$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$

Range
$\sin, \cos:\ -1 \le y \le 1 \quad \tan:\ y \in \mathbb{R}$

Combined form
$y = af(b\theta - q) = af\left(b\left(\theta - \frac{q}{b}\right)\right)$

Transform order
shift $\frac{q}{b}$ → horizontal scale $\frac{1}{|b|}$ → vertical scale $|a|$

Concept 3 | Properties and Graphs of Trigonometric Functions

F02 | V01

Q068 | WRITTEN

C05-K03-5-9-GW03

For the parent sine graph, find the x-values where $y = 0$ in $[0, 2\pi]$.

WORKING AREA

FORMULA BANK

Parent graphs
$y = \sin\theta,\ y = \cos\theta,\ y = \tan\theta$

Periods
$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$

Range
$\sin,\cos:\ -1 \le y \le 1 \quad \tan:\ y \in \mathbb{R}$

Combined form
$y = af(b\theta - q) = af\left(b\left(\theta - \frac{q}{b}\right)\right)$

Transform order
shift $\frac{q}{b}$ $\rightarrow$ horizontal scale $\frac{1}{|b|}$ $\rightarrow$ vertical scale $|a|$

Concept 3 | Properties and Graphs of Trigonometric Functions

F02 | V01

Q069 | WRITTEN

C05-K03-5-9-GW04

For the parent cosine graph, find the x-values where $y = 0$ in $[0, 2\pi]$.

WORKING AREA

FORMULA BANK

Parent graphs
$y = \sin\theta,\ y = \cos\theta,\ y = \tan\theta$

Periods
$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$

Range
$\sin,\cos:\ -1 \le y \le 1 \quad \tan:\ y \in \mathbb{R}$

Combined form
$y = af(b\theta - q) = af\left(b\left(\theta - \frac{q}{b}\right)\right)$

Transform order
shift $\frac{q}{b}$ → horizontal scale $\frac{1}{|b|}$ → vertical scale $|a|$

Concept 3 | Properties and Graphs of Trigonometric Functions

F03 | V01

Q070 | WRITTEN

C05-K03-5-10-GW01

Find the period and range of $y = 4\cos\theta$.

WORKING AREA

FORMULA BANK

Parent graphs
$y = \sin\theta,\ y = \cos\theta,\ y = \tan\theta$

Periods
$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$

Range
$\sin,\cos:\ -1 \le y \le 1 \quad \tan:\ y \in \mathbb{R}$

Combined form
$y = af(b\theta - q) = af\left(b\left(\theta - \frac{q}{b}\right)\right)$

Transform order
shift $\frac{q}{b}$ → horizontal scale $\frac{1}{|b|}$ → vertical scale $|a|$

Concept 3 | Properties and Graphs of Trigonometric Functions

F03 | V01

Q071 | WRITTEN

C05-K03-5-10-GW02

Find the period and range of $y = \sin(5\theta)$.

WORKING AREA

FORMULA BANK

Parent graphs

$y = \sin\theta,\ y = \cos\theta,\ y = \tan\theta$

Periods

$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$

Range

$\sin,\cos:\ -1 \le y \le 1 \quad \tan:\ y \in \mathbb{R}$

Combined form

$y = af(b\theta - q) = af\left(b\left(\theta - \frac{q}{b}\right)\right)$

Transform order

shift $\frac{q}{b}$ → horizontal scale $\frac{1}{|b|}$ → vertical scale $|a|$

Concept 3 | Properties and Graphs of Trigonometric Functions

F03 | V01

Q072 | WRITTEN C05-K03-5-10-GW03

Find the period and range of $y = \tan(\theta/4)$.

WORKING AREA

FORMULA BANK

Parent graphs

$y = \sin\theta,\ y = \cos\theta,\ y = \tan\theta$

Periods

$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$

Range

$\sin, \cos:\ -1 \le y \le 1 \quad \tan:\ y \in \mathbb{R}$

Combined form

$y = af(b\theta - q) = af\left(b\left(\theta - \frac{q}{b}\right)\right)$

Transform order

shift $\frac{q}{b}$ → horizontal scale $\frac{1}{|b|}$ → vertical scale $|a|$

Concept 3 | Properties and Graphs of Trigonometric Functions

F03 | V01

Q073 | WRITTEN

C05-K03-5-10-GW04

Describe the transformations from $y = \cos\theta$ to $y = -\frac{1}{2}\cos(2\theta)$.

WORKING AREA

FORMULA BANK

Parent graphs
$y = \sin\theta,\ y = \cos\theta,\ y = \tan\theta$

Periods
$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$

Range
$\sin,\cos:\ -1 \le y \le 1 \quad \tan:\ y \in \mathbb{R}$

Combined form
$y = af(b\theta - q) = af\left(b\left(\theta - \frac{q}{b}\right)\right)$

Transform order
shift $\frac{q}{b}$ → horizontal scale $\frac{1}{|b|}$ → vertical scale $|a|$

Concept 3 | Properties and Graphs of Trigonometric Functions

F04 | V01

Q074 | WRITTEN

C05-K03-5-11-GW01

Find the shift and period of $y = \sin(\theta + \frac{\pi}{6})$.

WORKING AREA

FORMULA BANK

Parent graphs
$y = \sin\theta,\ y = \cos\theta,\ y = \tan\theta$

Periods
$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$

Range
$\sin, \cos: \ -1 \le y \le 1 \quad \tan: \ y \in \mathbb{R}$

Combined form
$y = af(b\theta - q) = af\left(b\left(\theta - \frac{q}{b}\right)\right)$

Transform order
shift $\frac{q}{b}$ → horizontal scale $\frac{1}{|b|}$ → vertical scale $|a|$

Concept 3 | Properties and Graphs of Trigonometric Functions

F04 | V01

Q075 | WRITTEN

C05-K03-5-11-GW02

Find the shift and period of $y = \cos(\theta - \frac{\pi}{2})$.

WORKING AREA

FORMULA BANK

Parent graphs
$y = \sin\theta,\ y = \cos\theta,\ y = \tan\theta$

Periods
$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$

Range
$\sin,\cos:\ -1 \le y \le 1 \quad \tan:\ y \in \mathbb{R}$

Combined form
$y = af(b\theta - q) = af\left(b\left(\theta - \frac{q}{b}\right)\right)$

Transform order
shift $\frac{q}{b}$ → horizontal scale $\frac{1}{|b|}$ → vertical scale $|a|$

Concept 3 | Properties and Graphs of Trigonometric Functions

F04 | V01

Q076 | WRITTEN

C05-K03-5-11-GW03

Find the central zero and period of $y = \tan(\theta + \frac{\pi}{4})$.

WORKING AREA

FORMULA BANK

Parent graphs
$y = \sin\theta,\ y = \cos\theta,\ y = \tan\theta$

Periods
$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$

Range
$\sin,\cos:\ -1 \le y \le 1 \quad \tan:\ y \in \mathbb{R}$

Combined form
$y = af(b\theta - q) = af\left(b\left(\theta - \frac{q}{b}\right)\right)$

Transform order
shift $\frac{q}{b}$ → horizontal scale $\frac{1}{|b|}$ → vertical scale $|a|$

Concept 3 | Properties and Graphs of Trigonometric Functions

F05 | V01

Q077 | WRITTEN

C05-K03-5-12-GW01

For $y = 2\sin(4\theta - \pi)$, state the shift, period and range.

WORKING AREA

FORMULA BANK

Parent graphs
$y = \sin\theta,\ y = \cos\theta,\ y = \tan\theta$

Periods
$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$

Range
$\sin,\cos:\ -1 \le y \le 1 \quad \tan:\ y \in \mathbb{R}$

Combined form
$y = af(b\theta - q) = af\left(b\left(\theta - \frac{q}{b}\right)\right)$

Transform order
shift $\frac{q}{b}$ → horizontal scale $\frac{1}{|b|}$ → vertical scale $|a|$

Concept 3 | Properties and Graphs of Trigonometric Functions

F05 | V01

Q078 | WRITTEN

C05-K03-5-12-GW02

For $y = \cos(\theta/3 + \pi/6)$, state the shift and period.

WORKING AREA

FORMULA BANK

Parent graphs
$y = \sin\theta,\ y = \cos\theta,\ y = \tan\theta$

Periods
$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$

Range
$\sin,\cos:\ -1 \le y \le 1 \quad \tan:\ y \in \mathbb{R}$

Combined form
$y = af(b\theta - q) = af\left(b\left(\theta - \frac{q}{b}\right)\right)$

Transform order
shift $\frac{q}{b}$ → horizontal scale $\frac{1}{|b|}$ → vertical scale $|a|$

Concept 3 | Properties and Graphs of Trigonometric Functions

F05 | V01

Q079 | WRITTEN

C05-K03-5-12-GW03

For $y = 3\tan(2\theta + \pi)$, state the shift and period.

WORKING AREA

FORMULA BANK

Parent graphs
$y = \sin\theta,\ y = \cos\theta,\ y = \tan\theta$

Periods
$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$

Range
$\sin,\cos:\ -1 \le y \le 1 \quad \tan:\ y \in \mathbb{R}$

Combined form
$y = af(b\theta - q) = af\left(b\left(\theta - \frac{q}{b}\right)\right)$

Transform order
shift $\frac{q}{b}$ → horizontal scale $\frac{1}{|b|}$ → vertical scale $|a|$

Concept 3 | Properties and Graphs of Trigonometric Functions

F05 | V01

Q080 | WRITTEN

C05-K03-5-12-GW04

For $y = 2\cos(\theta/2 - \pi/8)$, state the shift, period and range.

WORKING AREA

FORMULA BANK

Parent graphs
$y = \sin\theta,\ y = \cos\theta,\ y = \tan\theta$

Periods
$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$

Range
$\sin,\cos:\ -1 \le y \le 1 \quad \tan:\ y \in \mathbb{R}$

Combined form
$y = af(b\theta - q) = af\left(b\left(\theta - \frac{q}{b}\right)\right)$

Transform order
shift $\frac{q}{b}$ → horizontal scale $\frac{1}{|b|}$ → vertical scale $|a|$

Concept 3 | Properties and Graphs of Trigonometric Functions

F05 | V01

Quiz M1 | QUIZ MCQ

C05-K03-Q-M01

The period of $y = \sin(2\theta - \frac{\pi}{3})$ is:

A $\pi/2$

B π

C 2π

D 4π

WORKING AREA

FORMULA BANK

Parent graphs

$y = \sin\theta,\ y = \cos\theta,\ y = \tan\theta$

Periods

$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$

Range

$\sin, \cos:\ -1 \leq y \leq 1 \quad \tan:\ y \in \mathbb{R}$

Combined form

$y = af(b\theta - q) = af\left(b\left(\theta - \frac{q}{b}\right)\right)$

Transform order

shift $\frac{q}{b}$ $\rightarrow$ horizontal scale $\frac{1}{|b|}$ $\rightarrow$ vertical scale $|a|$

Concept 3 | Properties and Graphs of Trigonometric Functions

F01 | V01

Quiz M2 | QUIZ MCQ

C05-K03-Q-M02

$\sin(\theta + \pi)$ equals:

A $\sin\theta$

B $-\sin\theta$

C $\cos\theta$

D $-\cos\theta$

WORKING AREA

FORMULA BANK

Parent graphs
$y = \sin\theta,\ y = \cos\theta,\ y = \tan\theta$

Periods
$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$

Range
$\sin,\cos:\ -1 \le y \le 1 \quad \tan:\ y \in \mathbb{R}$

Combined form
$y = af(b\theta - q) = af\left(b\left(\theta - \frac{q}{b}\right)\right)$

Transform order
shift $\frac{q}{b}$ $\to$ horizontal scale $\frac{1}{|b|}$ $\to$ vertical scale $|a|$

Concept 3 | Properties and Graphs of Trigonometric Functions

F03 | V01

Quiz M3 | QUIZ MCQ

C05-K03-Q-M03

The range of $y = 2\cos\theta$ is:

A [-1,1]

B [-2,2]

C [0,2]

D $\mathbb{R}$

WORKING AREA

FORMULA BANK

Parent graphs
$y = \sin\theta,\ y = \cos\theta,\ y = \tan\theta$

Periods
$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$

Range
$\sin, \cos:\ -1 \le y \le 1 \quad \tan:\ y \in \mathbb{R}$

Combined form
$y = af(b\theta - q) = af\left(b\left(\theta - \frac{q}{b}\right)\right)$

Transform order
shift $\frac{q}{b}$ → horizontal scale $\frac{1}{|b|}$ → vertical scale $|a|$

Concept 3 | Properties and Graphs of Trigonometric Functions

F04 | V01

Quiz M4 | QUIZ MCQ C05-K03-Q-M04

The graph of $y = \cos(\theta - \frac{\pi}{4})$ is shifted:

A $left by \frac{\pi}{4}$

B $right by \frac{\pi}{4}$

C $up by \frac{\pi}{4}$

D $down by \frac{\pi}{4}$

WORKING AREA

FORMULA BANK

Parent graphs
$y = \sin\theta,\ y = \cos\theta,\ y = \tan\theta$

Periods
$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$

Range
$\sin, \cos:\ -1 \leq y \leq 1 \quad \tan:\ y \in \mathbb{R}$

Combined form
$y = af(b\theta - q) = af\left(b\left(\theta - \frac{q}{b}\right)\right)$

Transform order
shift $\frac{q}{b}$ → horizontal scale $\frac{1}{|b|}$ → vertical scale $|a|$

Concept 3 | Properties and Graphs of Trigonometric Functions

F05 | V01

Quiz W1 | QUIZ WRITTEN

C05-K03-Q-W01

For $y = 3\sin(2\theta - \pi)$, state the horizontal shift, period and range.

WORKING AREA

FORMULA BANK

Parent graphs
$y = \sin\theta,\ y = \cos\theta,\ y = \tan\theta$

Periods
$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$

Range
$\sin,\cos:\ -1 \le y \le 1 \quad \tan:\ y \in \mathbb{R}$

Combined form
$y = af(b\theta - q) = af\left(b\left(\theta - \frac{q}{b}\right)\right)$

Transform order
shift $\frac{q}{b}$ → horizontal scale $\frac{1}{|b|}$ → vertical scale $|a|$

Concept 3 | Properties and Graphs of Trigonometric Functions

F01 | V01

Quiz W2 | QUIZ WRITTEN

C05-K03-Q-W02

Simplify $\sin(\theta + \frac{\pi}{2}) + \sin(\theta + \pi)$.

WORKING AREA

FORMULA BANK

Parent graphs
$y = \sin\theta,\ y = \cos\theta,\ y = \tan\theta$

Periods
$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$

Range
$\sin,\cos:\ -1 \le y \le 1 \quad \tan:\ y \in \mathbb{R}$

Combined form
$y = af(b\theta - q) = af\left(b\left(\theta - \frac{q}{b}\right)\right)$

Transform order
shift $\frac{q}{b}$ $\rightarrow$ horizontal scale $\frac{1}{|b|}$ $\rightarrow$ vertical scale $|a|$

Concept 3 | Properties and Graphs of Trigonometric Functions

F04 | V02

Quiz W3 | QUIZ WRITTEN C05-K03-Q-W03

Draw one cycle of $y = \tan(\theta - \frac{\pi}{3})$ and state its period and asymptotes.

WORKING AREA

FORMULA BANK

Parent graphs
$y = \sin\theta,\ y = \cos\theta,\ y = \tan\theta$

Periods
$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$

Range
$\sin, \cos: \ -1 \le y \le 1 \quad \tan: \ y \in \mathbb{R}$

Combined form
$y = af(b\theta - q) = af\left(b\left(\theta - \frac{q}{b}\right)\right)$

Transform order
shift $\frac{q}{b}$ → horizontal scale $\frac{1}{|b|}$ → vertical scale $|a|$

Concept 3 | Properties and Graphs of Trigonometric Functions

F03 | V04

Quiz W4 | QUIZ WRITTEN

C05-K03-Q-W04

For $y = -2|\sin(3x)|$, state its period and range and describe the reflection.

WORKING AREA

FORMULA BANK

Parent graphs

$y = \sin\theta,\ y = \cos\theta,\ y = \tan\theta$

Periods

$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$

Range

$\sin, \cos: \ -1 \le y \le 1 \quad \tan: \ y \in \mathbb{R}$

Combined form

$y = af(b\theta - q) = af\left(b\left(\theta - \frac{q}{b}\right)\right)$

Transform order

shift $\frac{q}{b}$ $\to$ horizontal scale $\frac{1}{|b|}$ $\to$ vertical scale $|a|$

EGYPTIAN BACCALAUREATE • 2026

MATHEMATICS

CLASSROOM WORKING EDITION

Chapter 05

Trigonometric Functions

1 Concept | 42 questions

B5 Landscape | one question per teaching page

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Chapter 05 | Trigonometric Functions

CLASSROOM ROUTE

Concept 4 | Solving Equations and Inequalities Involving Trigonometric Functions

42 questions

Official lessons: 5-13, 5-14, 5-15, 5-16

Formula Bank changes at each Concept boundary.

Concept 4 | Solving Equations and Inequalities Involving Trigonometric Functions **OFFICIAL LESSONS 5-13, 5-14, 5-15, 5-16**

Solving Equations and Inequalities Involving Trigonometric Functions

Equation-first recall | use the same rail beside every question in this Concept

Single-function form

$$\sin^2\theta + \cos^2\theta = 1$$

Periods

$$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$$

Range check

$$-1 \leq \sin\theta, \cos\theta \leq 1$$

Inequality

solve equality → shade arcs → check endpoints

Shifted argument

$t = \theta + \alpha$ **find the range of** t

Start with the family cue, write the first move, then use the working grid.

Concept 4 | Solving Equations and Inequalities Involving Trigonometric Functions

F01 | V01

Q001 | WRITTEN

C05-K04-5-13-WU

Solve (a) $-2\cos\theta = \sqrt{2}$ and (b) $\tan\theta + \sqrt{3} = 0$, giving the general solution for (b).

WORKING AREA

FORMULA BANK

Single-function form

$$\sin^2\theta + \cos^2\theta = 1$$

Periods

$$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$$

Range check

$$-1 \le \sin\theta, \cos\theta \le 1$$

Inequality

solve equality → shade arcs → check endpoints

Shifted argument

$t = \theta + \alpha$ find the range of t

Concept 4 | Solving Equations and Inequalities Involving Trigonometric Functions

F01 | V02

Q002 | WRITTEN

C05-K04-5-13-TRY

Solve (a) $2\cos\theta - 1 = 0$ and (b) $\tan\theta - 1 = 0$ for $0 \le \theta \le 2\pi$.

WORKING AREA

FORMULA BANK

Single-function form
$\sin^2\theta + \cos^2\theta = 1$

Periods
$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$

Range check
$-1 \le \sin\theta, \cos\theta \le 1$

Inequality
solve equality → shade arcs → check endpoints

Shifted argument
$t = \theta + \alpha$ find the range of t

Concept 4 | Solving Equations and Inequalities Involving Trigonometric Functions

F01 | V03

Q003 | WRITTEN C05-K04-5-13-EX

Solve all given equations on $0 \le \theta \le 2\pi$: (a) $2\sin\theta + 1 = 0$, (b) $2\sin\theta = \sqrt{3}$, (c) $3\sin\theta + 3 = 0$, (d) $\sqrt{2}\cos\theta + 1 = 0$, (e) $2\cos\theta = \sqrt{3}$, (f) $2\cos\theta - \sqrt{2} = 0$, (g) $\tan\theta + 1 = 0$, (h) $\sqrt{3}\tan\theta = 1$, (i) $\sqrt{3}\tan\theta = 3$, and give general solutions for (j) $2\sin\theta + \sqrt{3} = 0$, (k) $\cos\theta + 1 = 0$, (l) $\sqrt{3}\tan\theta + 3 = 0$.

WORKING AREA

FORMULA BANK

Single-function form

$$\sin^2\theta + \cos^2\theta = 1$$

Periods

$$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$$

Range check

$$-1 \le \sin\theta, \cos\theta \le 1$$

Inequality

solve equality → shade arcs → check endpoints

Shifted argument

$t = \theta + \alpha$ find the range of t

Concept 4 | Solving Equations and Inequalities Involving Trigonometric Functions

F01 | V04

Q004 | WRITTEN

C05-K04-5-13-CH

Solve $\tan^2\theta - (2+\sqrt{3})\tan\theta + 2\sqrt{3} = 0$ and give the general solution.

WORKING AREA

FORMULA BANK

Single-function form
$$\sin^2\theta + \cos^2\theta = 1$$

Periods
$$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$$

Range check
$$-1 \le \sin\theta, \cos\theta \le 1$$

Inequality
solve equality → shade arcs → check endpoints

Shifted argument
$t = \theta + \alpha$ find the range of t

Concept 4 | Solving Equations and Inequalities Involving Trigonometric Functions

F02 | V01

Q005 | WRITTEN

C05-K04-5-14-WU

Solve for $0 \le \theta < 2\pi$: (a) $2\cos\theta > -\sqrt{3}$, (b) $\tan\theta \le \frac{1}{\sqrt{3}}$.

WORKING AREA

FORMULA BANK

Single-function form

$\sin^2\theta + \cos^2\theta = 1$

Periods

$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$

Range check

$-1 \le \sin\theta, \cos\theta \le 1$

Inequality

solve equality → shade arcs → check endpoints

Shifted argument

$t = \theta + \alpha$ find the range of t

Concept 4 | Solving Equations and Inequalities Involving Trigonometric Functions

F02 | V02

Q006 | WRITTEN

C05-K04-5-14-TRY

Solve for $0 \leq \theta < 2\pi$: (a) $2\sin\theta \leq -1$, (b) $\cos\theta > \frac{1}{2}$, (c) $\tan\theta > 1$.

WORKING AREA

FORMULA BANK

Single-function form

$$\sin^2\theta + \cos^2\theta = 1$$

Periods

$$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$$

Range check

$$-1 \leq \sin\theta, \cos\theta \leq 1$$

Inequality

solve equality → shade arcs → check endpoints

Shifted argument

$t = \theta + \alpha$ find the range of t

Concept 4 | Solving Equations and Inequalities Involving Trigonometric Functions

F02 | V03

Q007 | WRITTEN

C05-K04-5-14-EX

Solve all given inequalities for $0 \le \theta < 2\pi$: (a) $\sin\theta > \frac{\sqrt{3}}{2}$, (b) $\sin\theta - \frac{\sqrt{2}}{2} > 0$, (c) $2\sin\theta + \sqrt{3} \le 0$, (d) $\cos\theta > 0$, (e) $\cos\theta - \frac{1}{2} \le 0$, (f) $2\cos\theta + \sqrt{2} \le 0$, (g) $\tan\theta \le -\frac{1}{\sqrt{3}}$, (h) $\tan\theta + \sqrt{3} > 0$, (i) $\tan\theta + 1 \ge 0$.

WORKING AREA

FORMULA BANK

Single-function form
$\sin^2\theta + \cos^2\theta = 1$

Periods
$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$

Range check
$-1 \le \sin\theta, \cos\theta \le 1$

Inequality
solve equality → shade arcs → check endpoints

Shifted argument
$t = \theta + \alpha$ find the range of t

Concept 4 | Solving Equations and Inequalities Involving Trigonometric Functions

F02 | V04

Q008 | WRITTEN

C05-K04-5-14-CH

Solve $2\sin^2\theta - \cos\theta - 1 > 0$ for $0 \le \theta < 2\pi$.

WORKING AREA

FORMULA BANK

Single-function form

$$\sin^2\theta + \cos^2\theta = 1$$

Periods

$$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$$

Range check

$$-1 \le \sin\theta, \cos\theta \le 1$$

Inequality

solve equality → shade arcs → check endpoints

Shifted argument

$t = \theta + \alpha$ find the range of t

Concept 4 | Solving Equations and Inequalities Involving Trigonometric Functions

F03 | V01

Q009 | WRITTEN | C05-K04-5-15-WU

Solve for $0 \le \theta < 2\pi$: (a) $2\cos^2\theta - 3\sin\theta = 0$, (b) $2\sin^2\theta - 3\cos\theta - 3 \le 0$.

WORKING AREA

FORMULA BANK

Single-function form

$$\sin^2\theta + \cos^2\theta = 1$$

Periods

$$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$$

Range check

$$-1 \le \sin\theta, \cos\theta \le 1$$

Inequality

solve equality → shade arcs → check endpoints

Shifted argument

$t = \theta + \alpha$ find the range of t

Concept 4 | Solving Equations and Inequalities Involving Trigonometric Functions

F03 | V02

Q010 | WRITTEN

C05-K04-5-15-TRY

Solve for $0 \leq \theta < 2\pi$: (a) $2\sin^2\theta - \sin\theta = 0$, (b) $2\cos^2\theta - \sin\theta = 2$, (c) $2\sin^2\theta - 4 < 5\cos\theta$.

WORKING AREA

FORMULA BANK

Single-function form

$$\sin^2\theta + \cos^2\theta = 1$$

Periods

$$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$$

Range check

$$-1 \leq \sin\theta, \cos\theta \leq 1$$

Inequality

solve equality → shade arcs → check endpoints

Shifted argument

$t = \theta + \alpha$ **find the range of** t

Concept 4 | Solving Equations and Inequalities Involving Trigonometric Functions

F03 | V03

Q011 | WRITTEN — C05-K04-5-15-EX

Solve all given equations and inequalities for $0 \le \theta < 2\pi$: (a) $2\sin^2\theta + \sin\theta - 1 = 0$, (b) $2\sin^2\theta + 3\cos\theta = 0$, (c) $2\sin^2\theta - \cos\theta = 2$, (d) $2\cos^2\theta - 5\cos\theta - 3 = 0$, (e) $2\cos^2\theta + \sin\theta - 1 = 0$, (f) $2\cos^2\theta + 3\sin\theta = 3$, (g) $2\cos^2\theta + 3\sin\theta \le 0$, (h) $2\sin^2\theta + \cos\theta > 1$, (i) $2\cos^2\theta - 4 \le 5\sin\theta$.

WORKING AREA

FORMULA BANK

Single-function form

$$\sin^2\theta + \cos^2\theta = 1$$

Periods

$$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$$

Range check

$$-1 \le \sin\theta, \cos\theta \le 1$$

Inequality

solve equality → shade arcs → check endpoints

Shifted argument

$t = \theta + \alpha$ find the range of t

Concept 4 | Solving Equations and Inequalities Involving Trigonometric Functions

F03 | V04

Q012 | WRITTEN

C05-K04-5-15-CH

Solve $\frac{1-2\sin^2\theta+\sin\theta}{\cos^2\theta} \geq 1$ for $0 \leq \theta < 2\pi$.

WORKING AREA

FORMULA BANK

Single-function form

$$\sin^2\theta + \cos^2\theta = 1$$

Periods

$$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$$

Range check

$$-1 \leq \sin\theta, \cos\theta \leq 1$$

Inequality

solve equality → shade arcs → check endpoints

Shifted argument

$t = \theta + \alpha$ find the range of t

Concept 4 | Solving Equations and Inequalities Involving Trigonometric Functions

F04 | V01

Q013 | WRITTEN

C05-K04-5-16-WU

Solve for $0 \leq \theta < 2\pi$: (a) $\sin(\theta + \frac{\pi}{3}) = \frac{1}{2}$, (b) $\sin(\theta + \frac{\pi}{6}) > -\frac{1}{2}$.

WORKING AREA

FORMULA BANK

Single-function form
$$\sin^2\theta + \cos^2\theta = 1$$

Periods
$$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$$

Range check
$$-1 \leq \sin\theta, \cos\theta \leq 1$$

Inequality
solve equality → shade arcs → check endpoints

Shifted argument
$t = \theta + \alpha$ find the range of t

Concept 4 | Solving Equations and Inequalities Involving Trigonometric Functions

F04 | V02

Q014 | WRITTEN C05-K04-5-16-TRY

Solve for $0 \le \theta < 2\pi$: (a) $\sin(\theta + \frac{\pi}{6}) = \frac{1}{2}$, (b) $\tan(\theta + \frac{\pi}{4}) = \sqrt{3}$, (c) $\cos(\theta + \frac{\pi}{4}) < -\frac{\sqrt{2}}{2}$.

WORKING AREA

FORMULA BANK

Single-function form

$$\sin^2\theta + \cos^2\theta = 1$$

Periods

$$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$$

Range check

$$-1 \le \sin\theta, \cos\theta \le 1$$

Inequality

solve equality → shade arcs → check endpoints

Shifted argument

$t = \theta + \alpha$ find the range of t

Concept 4 | Solving Equations and Inequalities Involving Trigonometric Functions

F04 | V03

Q015 | WRITTEN

C05-K04-5-16-EX

Solve for $0 \le \theta < 2\pi$: (a) $\cos(\theta + \frac{\pi}{3}) = -\frac{\sqrt{3}}{2}$, (b) $\cos(\theta + \frac{\pi}{6}) = -\frac{1}{2}$, (c) $\sin(\theta + \frac{\pi}{4}) = \frac{\sqrt{3}}{2}$, (d) $\tan(\theta + \frac{\pi}{4}) = -1$, (e) $\tan(\theta + \frac{\pi}{3}) = \frac{1}{\sqrt{3}}$, (f) $\tan(\theta + \frac{\pi}{6}) = 1$, (g) $\sin(\theta + \frac{\pi}{6}) \le \frac{\sqrt{3}}{2}$, (h) $\sin(\theta + \frac{\pi}{4}) \ge \frac{\sqrt{2}}{2}$, (i) $\cos(\theta + \frac{\pi}{3}) > \frac{1}{2}$.

WORKING AREA

FORMULA BANK

Single-function form

$$\sin^2\theta + \cos^2\theta = 1$$

Periods

$$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$$

Range check

$$-1 \le \sin\theta, \cos\theta \le 1$$

Inequality

solve equality → shade arcs → check endpoints

Shifted argument

$t = \theta + \alpha$ find the range of t

Concept 4 | Solving Equations and Inequalities Involving Trigonometric Functions

F04 | V04

Q016 | WRITTEN

C05-K04-5-16-CH

Solve $4\sin^2(\theta + \frac{\pi}{6}) \geq 1$ for $0 \leq \theta < 2\pi$.

WORKING AREA

FORMULA BANK

Single-function form

$$\sin^2\theta + \cos^2\theta = 1$$

Periods

$$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$$

Range check

$$-1 \leq \sin\theta, \cos\theta \leq 1$$

Inequality

solve equality → shade arcs → check endpoints

Shifted argument

$t = \theta + \alpha$ find the range of t

Concept 4 | Solving Equations and Inequalities Involving Trigonometric Functions

F04 | V04

Q017 | WRITTEN

C05-K04-5-16-REF

A student divides $\tan x \times \cos x = \frac{1}{2}$ by $\cos x$, obtaining $\tan x = \frac{1}{2\cos x}$. Evaluate this strategy and propose a better substitution on $[0°, 360°[$.

WORKING AREA

FORMULA BANK

Single-function form
$\sin^2\theta + \cos^2\theta = 1$

Periods
$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$

Range check
$-1 \le \sin\theta, \cos\theta \le 1$

Inequality
solve equality → shade arcs → check endpoints

Shifted argument
$t = \theta + \alpha$ find the range of t

Concept 4 | Solving Equations and Inequalities Involving Trigonometric Functions

F01 | V01

Q018 | MCQ

C05-K04-5-13-GM01

The period used for a general tangent solution is:

A 2π

B π

C $\pi/2$

D 4π

WORKING AREA

FORMULA BANK

Single-function form

$$\sin^2\theta + \cos^2\theta = 1$$

Periods

$$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$$

Range check

$$-1 \le \sin\theta, \cos\theta \le 1$$

Inequality

solve equality → shade arcs → check endpoints

Shifted argument

$t = \theta + \alpha$ find the range of t

Concept 4 | Solving Equations and Inequalities Involving Trigonometric Functions

F01 | V01

Q019 | MCQ

C05-K04-5-13-GM02

The solutions of $2\cos\theta = 1$ on $[0, 2\pi]$ are:

A $\pi/6, 11\pi/6$

B $\pi/3, 5\pi/3$

C $2\pi/3, 4\pi/3$

D $\pi/4, 7\pi/4$

WORKING AREA

FORMULA BANK

Single-function form

$$\sin^2\theta + \cos^2\theta = 1$$

Periods

$$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$$

Range check

$$-1 \le \sin\theta, \cos\theta \le 1$$

Inequality

solve equality → shade arcs → check endpoints

Shifted argument

$t = \theta + \alpha$ find the range of t

Concept 4 | Solving Equations and Inequalities Involving Trigonometric Functions

F01 | V01

Q020 | MCQ

C05-K04-5-13-GM03

Which root is impossible when solving $2\sin^2\theta + 3\sin\theta - 2 = 0$?:

A $\sin\theta = 1/2$

B $\sin\theta = -2$

C $\sin\theta = 0$

D $\sin\theta = -1/2$

WORKING AREA

FORMULA BANK

Single-function form

$$\sin^2\theta + \cos^2\theta = 1$$

Periods

$$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$$

Range check

$$-1 \le \sin\theta, \cos\theta \le 1$$

Inequality

solve equality → shade arcs → check endpoints

Shifted argument

$t = \theta + \alpha$ find the range of t

Concept 4 | Solving Equations and Inequalities Involving Trigonometric Functions

F01 | V01

Q021 | MCQ

C05-K04-5-13-GM04

The quadratic challenge factors into roots:

A 1 and $\sqrt{3}$

B 2 and $\sqrt{3}$

C 2 and 3

D $\pi/3$ and 2

WORKING AREA

FORMULA BANK

Single-function form

$$\sin^2\theta + \cos^2\theta = 1$$

Periods

$$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$$

Range check

$$-1 \le \sin\theta, \cos\theta \le 1$$

Inequality

solve equality → shade arcs → check endpoints

Shifted argument

$t = \theta + \alpha$ **find the range of** t

Concept 4 | Solving Equations and Inequalities Involving Trigonometric Functions

F02 | V01

Q022 | MCQ — C05-K04-5-14-GM01

For $2\cos\theta > -\sqrt{3}$, the excluded arc is:

A $[0, 5\pi/6]$

B $[5\pi/6, 7\pi/6]$

C $[7\pi/6, 2\pi)$

D $(\pi/6, 5\pi/6)$

WORKING AREA

FORMULA BANK

Single-function form
$$\sin^2\theta + \cos^2\theta = 1$$

Periods
$$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$$

Range check
$$-1 \le \sin\theta, \cos\theta \le 1$$

Inequality
solve equality → shade arcs → check endpoints

Shifted argument
$t = \theta + \alpha$ find the range of t

Concept 4 | Solving Equations and Inequalities Involving Trigonometric Functions

F02 | V01

Q023 | MCQ

C05-K04-5-14-GM02

The inequality $\tan\theta > 1$ on $[0, 2\pi)$ holds on:

A $(\pi/4, \pi/2) \cup (5\pi/4, 3\pi/2)$

B $[0, \pi/4]$

C $(\pi/2, \pi)$

D $[\pi, 2\pi)$

WORKING AREA

FORMULA BANK

Single-function form

$$\sin^2\theta + \cos^2\theta = 1$$

Periods

$$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$$

Range check

$$-1 \le \sin\theta, \cos\theta \le 1$$

Inequality

solve equality → shade arcs → check endpoints

Shifted argument

$t = \theta + \alpha$ find the range of t

Concept 4 | Solving Equations and Inequalities Involving Trigonometric Functions

F02 | V01

Q024 | MCQ

C05-K04-5-14-GM03

A closed endpoint in a trig inequality is used when the sign is:

A strictly greater

B strictly less

C greater than or equal to

D never

WORKING AREA

FORMULA BANK

Single-function form

$$\sin^2\theta + \cos^2\theta = 1$$

Periods

$$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$$

Range check

$$-1 \le \sin\theta, \cos\theta \le 1$$

Inequality

solve equality → shade arcs → check endpoints

Shifted argument

$t = \theta + \alpha$ find the range of t

Concept 4 | Solving Equations and Inequalities Involving Trigonometric Functions

F02 | V01

Q025 | MCQ

C05-K04-5-14-GM04

The 5-14 challenge reduces to which cosine condition?:

A $\cos\theta > 1/2$

B $-1 < \cos\theta < 1/2$

C $\cos\theta \leq -1$

D $\cos\theta \geq 1$

WORKING AREA

FORMULA BANK

Single-function form

$$\sin^2\theta + \cos^2\theta = 1$$

Periods

$$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$$

Range check

$$-1 \leq \sin\theta, \cos\theta \leq 1$$

Inequality

solve equality → shade arcs → check endpoints

Shifted argument

$t = \theta + \alpha$ find the range of t

Concept 4 | Solving Equations and Inequalities Involving Trigonometric Functions

F03 | V01

Q026 | MCQ

C05-K04-5-15-GM01

When a quadratic gives $\sin\theta = -2$, we:

A keep it

B reject it

C square it

D divide by 2

WORKING AREA

FORMULA BANK

Single-function form

$$\sin^2\theta + \cos^2\theta = 1$$

Periods

$$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$$

Range check

$$-1 \leq \sin\theta, \cos\theta \leq 1$$

Inequality

solve equality → shade arcs → check endpoints

Shifted argument

$t = \theta + \alpha$ find the range of t

Concept 4 | Solving Equations and Inequalities Involving Trigonometric Functions

F03 | V01

Q027 | MCQ

C05-K04-5-15-GM02

The first step for $2\cos^2\theta - 3\sin\theta = 0$ is to use:

A $\sin^2\theta + \cos^2\theta = 1$

B $\tan\theta = \sin\theta$

C $\cos\theta = 1$

D the tangent period

WORKING AREA

FORMULA BANK

Single-function form

$$\sin^2\theta + \cos^2\theta = 1$$

Periods

$$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$$

Range check

$$-1 \leq \sin\theta, \cos\theta \leq 1$$

Inequality

solve equality → shade arcs → check endpoints

Shifted argument

$t = \theta + \alpha$ find the range of t

Concept 4 | Solving Equations and Inequalities Involving Trigonometric Functions

F03 | V01

Q028 | MCQ C05-K04-5-15-GM03

The solution of $2\sin^2\theta + \sin\theta - 1 = 0$ includes:

A $\theta = \pi/6$

B $\theta = \pi/4$

C $\theta = \pi/3$

D $\theta = 2\pi/3$

WORKING AREA

FORMULA BANK

Single-function form
$\sin^2\theta + \cos^2\theta = 1$

Periods
$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$

Range check
$-1 \le \sin\theta, \cos\theta \le 1$

Inequality
solve equality → shade arcs → check endpoints

Shifted argument
$t = \theta + \alpha$ find the range of t

Concept 4 | Solving Equations and Inequalities Involving Trigonometric Functions

F03 | V01

Q029 | MCQ

C05-K04-5-15-GM04

In the rational challenge, which angle must be excluded?:

A 0

B $\pi/4$

C $\pi/2$

D π

WORKING AREA

FORMULA BANK

Single-function form

$$\sin^2\theta + \cos^2\theta = 1$$

Periods

$$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$$

Range check

$$-1 \le \sin\theta, \cos\theta \le 1$$

Inequality

solve equality → shade arcs → check endpoints

Shifted argument

$t = \theta + \alpha$ find the range of t

Concept 4 | Solving Equations and Inequalities Involving Trigonometric Functions

F04 | V01

Q030 | MCQ

C05-K04-5-16-GM01

After setting $t = \theta + \pi/3$ with $0 \le \theta < 2\pi$, the t-range is:

A $[0, 2\pi)$

B $[\pi/3, 7\pi/3)$

C $[-\pi/3, 5\pi/3)$

D $[\pi, 3\pi)$

WORKING AREA

FORMULA BANK

Single-function form
$$\sin^2\theta + \cos^2\theta = 1$$

Periods
$$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$$

Range check
$$-1 \le \sin\theta, \cos\theta \le 1$$

Inequality
solve equality → shade arcs → check endpoints

Shifted argument
$t = \theta + \alpha$ find the range of t

Concept 4 | Solving Equations and Inequalities Involving Trigonometric Functions

F04 | V01

Q031 | MCQ

C05-K04-5-16-GM02

The solutions of $\tan(\theta + \pi/4) = \sqrt{3}$ include:

A $\pi/12$

B $\pi/6$

C $\pi/3$

D $\pi/2$

WORKING AREA

FORMULA BANK

Single-function form
$$\sin^2\theta + \cos^2\theta = 1$$

Periods
$$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$$

Range check
$$-1 \le \sin\theta, \cos\theta \le 1$$

Inequality
solve equality → shade arcs → check endpoints

Shifted argument
$t = \theta + \alpha$ find the range of t

Concept 4 | Solving Equations and Inequalities Involving Trigonometric Functions

F04 | V01

Q032 | MCQ

C05-K04-5-16-GM03

For $\sin(\theta + \pi/4) \geq \sqrt{2}/2$, the θ-solution is:

A $[0, \pi/2]$

B $(0, \pi/2)$

C $[\pi/2, \pi]$

D $[0, 2\pi)$

WORKING AREA

FORMULA BANK

Single-function form
$$\sin^2\theta + \cos^2\theta = 1$$

Periods
$$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$$

Range check
$$-1 \leq \sin\theta, \cos\theta \leq 1$$

Inequality
solve equality → shade arcs → check endpoints

Shifted argument
$t = \theta + \alpha$ find the range of t

Concept 4 | Solving Equations and Inequalities Involving Trigonometric Functions

F04 | V01

Q033 | MCQ

C05-K04-5-16-GM04

The 5-16 challenge is equivalent to:

A $|\sin t| \geq 1/2$

B $\sin t = 0$

C $\cos t > 1/2$

D $\tan t = 1$

WORKING AREA

FORMULA BANK

Single-function form

$$\sin^2\theta + \cos^2\theta = 1$$

Periods

$$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$$

Range check

$$-1 \leq \sin\theta, \cos\theta \leq 1$$

Inequality

solve equality → shade arcs → check endpoints

Shifted argument

$t = \theta + \alpha$ find the range of t

Concept 4 | Solving Equations and Inequalities Involving Trigonometric Functions

F04 | V01

Q034 | MCQ

C05-K04-5-16-GM05

In the Reflect problem, $\tan x \cos x$ simplifies to:

A $\cos x$

B $\sin x$

C $\tan^2 x$

D 1

WORKING AREA

FORMULA BANK

Single-function form

$$\sin^2\theta + \cos^2\theta = 1$$

Periods

$$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$$

Range check

$$-1 \le \sin\theta, \cos\theta \le 1$$

Inequality

solve equality → shade arcs → check endpoints

Shifted argument

$t = \theta + \alpha$ find the range of t

Concept 4 | Solving Equations and Inequalities Involving Trigonometric Functions

F01 | V01

Quiz M1 | QUIZ MCQ — C05-K04-Q-M01

The general solution of $\tan\theta = -1$ is:

A $-\pi/4 + 2n\pi$

B $-\pi/4 + n\pi$

C $\pi/4 + 2n\pi$

D $\pi/2 + n\pi$

WORKING AREA

FORMULA BANK

Single-function form
$$\sin^2\theta + \cos^2\theta = 1$$

Periods
$$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$$

Range check
$$-1 \le \sin\theta, \cos\theta \le 1$$

Inequality
solve equality → shade arcs → check endpoints

Shifted argument
$t = \theta + \alpha$ find the range of t

Concept 4 | Solving Equations and Inequalities Involving Trigonometric Functions

F02 | V01

Quiz M2 | QUIZ MCQ

C05-K04-Q-M02

The solution of $\cos\theta > 0$ on $[0, 2\pi)$ is:

A $(\pi/2, 3\pi/2)$

B $[0, \pi/2) \cup (3\pi/2, 2\pi)$

C $[0, \pi)$

D $(\pi, 2\pi)$

WORKING AREA

FORMULA BANK

Single-function form
$$\sin^2\theta + \cos^2\theta = 1$$

Periods
$$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$$

Range check
$$-1 \leq \sin\theta, \cos\theta \leq 1$$

Inequality
solve equality → shade arcs → check endpoints

Shifted argument
$t = \theta + \alpha$ **find the range of** t

Concept 4 | Solving Equations and Inequalities Involving Trigonometric Functions

F03 | V01

Quiz M3 | QUIZ MCQ

C05-K04-Q-M03

The valid sine root of $2\sin^2\theta + 3\sin\theta - 2 = 0$ is:

A $\sin\theta = -2$

B $\sin\theta = 1/2$

C $\sin\theta = 2$

D $\sin\theta = -1/2$

WORKING AREA

FORMULA BANK

Single-function form

$$\sin^2\theta + \cos^2\theta = 1$$

Periods

$$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$$

Range check

$$-1 \le \sin\theta, \cos\theta \le 1$$

Inequality

solve equality → shade arcs → check endpoints

Shifted argument

$t = \theta + \alpha$ find the range of t

Concept 4 | Solving Equations and Inequalities Involving Trigonometric Functions

F04 | V01

Quiz M4 | QUIZ MCQ

C05-K04-Q-M04

The first operation in a shifted-argument problem is to:

A divide by $\cos\theta$

B set t equal to the whole argument and find its range

C change radians to degrees

D square both sides

WORKING AREA

FORMULA BANK

Single-function form
$$\sin^2\theta + \cos^2\theta = 1$$

Periods
$$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$$

Range check
$$-1 \le \sin\theta, \cos\theta \le 1$$

Inequality
solve equality → shade arcs → check endpoints

Shifted argument
$t = \theta + \alpha$ **find the range of** t

Concept 4 | Solving Equations and Inequalities Involving Trigonometric Functions

F01 | V01

Quiz W1 | QUIZ WRITTEN

C05-K04-Q-W01

Solve $2\cos\theta = \sqrt{3}$ for $0 \le \theta < 2\pi$.

WORKING AREA

FORMULA BANK

Single-function form

$$\sin^2\theta + \cos^2\theta = 1$$

Periods

$$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$$

Range check

$$-1 \le \sin\theta, \cos\theta \le 1$$

Inequality

solve equality → shade arcs → check endpoints

Shifted argument

$t = \theta + \alpha$ find the range of t

Concept 4 | Solving Equations and Inequalities Involving Trigonometric Functions

F02 | V01

Quiz W2 | QUIZ WRITTEN

C05-K04-Q-W02

Solve $\tan\theta \le -1$ for $0 \le \theta < 2\pi$.

WORKING AREA

FORMULA BANK

Single-function form

$$\sin^2\theta + \cos^2\theta = 1$$

Periods

$$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$$

Range check

$$-1 \le \sin\theta, \cos\theta \le 1$$

Inequality

solve equality → shade arcs → check endpoints

Shifted argument

$t = \theta + \alpha$ find the range of t

Concept 4 | Solving Equations and Inequalities Involving Trigonometric Functions

F03 | V01

Quiz W3 | QUIZ WRITTEN

C05-K04-Q-W03

Solve $2\sin^2\theta + \sin\theta - 1 = 0$ for $0 \le \theta < 2\pi$.

WORKING AREA

FORMULA BANK

Single-function form

$\sin^2\theta + \cos^2\theta = 1$

Periods

$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$

Range check

$-1 \le \sin\theta, \cos\theta \le 1$

Inequality

solve equality → shade arcs → check endpoints

Shifted argument

$t = \theta + \alpha$ find the range of t

Concept 4 | Solving Equations and Inequalities Involving Trigonometric Functions

F04 | V01

Quiz W4 | QUIZ WRITTEN C05-K04-Q-W04

Solve $\cos(\theta + \frac{\pi}{3}) > \frac{1}{2}$ for $0 \le \theta < 2\pi$.

WORKING AREA

FORMULA BANK

Single-function form

$$\sin^2\theta + \cos^2\theta = 1$$

Periods

$$T_{\sin,\cos} = 2\pi \quad T_{\tan} = \pi$$

Range check

$$-1 \le \sin\theta, \cos\theta \le 1$$

Inequality

solve equality → shade arcs → check endpoints

Shifted argument

$t = \theta + \alpha$ find the range of t

EGYPTIAN BACCALAUREATE • 2026

MATHEMATICS

CLASSROOM WORKING EDITION

Chapter 05

Trigonometric Functions

1 Concepts | 62 questions

B5 Landscape | one question per teaching page

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Chapter 05 | Trigonometric Functions

CLASSROOM ROUTE

Concept 5 | Advanced Theorems and Applications

62 questions

Official lessons: 5-17, 5-18, 5-19, 5-20, 5-21, 5-22, 5-23, 5-24, 5-25

The Formula Bank changes at each Concept boundary.

Concept 5 | Advanced Theorems and Applications | **OFFICIAL LESSONS 5-17, 5-18, 5-19, 5-20, 5-21, 5-22, 5-23, 5-24, 5-25**

Advanced Theorems and Applications

Equation-first recall | use the same rail beside every question in this Concept

Extrema

$$y = -(t-h)^2 + k$$

Addition

$$\sin(\alpha+\beta) = \sin\alpha\cos\beta + \cos\alpha\sin\beta$$

Double angle

$$\sin 2\alpha = 2\sin\alpha\cos\alpha \qquad \cos 2\alpha = \cos^2\alpha - \sin^2\alpha$$

Harmonic form

$$a\sin\theta + b\cos\theta = r\sin(\theta+\alpha)$$

Angle / range

$$\tan\theta = \frac{m_1 - m_2}{1 + m_1 m_2} \qquad -1 \le \sin\theta, \cos\theta \le 1$$

Start with the family cue, write the first move, then use the working grid.

Concept 5 | Advanced Theorems and Applications

F01 | V01

Q001 | WRITTEN

C05-K05-5-17-WU

Find the maximum and minimum of $y = sin^2\theta + cos\theta + 1,\ 0le\theta < 2pi$, and the corresponding θ.

WORKING AREA

FORMULA BANK

Extrema

$$y = -(t-h)^2 + k$$

Addition

$$\sin(\alpha+\beta) = \sin\alpha\cos\beta + \cos\alpha\sin\beta$$

Double angle

$$\sin 2\alpha = 2\sin\alpha\cos\alpha \qquad \cos 2\alpha = \cos^2\alpha - \sin^2\alpha$$

Harmonic form

$$a\sin\theta + b\cos\theta = r\sin(\theta+\alpha)$$

Angle / range

$$\tan\theta = \frac{m_1 - m_2}{1 + m_1 m_2} \qquad -1 \le \sin\theta, \cos\theta \le 1$$

Concept 5 | Advanced Theorems and Applications

F01 | V02

Q002 | WRITTEN

C05-K05-5-17-TRY

Find the maximum and minimum of $y = cos^2\theta - sin\theta, 0 le\theta < 2pi$, and the corresponding θ.

WORKING AREA

FORMULA BANK

Extrema
$$y = -(t-h)^2 + k$$

Addition
$$\sin(\alpha+\beta) = \sin\alpha\cos\beta + \cos\alpha\sin\beta$$

Double angle
$$\sin 2\alpha = 2\sin\alpha\cos\alpha \quad \cos 2\alpha = \cos^2\alpha - \sin^2\alpha$$

Harmonic form
$$a\sin\theta + b\cos\theta = r\sin(\theta+\alpha)$$

Angle / range
$$\tan\theta = \frac{m_1 - m_2}{1 + m_1 m_2} \quad -1 \le \sin\theta, \cos\theta \le 1$$

Concept 5 | Advanced Theorems and Applications

F01 | V03

Q003 | WRITTEN

C05-K05-5-17-EX

Solve the four given maximum/minimum problems on $0le\theta < 2pi$, including $y = \sin^2\theta + \sin\theta - 1$, $y = \cos^2\theta - \cos\theta + 1$, $y = 2\sin\theta + \cos^2\theta - 1$, and $y = -2\sin^2\theta - 2\cos\theta + 1$. Include the given challenge $y = \cos^2\theta - \sin\theta$.

WORKING AREA

FORMULA BANK

Extrema
$$y = -(t-h)^2 + k$$

Addition
$$\sin(\alpha+\beta) = \sin\alpha\cos\beta + \cos\alpha\sin\beta$$

Double angle
$$\sin 2\alpha = 2\sin\alpha\cos\alpha \qquad \cos 2\alpha = \cos^2\alpha - \sin^2\alpha$$

Harmonic form
$$a\sin\theta + b\cos\theta = r\sin(\theta+\alpha)$$

Angle / range
$$\tan\theta = \frac{m_1 - m_2}{1 + m_1 m_2} \qquad -1 \le \sin\theta, \cos\theta \le 1$$

Concept 5 | Advanced Theorems and Applications

F02 | V01

Q004 | WRITTEN C05-K05-5-18-WU

Use addition theorems to find $\sin 75°$ and $\cos(\pi/12)$.

WORKING AREA

FORMULA BANK

Extrema
$$y = -(t-h)^2 + k$$

Addition
$$\sin(\alpha+\beta) = \sin\alpha\cos\beta + \cos\alpha\sin\beta$$

Double angle
$$\sin 2\alpha = 2\sin\alpha\cos\alpha \quad \cos 2\alpha = \cos^2\alpha - \sin^2\alpha$$

Harmonic form
$$a\sin\theta + b\cos\theta = r\sin(\theta+\alpha)$$

Angle / range
$$\tan\theta = \frac{m_1 - m_2}{1 + m_1 m_2} \quad -1 \le \sin\theta, \cos\theta \le 1$$

Concept 5 | Advanced Theorems and Applications

F02 | V02

Q005 | WRITTEN

C05-K05-5-18-TRY

Given $\sin\alpha = \frac{\sqrt{3}}{2}$ in QII and $\cos\beta = -\frac{3}{5}$ in QIII, find $\sin(\alpha+\beta)$ and $\cos(\alpha-\beta)$.

WORKING AREA

FORMULA BANK

Extrema
$$y = -(t-h)^2 + k$$

Addition
$$\sin(\alpha+\beta) = \sin\alpha\cos\beta + \cos\alpha\sin\beta$$

Double angle
$$\sin 2\alpha = 2\sin\alpha\cos\alpha \qquad \cos 2\alpha = \cos^2\alpha - \sin^2\alpha$$

Harmonic form
$$a\sin\theta + b\cos\theta = r\sin(\theta+\alpha)$$

Angle / range
$$\tan\theta = \frac{m_1 - m_2}{1 + m_1 m_2} \qquad -1 \le \sin\theta, \cos\theta \le 1$$

Concept 5 | Advanced Theorems and Applications

F02 | V03

Q006 | WRITTEN

C05-K05-5-18-EX

Complete the given addition-theorem exercises $exactvalues,\ quadrant - controlled\alpha,\ \beta,\ andthethree - anglechallenge(\sin\alpha = 0.6,\ \cos\beta = 0.8,\ \tan\gamma = 0.75)$.

WORKING AREA

FORMULA BANK

Extrema

$$y = -(t-h)^2 + k$$

Addition

$$\sin(\alpha+\beta) = \sin\alpha\cos\beta + \cos\alpha\sin\beta$$

Double angle

$$\sin 2\alpha = 2\sin\alpha\cos\alpha \quad \cos 2\alpha = \cos^2\alpha - \sin^2\alpha$$

Harmonic form

$$a\sin\theta + b\cos\theta = r\sin(\theta+\alpha)$$

Angle / range

$$\tan\theta = \frac{m_1 - m_2}{1 + m_1 m_2} \quad -1 \le \sin\theta, \cos\theta \le 1$$

Concept 5 | Advanced Theorems and Applications

F03 | V01

Q007 | WRITTEN

C05-K05-5-19-WU

Find $\tan 105°$, then find the angle between $y = 2x + 1$ and $y = -3x + 2$.

WORKING AREA

FORMULA BANK

Extrema

$$y = -(t-h)^2 + k$$

Addition

$$\sin(\alpha+\beta) = \sin\alpha\cos\beta + \cos\alpha\sin\beta$$

Double angle

$$\sin 2\alpha = 2\sin\alpha\cos\alpha \qquad \cos 2\alpha = \cos^2\alpha - \sin^2\alpha$$

Harmonic form

$$a\sin\theta + b\cos\theta = r\sin(\theta+\alpha)$$

Angle / range

$$\tan\theta = \frac{m_1 - m_2}{1 + m_1 m_2} \qquad -1 \le \sin\theta, \cos\theta \le 1$$

Concept 5 | Advanced Theorems and Applications

F03 | V02

Q008 | WRITTEN

C05-K05-5-19-TRY

Find the angle between each given pair of straight lines, including $y = \frac{1}{3}x$ with $y = 2x$, and $8x + 2y - 3 = 0$ with $-5x + 3y + 1 = 0$.

WORKING AREA

FORMULA BANK

Extrema

$$y = -(t-h)^2 + k$$

Addition

$$\sin(\alpha + \beta) = \sin\alpha\cos\beta + \cos\alpha\sin\beta$$

Double angle

$$\sin 2\alpha = 2\sin\alpha\cos\alpha \quad \cos 2\alpha = \cos^2\alpha - \sin^2\alpha$$

Harmonic form

$$a\sin\theta + b\cos\theta = r\sin(\theta + \alpha)$$

Angle / range

$$\tan\theta = \frac{m_1 - m_2}{1 + m_1 m_2} \quad -1 \le \sin\theta, \cos\theta \le 1$$

Concept 5 | Advanced Theorems and Applications

F03 | V03

Q009 | WRITTEN

C05-K05-5-19-EX

Find the exact angles for all given line pairs and solve the challenge: a line through $P(2, 3)$ makes acute angle $\tan\theta = 3/4$ with $x + 2y - 4 = 0$; give both line equations and the areas of the triangles with the y-axis.

WORKING AREA

FORMULA BANK

Extrema

$$y = -(t-h)^2 + k$$

Addition

$$\sin(\alpha+\beta) = \sin\alpha\cos\beta + \cos\alpha\sin\beta$$

Double angle

$$\sin 2\alpha = 2\sin\alpha\cos\alpha \quad \cos 2\alpha = \cos^2\alpha - \sin^2\alpha$$

Harmonic form

$$a\sin\theta + b\cos\theta = r\sin(\theta+\alpha)$$

Angle / range

$$\tan\theta = \frac{m_1 - m_2}{1 + m_1 m_2} \quad -1 \le \sin\theta, \cos\theta \le 1$$

Concept 5 | Advanced Theorems and Applications

F04 | V01

Q010 | WRITTEN

C05-K05-5-20-WU

Given $\sin\alpha = 1/3$ with α in QII, find $\sin 2\alpha,\ \cos 2\alpha,\ \tan 2\alpha$.

WORKING AREA

FORMULA BANK

Extrema

$y = -(t-h)^2 + k$

Addition

$\sin(\alpha+\beta) = \sin\alpha\cos\beta + \cos\alpha\sin\beta$

Double angle

$\sin 2\alpha = 2\sin\alpha\cos\alpha \quad \cos 2\alpha = \cos^2\alpha - \sin^2\alpha$

Harmonic form

$a\sin\theta + b\cos\theta = r\sin(\theta+\alpha)$

Angle / range

$\tan\theta = \dfrac{m_1 - m_2}{1 + m_1 m_2} \quad -1 \le \sin\theta, \cos\theta \le 1$

Concept 5 | Advanced Theorems and Applications

F04 | V02

Q011 | WRITTEN

C05-K05-5-20-TRY

Given $\sin\alpha = -2/3$ with $2\pi < \alpha < 3\pi$, find $\sin 2\alpha,\ \cos 2\alpha,\ \tan 2\alpha$.

WORKING AREA

FORMULA BANK

Extrema

$$y = -(t-h)^2 + k$$

Addition

$$\sin(\alpha+\beta) = \sin\alpha\cos\beta + \cos\alpha\sin\beta$$

Double angle

$$\sin 2\alpha = 2\sin\alpha\cos\alpha \qquad \cos 2\alpha = \cos^2\alpha - \sin^2\alpha$$

Harmonic form

$$a\sin\theta + b\cos\theta = r\sin(\theta+\alpha)$$

Angle / range

$$\tan\theta = \frac{m_1 - m_2}{1 + m_1 m_2} \qquad -1 \le \sin\theta, \cos\theta \le 1$$

Concept 5 | Advanced Theorems and Applications

F04 | V03

Q012 | WRITTEN C05-K05-5-20-EX

Solve the given double-angle exercises for given sine/cosine values and the challenge $\cos\theta = 0.6,\ \pi < \theta < 2\pi$, evaluating $\cos 3\theta \tan 2\theta$.

WORKING AREA

FORMULA BANK

Extrema

$$y = -(t-h)^2 + k$$

Addition

$$\sin(\alpha+\beta) = \sin\alpha\cos\beta + \cos\alpha\sin\beta$$

Double angle

$$\sin 2\alpha = 2\sin\alpha\cos\alpha \qquad \cos 2\alpha = \cos^2\alpha - \sin^2\alpha$$

Harmonic form

$$a\sin\theta + b\cos\theta = r\sin(\theta+\alpha)$$

Angle / range

$$\tan\theta = \frac{m_1 - m_2}{1 + m_1 m_2} \qquad -1 \le \sin\theta, \cos\theta \le 1$$

Concept 5 | Advanced Theorems and Applications

F05 | V01

Q013 | WRITTEN | C05-K05-5-21-WU

Given $\cos\alpha = 3/5, 0 < \alpha < \pi$, find $\sin(\alpha/2),\ \cos(\alpha/2),\ \tan(\alpha/2)$, and evaluate $\cos(5\pi/8)$.

WORKING AREA

FORMULA BANK

Extrema
$$y = -(t-h)^2 + k$$

Addition
$$\sin(\alpha+\beta) = \sin\alpha\cos\beta + \cos\alpha\sin\beta$$

Double angle
$$\sin 2\alpha = 2\sin\alpha\cos\alpha \qquad \cos 2\alpha = \cos^2\alpha - \sin^2\alpha$$

Harmonic form
$$a\sin\theta + b\cos\theta = r\sin(\theta+\alpha)$$

Angle / range
$$\tan\theta = \frac{m_1 - m_2}{1 + m_1 m_2} \qquad -1 \le \sin\theta, \cos\theta \le 1$$

Concept 5 | Advanced Theorems and Applications

F05 | V02

Q014 | WRITTEN

C05-K05-5-21-TRY

Given $\cos\alpha = -1/3$, $0 < \alpha < \pi$, find $\sin(\alpha/2)$, $\cos(\alpha/2)$, $\tan(\alpha/2)$, and evaluate $\sin(3\pi/8)$.

WORKING AREA

FORMULA BANK

Extrema

$$y = -(t-h)^2 + k$$

Addition

$$\sin(\alpha+\beta) = \sin\alpha\cos\beta + \cos\alpha\sin\beta$$

Double angle

$$\sin 2\alpha = 2\sin\alpha\cos\alpha \qquad \cos 2\alpha = \cos^2\alpha - \sin^2\alpha$$

Harmonic form

$$a\sin\theta + b\cos\theta = r\sin(\theta+\alpha)$$

Angle / range

$$\tan\theta = \frac{m_1 - m_2}{1 + m_1 m_2} \qquad -1 \le \sin\theta, \cos\theta \le 1$$

Concept 5 | Advanced Theorems and Applications

F05 | V03

Q015 | WRITTEN

C05-K05-5-21-EX

Solve the given half-angle exercises for the three quadrant intervals and the challenge with $\tan A = \frac{\sqrt{5}}{2}$, $4\pi < A < 5\pi$, $\cos B = -\frac{\sqrt{5}}{3}$, $5\pi < B < 6\pi$.

WORKING AREA

FORMULA BANK

Extrema
$$y = -(t-h)^2 + k$$

Addition
$$\sin(\alpha+\beta) = \sin\alpha\cos\beta + \cos\alpha\sin\beta$$

Double angle
$$\sin 2\alpha = 2\sin\alpha\cos\alpha \qquad \cos 2\alpha = \cos^2\alpha - \sin^2\alpha$$

Harmonic form
$$a\sin\theta + b\cos\theta = r\sin(\theta+\alpha)$$

Angle / range
$$\tan\theta = \frac{m_1 - m_2}{1 + m_1 m_2} \qquad -1 \le \sin\theta, \cos\theta \le 1$$

Concept 5 | Advanced Theorems and Applications

F06 | V01

Q016 | WRITTEN

C05-K05-5-22-WU

Solve on $0 \le \theta < 2\pi$: $a \sin 2\theta + \sin\theta = 0$, $b \cos 2\theta - 7\cos\theta + 4 \ge 0$.

WORKING AREA

FORMULA BANK

Extrema
$$y = -(t-h)^2 + k$$

Addition
$$\sin(\alpha+\beta) = \sin\alpha\cos\beta + \cos\alpha\sin\beta$$

Double angle
$$\sin 2\alpha = 2\sin\alpha\cos\alpha \qquad \cos 2\alpha = \cos^2\alpha - \sin^2\alpha$$

Harmonic form
$$a\sin\theta + b\cos\theta = r\sin(\theta+\alpha)$$

Angle / range
$$\tan\theta = \frac{m_1 - m_2}{1 + m_1 m_2} \qquad -1 \le \sin\theta, \cos\theta \le 1$$

Concept 5 | Advanced Theorems and Applications

F06 | V02

Q017 | WRITTEN

C05-K05-5-22-TRY

Solve the given equation $\sin 2\theta + \cos\theta = 0$ on $0 \le \theta < 2\pi$.

WORKING AREA

FORMULA BANK

Extrema
$$y = -(t-h)^2 + k$$

Addition
$$\sin(\alpha+\beta) = \sin\alpha\cos\beta + \cos\alpha\sin\beta$$

Double angle
$$\sin 2\alpha = 2\sin\alpha\cos\alpha \qquad \cos 2\alpha = \cos^2\alpha - \sin^2\alpha$$

Harmonic form
$$a\sin\theta + b\cos\theta = r\sin(\theta+\alpha)$$

Angle / range
$$\tan\theta = \frac{m_1 - m_2}{1 + m_1 m_2} \qquad -1 \le \sin\theta, \cos\theta \le 1$$

Concept 5 | Advanced Theorems and Applications

F06 | V03

Q018 | WRITTEN

C05-K05-5-22-EX

Solve all given double-angle equations and inequalities on $0 \leq \theta < 2\pi$, including the challenge $\cos 2\theta - 7\cos\theta + 4 \geq 0$.

WORKING AREA

FORMULA BANK

Extrema

$$y = -(t-h)^2 + k$$

Addition

$$\sin(\alpha+\beta) = \sin\alpha\cos\beta + \cos\alpha\sin\beta$$

Double angle

$$\sin 2\alpha = 2\sin\alpha\cos\alpha \qquad \cos 2\alpha = \cos^2\alpha - \sin^2\alpha$$

Harmonic form

$$a\sin\theta + b\cos\theta = r\sin(\theta+\alpha)$$

Angle / range

$$\tan\theta = \frac{m_1 - m_2}{1 + m_1 m_2} \qquad -1 \leq \sin\theta, \cos\theta \leq 1$$

Concept 5 | Advanced Theorems and Applications

F07 | V01

Q019 | WRITTEN

C05-K05-5-23-WU

Rewrite $\sin\theta - \sqrt{3}\cos\theta$ as $r\sin(\theta+\alpha)$, $r>0$, $-\pi<\alpha<\pi$.

WORKING AREA

FORMULA BANK

Extrema
$$y = -(t-h)^2 + k$$

Addition
$$\sin(\alpha+\beta) = \sin\alpha\cos\beta + \cos\alpha\sin\beta$$

Double angle
$$\sin 2\alpha = 2\sin\alpha\cos\alpha \qquad \cos 2\alpha = \cos^2\alpha - \sin^2\alpha$$

Harmonic form
$$a\sin\theta + b\cos\theta = r\sin(\theta+\alpha)$$

Angle / range
$$\tan\theta = \frac{m_1 - m_2}{1 + m_1 m_2} \qquad -1 \le \sin\theta, \cos\theta \le 1$$

Concept 5 | Advanced Theorems and Applications

F07 | V02

Q020 | WRITTEN — C05-K05-5-23-TRY

Rewrite the given expressions $\sin\theta - \cos\theta$, $\sin\theta + \cos\theta$, $3\sin\theta - \sqrt{3}\cos\theta$, and $3\sin\theta - \cos\theta$ in harmonic-addition form.

WORKING AREA

FORMULA BANK

Extrema

$$y = -(t-h)^2 + k$$

Addition

$$\sin(\alpha+\beta) = \sin\alpha\cos\beta + \cos\alpha\sin\beta$$

Double angle

$$\sin 2\alpha = 2\sin\alpha\cos\alpha \qquad \cos 2\alpha = \cos^2\alpha - \sin^2\alpha$$

Harmonic form

$$a\sin\theta + b\cos\theta = r\sin(\theta+\alpha)$$

Angle / range

$$\tan\theta = \frac{m_1 - m_2}{1 + m_1 m_2} \qquad -1 \le \sin\theta, \cos\theta \le 1$$

Concept 5 | Advanced Theorems and Applications

F07 | V03

Q021 | WRITTEN C05-K05-5-23-EX

Rewrite all given harmonic-addition exercises in the form $r\sin(\theta+\alpha)$, including $2\sin\theta+2\cos\theta$ and the challenge $\sin\theta-\sqrt{3}\cos\theta$.

WORKING AREA

FORMULA BANK

Extrema
$$y=-(t-h)^2+k$$

Addition
$$\sin(\alpha+\beta)=\sin\alpha\cos\beta+\cos\alpha\sin\beta$$

Double angle
$$\sin 2\alpha=2\sin\alpha\cos\alpha \qquad \cos 2\alpha=\cos^2\alpha-\sin^2\alpha$$

Harmonic form
$$a\sin\theta+b\cos\theta=r\sin(\theta+\alpha)$$

Angle / range
$$\tan\theta=\frac{m_1-m_2}{1+m_1m_2} \qquad -1\le\sin\theta,\cos\theta\le 1$$

Concept 5 | Advanced Theorems and Applications

F08 | V01

Q022 | WRITTEN

C05-K05-5-24-WU

Solve on $0 \leq \theta < 2\pi$: $a \sin\theta - \cos\theta = 1$, $b \sin\theta + \cos\theta < 3/2$.

WORKING AREA

FORMULA BANK

Extrema

$$y = -(t-h)^2 + k$$

Addition

$$\sin(\alpha+\beta) = \sin\alpha\cos\beta + \cos\alpha\sin\beta$$

Double angle

$$\sin 2\alpha = 2\sin\alpha\cos\alpha \qquad \cos 2\alpha = \cos^2\alpha - \sin^2\alpha$$

Harmonic form

$$a\sin\theta + b\cos\theta = r\sin(\theta+\alpha)$$

Angle / range

$$\tan\theta = \frac{m_1 - m_2}{1 + m_1 m_2} \qquad -1 \leq \sin\theta, \cos\theta \leq 1$$

Concept 5 | Advanced Theorems and Applications

F08 | V02

Q023 | WRITTEN

C05-K05-5-24-TRY

Solve $\sin\theta \le \cos\theta$ on $0 \le \theta < 2\pi$.

WORKING AREA

FORMULA BANK

Extrema

$$y = -(t-h)^2 + k$$

Addition

$$\sin(\alpha+\beta) = \sin\alpha\cos\beta + \cos\alpha\sin\beta$$

Double angle

$$\sin 2\alpha = 2\sin\alpha\cos\alpha \quad \cos 2\alpha = \cos^2\alpha - \sin^2\alpha$$

Harmonic form

$$a\sin\theta + b\cos\theta = r\sin(\theta+\alpha)$$

Angle / range

$$\tan\theta = \frac{m_1 - m_2}{1 + m_1 m_2} \quad -1 \le \sin\theta, \cos\theta \le 1$$

Concept 5 | Advanced Theorems and Applications

F08 | V03

Q024 | WRITTEN

C05-K05-5-24-EX

Solve all given harmonic-addition equations and inequalities on $0 \leq \theta < 2\pi$, including $\sin\theta + \cos\theta + 1 = 0$ and $\sin\theta - \cos\theta + 3/2 > 0$.

WORKING AREA

FORMULA BANK

Extrema

$$y = -(t-h)^2 + k$$

Addition

$$\sin(\alpha+\beta) = \sin\alpha\cos\beta + \cos\alpha\sin\beta$$

Double angle

$$\sin 2\alpha = 2\sin\alpha\cos\alpha \qquad \cos 2\alpha = \cos^2\alpha - \sin^2\alpha$$

Harmonic form

$$a\sin\theta + b\cos\theta = r\sin(\theta+\alpha)$$

Angle / range

$$\tan\theta = \frac{m_1 - m_2}{1 + m_1 m_2} \qquad -1 \leq \sin\theta, \cos\theta \leq 1$$

Concept 5 | Advanced Theorems and Applications

F09 | V01

Q025 | WRITTEN

C05-K05-5-25-WU

Find the maximum and minimum of $y = 2\sin(\theta - \pi/3)$, and of $y = 2\cos 2\theta + 4\cos\theta + 1$, $0 \le \theta < 2\pi$.

WORKING AREA

FORMULA BANK

Extrema

$$y = -(t-h)^2 + k$$

Addition

$$\sin(\alpha+\beta) = \sin\alpha\cos\beta + \cos\alpha\sin\beta$$

Double angle

$$\sin 2\alpha = 2\sin\alpha\cos\alpha \qquad \cos 2\alpha = \cos^2\alpha - \sin^2\alpha$$

Harmonic form

$$a\sin\theta + b\cos\theta = r\sin(\theta+\alpha)$$

Angle / range

$$\tan\theta = \frac{m_1 - m_2}{1 + m_1 m_2} \qquad -1 \le \sin\theta, \cos\theta \le 1$$

Concept 5 | Advanced Theorems and Applications

F09 | V02

Q026 | WRITTEN

C05-K05-5-25-TRY

Find the maximum and minimum of the given functions $y = -\frac{4}{3}\sin^2\theta + \cos\theta$ and $y = \cos 2\theta - 2\cos\theta$.

WORKING AREA

FORMULA BANK

Extrema

$$y = -(t-h)^2 + k$$

Addition

$$\sin(\alpha+\beta) = \sin\alpha\cos\beta + \cos\alpha\sin\beta$$

Double angle

$$\sin 2\alpha = 2\sin\alpha\cos\alpha \quad \cos 2\alpha = \cos^2\alpha - \sin^2\alpha$$

Harmonic form

$$a\sin\theta + b\cos\theta = r\sin(\theta+\alpha)$$

Angle / range

$$\tan\theta = \frac{m_1 - m_2}{1 + m_1 m_2} \quad -1 \le \sin\theta, \cos\theta \le 1$$

Concept 5 | Advanced Theorems and Applications

F09 | V03

Q027 | WRITTEN

C05-K05-5-25-EX

Solve all given maximum/minimum problems in the final lesson, including absolute-value and double-angle forms, and state the corresponding angles where requested.

WORKING AREA

FORMULA BANK

Extrema

$$y = -(t-h)^2 + k$$

Addition

$$\sin(\alpha+\beta) = \sin\alpha\cos\beta + \cos\alpha\sin\beta$$

Double angle

$$\sin 2\alpha = 2\sin\alpha\cos\alpha \qquad \cos 2\alpha = \cos^2\alpha - \sin^2\alpha$$

Harmonic form

$$a\sin\theta + b\cos\theta = r\sin(\theta+\alpha)$$

Angle / range

$$\tan\theta = \frac{m_1 - m_2}{1 + m_1 m_2} \qquad -1 \le \sin\theta, \cos\theta \le 1$$

Concept 5 | Advanced Theorems and Applications

F01 | C05-K05-F01-VMCQ

Q028 | MCQ

C05-K05-5-17-GM01

The substitution for $\sin^2\theta + \cos\theta + 1$ should be Which of the following is correct?:

A $t = sintheta$

B $t = costheta$

C $t = tantheta$

D $t = theta$

WORKING AREA

FORMULA BANK

Extrema
$$y = -(t-h)^2 + k$$

Addition
$$\sin(\alpha+\beta) = \sin\alpha\cos\beta + \cos\alpha\sin\beta$$

Double angle
$$\sin 2\alpha = 2\sin\alpha\cos\alpha \qquad \cos 2\alpha = \cos^2\alpha - \sin^2\alpha$$

Harmonic form
$$a\sin\theta + b\cos\theta = r\sin(\theta+\alpha)$$

Angle / range
$$\tan\theta = \frac{m_1 - m_2}{1 + m_1 m_2} \qquad -1 \le \sin\theta, \cos\theta \le 1$$

Concept 5 | Advanced Theorems and Applications

F01 | C05-K05-F01-VMCQ

Q029 | MCQ C05-K05-5-17-GM02

The maximum of $\sin^2\theta + \cos\theta + 1$ is:

A 0

B 1

C 9/4

D 4

WORKING AREA

FORMULA BANK

Extrema
$$y = -(t-h)^2 + k$$

Addition
$$\sin(\alpha+\beta) = \sin\alpha\cos\beta + \cos\alpha\sin\beta$$

Double angle
$$\sin 2\alpha = 2\sin\alpha\cos\alpha \qquad \cos 2\alpha = \cos^2\alpha - \sin^2\alpha$$

Harmonic form
$$a\sin\theta + b\cos\theta = r\sin(\theta+\alpha)$$

Angle / range
$$\tan\theta = \frac{m_1 - m_2}{1 + m_1 m_2} \qquad -1 \le \sin\theta, \cos\theta \le 1$$

Concept 5 | Advanced Theorems and Applications

F01 | C05-K05-F01-VMCQ

Q030 | MCQ

C05-K05-5-17-GM03

An extremal vertex is valid only if t lies in Which of the following is correct?:

A $[-2, 2]$

B $[-1, 1]$

C [0,1]

D all real numbers

WORKING AREA

FORMULA BANK

Extrema
$$y = -(t-h)^2 + k$$

Addition
$$\sin(\alpha+\beta) = \sin\alpha\cos\beta + \cos\alpha\sin\beta$$

Double angle
$$\sin 2\alpha = 2\sin\alpha\cos\alpha \quad \cos 2\alpha = \cos^2\alpha - \sin^2\alpha$$

Harmonic form
$$a\sin\theta + b\cos\theta = r\sin(\theta+\alpha)$$

Angle / range
$$\tan\theta = \frac{m_1 - m_2}{1 + m_1 m_2} \quad -1 \le \sin\theta, \cos\theta \le 1$$

Concept 5 | Advanced Theorems and Applications

F02 | C05-K05-F02-VMCQ

Q031 | MCQ C05-K05-5-18-GM04

The identity for sin(a+b) is:

A $sinalphacosbeta + cosalphasinbeta$

B sin alpha sin beta

C cos alpha cos beta

D $tanalpha + tanbeta$

WORKING AREA

FORMULA BANK

Extrema
$$y = -(t-h)^2 + k$$

Addition
$$\sin(\alpha+\beta) = \sin\alpha\cos\beta + \cos\alpha\sin\beta$$

Double angle
$$\sin 2\alpha = 2\sin\alpha\cos\alpha \quad \cos 2\alpha = \cos^2\alpha - \sin^2\alpha$$

Harmonic form
$$a\sin\theta + b\cos\theta = r\sin(\theta+\alpha)$$

Angle / range
$$\tan\theta = \frac{m_1 - m_2}{1 + m_1 m_2} \quad -1 \le \sin\theta, \cos\theta \le 1$$

Concept 5 | Advanced Theorems and Applications

F02 | C05-K05-F02-VMCQ

Q032 | MCQ

C05-K05-5-18-GM05

sin 75 degrees equals Which of the following is correct?:

A $(sqrt6 - sqrt2)/4$

B $(sqrt6 + sqrt2)/4$

C $1/2$

D $sqrt3/2$

WORKING AREA

FORMULA BANK

Extrema
$$y = -(t-h)^2 + k$$

Addition
$$\sin(\alpha+\beta) = \sin\alpha\cos\beta + \cos\alpha\sin\beta$$

Double angle
$$\sin 2\alpha = 2\sin\alpha\cos\alpha \quad \cos 2\alpha = \cos^2\alpha - \sin^2\alpha$$

Harmonic form
$$a\sin\theta + b\cos\theta = r\sin(\theta+\alpha)$$

Angle / range
$$\tan\theta = \frac{m_1 - m_2}{1 + m_1 m_2} \quad -1 \le \sin\theta, \cos\theta \le 1$$

Concept 5 | Advanced Theorems and Applications

F02 | C05-K05-F02-VMCQ

Q033 | MCQ

C05-K05-5-18-GM06

A missing trig value is selected using:

A the quadrant

B the denominator

C the period only

D a graph scale

WORKING AREA

FORMULA BANK

Extrema
$$y = -(t-h)^2 + k$$

Addition
$$\sin(\alpha+\beta) = \sin\alpha\cos\beta + \cos\alpha\sin\beta$$

Double angle
$$\sin 2\alpha = 2\sin\alpha\cos\alpha \quad \cos 2\alpha = \cos^2\alpha - \sin^2\alpha$$

Harmonic form
$$a\sin\theta + b\cos\theta = r\sin(\theta+\alpha)$$

Angle / range
$$\tan\theta = \frac{m_1 - m_2}{1 + m_1 m_2} \quad -1 \le \sin\theta, \cos\theta \le 1$$

Concept 5 | Advanced Theorems and Applications

F03 | C05-K05-F03-VMCQ

Q034 | MCQ

C05-K05-5-19-GM07

The slope of y=mx+k is:

A k

B m

C $1/m$

D $m+k$

WORKING AREA

FORMULA BANK

Extrema
$$y=-(t-h)^2+k$$

Addition
$$\sin(\alpha+\beta)=\sin\alpha\cos\beta+\cos\alpha\sin\beta$$

Double angle
$$\sin 2\alpha = 2\sin\alpha\cos\alpha \qquad \cos 2\alpha=\cos^2\alpha-\sin^2\alpha$$

Harmonic form
$$a\sin\theta+b\cos\theta=r\sin(\theta+\alpha)$$

Angle / range
$$\tan\theta=\frac{m_1-m_2}{1+m_1m_2} \qquad -1\le\sin\theta,\cos\theta\le 1$$

Concept 5 | Advanced Theorems and Applications

F03 | C05-K05-F03-VMCQ

Q035 | MCQ

C05-K05-5-19-GM08

The angle formula uses denominator Which of the following is correct?:

A $1 + m1m2$

B $m1 - m2$

C $m1 + m2$

D m1m2

WORKING AREA

FORMULA BANK

Extrema
$$y = -(t-h)^2 + k$$

Addition
$$\sin(\alpha+\beta) = \sin\alpha\cos\beta + \cos\alpha\sin\beta$$

Double angle
$$\sin 2\alpha = 2\sin\alpha\cos\alpha \quad \cos 2\alpha = \cos^2\alpha - \sin^2\alpha$$

Harmonic form
$$a\sin\theta + b\cos\theta = r\sin(\theta+\alpha)$$

Angle / range
$$\tan\theta = \frac{m_1 - m_2}{1 + m_1 m_2} \quad -1 \le \sin\theta, \cos\theta \le 1$$

Concept 5 | Advanced Theorems and Applications

F03 | C05-K05-F03-VMCQ

Q036 | MCQ C05-K05-5-19-GM09

For an acute angle, if tan theta1 is negative we use Which of the following is correct?:

A theta1

B $pi - theta1$

C 2theta1

D $-theta1only$

WORKING AREA

FORMULA BANK

Extrema
$$y = -(t-h)^2 + k$$

Addition
$$\sin(\alpha+\beta) = \sin\alpha\cos\beta + \cos\alpha\sin\beta$$

Double angle
$$\sin 2\alpha = 2\sin\alpha\cos\alpha \quad \cos 2\alpha = \cos^2\alpha - \sin^2\alpha$$

Harmonic form
$$a\sin\theta + b\cos\theta = r\sin(\theta+\alpha)$$

Angle / range
$$\tan\theta = \frac{m_1 - m_2}{1 + m_1 m_2} \quad -1 \le \sin\theta, \cos\theta \le 1$$

Concept 5 | Advanced Theorems and Applications

F04 | C05-K05-F04-VMCQ

Q037 | MCQ

C05-K05-5-20-GM10

sin(2a) equals Which of the following is correct?:

A sin alpha

B 2 sin alpha cos alpha

C cos squared alpha

D tan alpha

WORKING AREA

FORMULA BANK

Extrema
$$y = -(t-h)^2 + k$$

Addition
$$\sin(\alpha+\beta) = \sin\alpha\cos\beta + \cos\alpha\sin\beta$$

Double angle
$$\sin 2\alpha = 2\sin\alpha\cos\alpha \qquad \cos 2\alpha = \cos^2\alpha - \sin^2\alpha$$

Harmonic form
$$a\sin\theta + b\cos\theta = r\sin(\theta+\alpha)$$

Angle / range
$$\tan\theta = \frac{m_1 - m_2}{1 + m_1 m_2} \qquad -1 \le \sin\theta, \cos\theta \le 1$$

Concept 5 | Advanced Theorems and Applications

F04 | C05-K05-F04-VMCQ

Q038 | MCQ

C05-K05-5-20-GM11

cos(2a) can be written as Which of the following is correct?:

A $cossquaredalpha - sinsquaredalpha$

B 2sin alpha

C tan alpha

D $1 + sinalpha$

WORKING AREA

FORMULA BANK

Extrema
$$y = -(t-h)^2 + k$$

Addition
$$\sin(\alpha+\beta) = \sin\alpha\cos\beta + \cos\alpha\sin\beta$$

Double angle
$$\sin 2\alpha = 2\sin\alpha\cos\alpha \qquad \cos 2\alpha = \cos^2\alpha - \sin^2\alpha$$

Harmonic form
$$a\sin\theta + b\cos\theta = r\sin(\theta+\alpha)$$

Angle / range
$$\tan\theta = \frac{m_1 - m_2}{1 + m_1 m_2} \qquad -1 \le \sin\theta, \cos\theta \le 1$$

Concept 5 | Advanced Theorems and Applications

F04 | C05-K05-F04-VMCQ

Q039 | MCQ

C05-K05-5-20-GM12

For theta in QIV, sin theta is:

A positive

B negative

C zero always

D undefined

WORKING AREA

FORMULA BANK

Extrema

$$y = -(t-h)^2 + k$$

Addition

$$\sin(\alpha+\beta) = \sin\alpha\cos\beta + \cos\alpha\sin\beta$$

Double angle

$$\sin 2\alpha = 2\sin\alpha\cos\alpha \qquad \cos 2\alpha = \cos^2\alpha - \sin^2\alpha$$

Harmonic form

$$a\sin\theta + b\cos\theta = r\sin(\theta+\alpha)$$

Angle / range

$$\tan\theta = \frac{m_1 - m_2}{1 + m_1 m_2} \qquad -1 \le \sin\theta, \cos\theta \le 1$$

Concept 5 | Advanced Theorems and Applications

F05 | C05-K05-F05-VMCQ

Q040 | MCQ C05-K05-5-21-GM13

The half-angle formula for sin(a/2) uses Which of the following is correct?:

A $(1 - cosalpha)/2$

B $(1 + cosalpha)/2$

C tan alpha

D cos 2alpha

WORKING AREA

FORMULA BANK

Extrema
$$y = -(t-h)^2 + k$$

Addition
$$\sin(\alpha+\beta) = \sin\alpha\cos\beta + \cos\alpha\sin\beta$$

Double angle
$$\sin 2\alpha = 2\sin\alpha\cos\alpha \qquad \cos 2\alpha = \cos^2\alpha - \sin^2\alpha$$

Harmonic form
$$a\sin\theta + b\cos\theta = r\sin(\theta+\alpha)$$

Angle / range
$$\tan\theta = \frac{m_1 - m_2}{1 + m_1 m_2} \qquad -1 \le \sin\theta, \cos\theta \le 1$$

Concept 5 | Advanced Theorems and Applications

F05 | C05-K05-F05-VMCQ

Q041 | MCQ

C05-K05-5-21-GM14

The sign of a half-angle root is chosen from Which of the following is correct?:

A the original quadrant interval

B the coefficient only

C the denominator

D the period of tan

WORKING AREA

FORMULA BANK

Extrema

$$y = -(t-h)^2 + k$$

Addition

$$\sin(\alpha+\beta) = \sin\alpha\cos\beta + \cos\alpha\sin\beta$$

Double angle

$$\sin 2\alpha = 2\sin\alpha\cos\alpha \qquad \cos 2\alpha = \cos^2\alpha - \sin^2\alpha$$

Harmonic form

$$a\sin\theta + b\cos\theta = r\sin(\theta+\alpha)$$

Angle / range

$$\tan\theta = \frac{m_1 - m_2}{1 + m_1 m_2} \qquad -1 \le \sin\theta, \cos\theta \le 1$$

Concept 5 | Advanced Theorems and Applications

F05 | C05-K05-F05-VMCQ

Q042 | MCQ C05-K05-5-21-GM15

If cos a=-1/3 and 0<a<pi, cos(a/2) is:

A negative

B positive

C zero

D undefined

WORKING AREA

FORMULA BANK

Extrema
$$y = -(t-h)^2 + k$$

Addition
$$\sin(\alpha+\beta) = \sin\alpha\cos\beta + \cos\alpha\sin\beta$$

Double angle
$$\sin 2\alpha = 2\sin\alpha\cos\alpha \quad \cos 2\alpha = \cos^2\alpha - \sin^2\alpha$$

Harmonic form
$$a\sin\theta + b\cos\theta = r\sin(\theta+\alpha)$$

Angle / range
$$\tan\theta = \frac{m_1 - m_2}{1 + m_1 m_2} \quad -1 \le \sin\theta, \cos\theta \le 1$$

Concept 5 | Advanced Theorems and Applications

F06 | C05-K05-F06-VMCQ

Q043 | MCQ C05-K05-5-22-GM16

For double-angle equations, first rewrite in Which of the following is correct?:

A one trig function

B two unrelated angles

C degrees only

D a logarithm

WORKING AREA

FORMULA BANK

Extrema
$$y = -(t-h)^2 + k$$

Addition
$$\sin(\alpha+\beta) = \sin\alpha\cos\beta + \cos\alpha\sin\beta$$

Double angle
$$\sin 2\alpha = 2\sin\alpha\cos\alpha \qquad \cos 2\alpha = \cos^2\alpha - \sin^2\alpha$$

Harmonic form
$$a\sin\theta + b\cos\theta = r\sin(\theta+\alpha)$$

Angle / range
$$\tan\theta = \frac{m_1 - m_2}{1 + m_1 m_2} \qquad -1 \le \sin\theta, \cos\theta \le 1$$

Concept 5 | Advanced Theorems and Applications

F06 | C05-K05-F06-VMCQ

Q044 | MCQ

C05-K05-5-22-GM17

The factorisation of sin(2 theta)+sin theta is:

A $sintheta(2costheta+1)$

B $costheta(2sintheta-1)$

C $sintheta+costheta$

D 2sin theta

WORKING AREA

FORMULA BANK

Extrema
$$y=-(t-h)^2+k$$

Addition
$$\sin(\alpha+\beta)=\sin\alpha\cos\beta+\cos\alpha\sin\beta$$

Double angle
$$\sin 2\alpha=2\sin\alpha\cos\alpha \quad \cos 2\alpha=\cos^2\alpha-\sin^2\alpha$$

Harmonic form
$$a\sin\theta+b\cos\theta=r\sin(\theta+\alpha)$$

Angle / range
$$\tan\theta=\frac{m_1-m_2}{1+m_1m_2} \quad -1\le\sin\theta,\cos\theta\le 1$$

Concept 5 | Advanced Theorems and Applications

F06 | C05-K05-F06-VMCQ

Q045 | MCQ

C05-K05-5-22-GM18

A tangent asymptote is always Which of the following is correct?:

A included

B excluded

C a maximum

D an endpoint

WORKING AREA

FORMULA BANK

Extrema
$$y = -(t-h)^2 + k$$

Addition
$$\sin(\alpha+\beta) = \sin\alpha\cos\beta + \cos\alpha\sin\beta$$

Double angle
$$\sin 2\alpha = 2\sin\alpha\cos\alpha \quad \cos 2\alpha = \cos^2\alpha - \sin^2\alpha$$

Harmonic form
$$a\sin\theta + b\cos\theta = r\sin(\theta+\alpha)$$

Angle / range
$$\tan\theta = \frac{m_1 - m_2}{1 + m_1 m_2} \quad -1 \le \sin\theta, \cos\theta \le 1$$

Concept 5 | Advanced Theorems and Applications

F07 | C05-K05-F07-VMCQ

Q046 | MCQ

C05-K05-5-23-GM19

For a sin theta+b cos theta, r is:

A $a + b$

B $sqrt(a^2 + b^2)$

C $a - b$

D ab

WORKING AREA

FORMULA BANK

Extrema
$$y = -(t - h)^2 + k$$

Addition
$$\sin(\alpha + \beta) = \sin\alpha\cos\beta + \cos\alpha\sin\beta$$

Double angle
$$\sin 2\alpha = 2\sin\alpha\cos\alpha \quad \cos 2\alpha = \cos^2\alpha - \sin^2\alpha$$

Harmonic form
$$a\sin\theta + b\cos\theta = r\sin(\theta + \alpha)$$

Angle / range
$$\tan\theta = \frac{m_1 - m_2}{1 + m_1 m_2} \quad -1 \leq \sin\theta, \cos\theta \leq 1$$

Concept 5 | Advanced Theorems and Applications

F07 | C05-K05-F07-VMCQ

Q047 | MCQ

C05-K05-5-23-GM20

The harmonic target form is:

A $rsin(theta + alpha)$

B r cos theta

C a tan theta

D sin theta only

WORKING AREA

FORMULA BANK

Extrema

$$y = -(t-h)^2 + k$$

Addition

$\sin(\alpha+\beta) = \sin\alpha\cos\beta + \cos\alpha\sin\beta$

Double angle

$\sin 2\alpha = 2\sin\alpha\cos\alpha$ $\cos 2\alpha = \cos^2\alpha - \sin^2\alpha$

Harmonic form

$a\sin\theta + b\cos\theta = r\sin(\theta+\alpha)$

Angle / range

$\tan\theta = \dfrac{m_1 - m_2}{1 + m_1 m_2}$ $-1 \le \sin\theta, \cos\theta \le 1$

Concept 5 | Advanced Theorems and Applications

F07 | C05-K05-F07-VMCQ

Q048 | MCQ

C05-K05-5-23-GM21

The angle a is located from point Which of the following is correct?:

A P(a,b)

B P(r,theta)

C the origin only

D the x intercept

WORKING AREA

FORMULA BANK

Extrema

$$y = -(t-h)^2 + k$$

Addition

$$\sin(\alpha+\beta) = \sin\alpha\cos\beta + \cos\alpha\sin\beta$$

Double angle

$$\sin 2\alpha = 2\sin\alpha\cos\alpha \qquad \cos 2\alpha = \cos^2\alpha - \sin^2\alpha$$

Harmonic form

$$a\sin\theta + b\cos\theta = r\sin(\theta+\alpha)$$

Angle / range

$$\tan\theta = \frac{m_1 - m_2}{1 + m_1 m_2} \qquad -1 \le \sin\theta, \cos\theta \le 1$$

Concept 5 | Advanced Theorems and Applications

F08 | C05-K05-F08-VMCQ

Q049 | MCQ

C05-K05-5-24-GM22

After synthesis, t is usually Which of the following is correct?:

A $theta + alpha$

B $theta^2$

C tan theta

D cos theta

WORKING AREA

FORMULA BANK

Extrema
$$y = -(t-h)^2 + k$$

Addition
$$\sin(\alpha+\beta) = \sin\alpha\cos\beta + \cos\alpha\sin\beta$$

Double angle
$$\sin 2\alpha = 2\sin\alpha\cos\alpha \qquad \cos 2\alpha = \cos^2\alpha - \sin^2\alpha$$

Harmonic form
$$a\sin\theta + b\cos\theta = r\sin(\theta+\alpha)$$

Angle / range
$$\tan\theta = \frac{m_1 - m_2}{1 + m_1 m_2} \qquad -1 \le \sin\theta, \cos\theta \le 1$$

Concept 5 | Advanced Theorems and Applications

F08 | C05-K05-F08-VMCQ

Q050 | MCQ

C05-K05-5-24-GM23

If max(sin theta+cos theta)=sqrt2, then sin theta+cos theta<3/2 is:

A never true

B always true

C true only at pi

D undefined

WORKING AREA

FORMULA BANK

Extrema
$$y = -(t-h)^2 + k$$

Addition
$$\sin(\alpha+\beta) = \sin\alpha\cos\beta + \cos\alpha\sin\beta$$

Double angle
$$\sin 2\alpha = 2\sin\alpha\cos\alpha \qquad \cos 2\alpha = \cos^2\alpha - \sin^2\alpha$$

Harmonic form
$$a\sin\theta + b\cos\theta = r\sin(\theta+\alpha)$$

Angle / range
$$\tan\theta = \frac{m_1 - m_2}{1 + m_1 m_2} \qquad -1 \le \sin\theta, \cos\theta \le 1$$

Concept 5 | Advanced Theorems and Applications

F08 | C05-K05-F08-VMCQ

Q051 | MCQ

C05-K05-5-24-GM24

In a strict inequality, equality endpoints are:

A included

B excluded

C doubled

D random

WORKING AREA

FORMULA BANK

Extrema

$$y = -(t-h)^2 + k$$

Addition

$$\sin(\alpha+\beta) = \sin\alpha\cos\beta + \cos\alpha\sin\beta$$

Double angle

$$\sin 2\alpha = 2\sin\alpha\cos\alpha \qquad \cos 2\alpha = \cos^2\alpha - \sin^2\alpha$$

Harmonic form

$$a\sin\theta + b\cos\theta = r\sin(\theta+\alpha)$$

Angle / range

$$\tan\theta = \frac{m_1 - m_2}{1 + m_1 m_2} \qquad -1 \le \sin\theta, \cos\theta \le 1$$

Concept 5 | Advanced Theorems and Applications

F09 | C05-K05-F09-VMCQ

Q052 | MCQ — C05-K05-5-25-GM25

The range of 2sin(theta-pi/3) is:

A $[-1, 1]$

B $[-2, 2]$

C [0,2]

D $[-3, 3]$

WORKING AREA

FORMULA BANK

Extrema
$$y = -(t-h)^2 + k$$

Addition
$$\sin(\alpha+\beta) = \sin\alpha\cos\beta + \cos\alpha\sin\beta$$

Double angle
$$\sin 2\alpha = 2\sin\alpha\cos\alpha \qquad \cos 2\alpha = \cos^2\alpha - \sin^2\alpha$$

Harmonic form
$$a\sin\theta + b\cos\theta = r\sin(\theta+\alpha)$$

Angle / range
$$\tan\theta = \frac{m_1 - m_2}{1 + m_1 m_2} \qquad -1 \le \sin\theta, \cos\theta \le 1$$

Concept 5 | Advanced Theorems and Applications

F09 | C05-K05-F09-VMCQ

Q053 | MCQ

C05-K05-5-25-GM26

For a double-angle maximum problem, use Which of the following is correct?:

A $thedouble-angleformulathenaboundedvariable$

B a logarithm

C only a graph

D no substitution

WORKING AREA

FORMULA BANK

Extrema
$$y = -(t-h)^2 + k$$

Addition
$$\sin(\alpha+\beta) = \sin\alpha\cos\beta + \cos\alpha\sin\beta$$

Double angle
$$\sin 2\alpha = 2\sin\alpha\cos\alpha \quad \cos 2\alpha = \cos^2\alpha - \sin^2\alpha$$

Harmonic form
$$a\sin\theta + b\cos\theta = r\sin(\theta+\alpha)$$

Angle / range
$$\tan\theta = \frac{m_1 - m_2}{1 + m_1 m_2} \quad -1 \le \sin\theta, \cos\theta \le 1$$

Concept 5 | Advanced Theorems and Applications

F09 | C05-K05-F09-VMCQ

Q054 | MCQ

C05-K05-5-25-GM27

A vertex outside [-1,1] is:

A an allowed extremum

B rejected

C always the maximum

D always the minimum

WORKING AREA

FORMULA BANK

Extrema
$$y = -(t-h)^2 + k$$

Addition
$$\sin(\alpha+\beta) = \sin\alpha\cos\beta + \cos\alpha\sin\beta$$

Double angle
$$\sin 2\alpha = 2\sin\alpha\cos\alpha \quad \cos 2\alpha = \cos^2\alpha - \sin^2\alpha$$

Harmonic form
$$a\sin\theta + b\cos\theta = r\sin(\theta+\alpha)$$

Angle / range
$$\tan\theta = \frac{m_1 - m_2}{1 + m_1 m_2} \quad -1 \le \sin\theta, \cos\theta \le 1$$

Concept 5 | Advanced Theorems and Applications

F09 | C05-K05-F09-VQUIZ

Quiz M1 | QUIZ MCQ — C05-K05-Q-M01

The maximum of $2\sin(\theta - \pi/3)$ is:

A -2	**B** -1
C 1	**D** 2

WORKING AREA

FORMULA BANK

Extrema
$$y = -(t-h)^2 + k$$

Addition
$$\sin(\alpha+\beta) = \sin\alpha\cos\beta + \cos\alpha\sin\beta$$

Double angle
$$\sin 2\alpha = 2\sin\alpha\cos\alpha \qquad \cos 2\alpha = \cos^2\alpha - \sin^2\alpha$$

Harmonic form
$$a\sin\theta + b\cos\theta = r\sin(\theta+\alpha)$$

Angle / range
$$\tan\theta = \frac{m_1 - m_2}{1 + m_1 m_2} \qquad -1 \le \sin\theta, \cos\theta \le 1$$

Concept 5 | Advanced Theorems and Applications

F02 | C05-K05-F02-VQUIZ

Quiz M2 | QUIZ MCQ

C05-K05-Q-M02

The addition-theorem value $\sin 75°$ is:

A $(\sqrt{6}+\sqrt{2})/4$

B $(\sqrt{6}-\sqrt{2})/4$

C $1/2$

D 1

WORKING AREA

FORMULA BANK

Extrema
$$y = -(t-h)^2 + k$$

Addition
$$\sin(\alpha+\beta) = \sin\alpha\cos\beta + \cos\alpha\sin\beta$$

Double angle
$$\sin 2\alpha = 2\sin\alpha\cos\alpha \qquad \cos 2\alpha = \cos^2\alpha - \sin^2\alpha$$

Harmonic form
$$a\sin\theta + b\cos\theta = r\sin(\theta+\alpha)$$

Angle / range
$$\tan\theta = \frac{m_1 - m_2}{1 + m_1 m_2} \qquad -1 \le \sin\theta, \cos\theta \le 1$$

Concept 5 | Advanced Theorems and Applications

F09 | C05-K05-F09-VQUIZ

Quiz M3 | QUIZ MCQ

C05-K05-Q-M03

For $y = 2x^2 + 4x + 1$, the vertex x-value is:

A -2

B -1

C 1

D 2

WORKING AREA

FORMULA BANK

Extrema
$$y = -(t-h)^2 + k$$

Addition
$$\sin(\alpha+\beta) = \sin\alpha\cos\beta + \cos\alpha\sin\beta$$

Double angle
$$\sin 2\alpha = 2\sin\alpha\cos\alpha \qquad \cos 2\alpha = \cos^2\alpha - \sin^2\alpha$$

Harmonic form
$$a\sin\theta + b\cos\theta = r\sin(\theta+\alpha)$$

Angle / range
$$\tan\theta = \frac{m_1 - m_2}{1 + m_1 m_2} \qquad -1 \le \sin\theta, \cos\theta \le 1$$

Concept 5 | Advanced Theorems and Applications

F03 | C05-K05-F03-VQUIZ

Quiz M4 | QUIZ MCQ

C05-K05-Q-M04

The angle between slopes 2 and -3 is:

A $\pi/6$

B $\pi/4$

C $\pi/3$

D $\pi/2$

WORKING AREA

FORMULA BANK

Extrema

$$y = -(t-h)^2 + k$$

Addition

$$\sin(\alpha+\beta) = \sin\alpha\cos\beta + \cos\alpha\sin\beta$$

Double angle

$$\sin 2\alpha = 2\sin\alpha\cos\alpha \quad \cos 2\alpha = \cos^2\alpha - \sin^2\alpha$$

Harmonic form

$$a\sin\theta + b\cos\theta = r\sin(\theta+\alpha)$$

Angle / range

$$\tan\theta = \frac{m_1 - m_2}{1 + m_1 m_2} \quad -1 \le \sin\theta, \cos\theta \le 1$$

Concept 5 | Advanced Theorems and Applications

F01 | C05-K05-F01-VQUIZ

Quiz W1 | QUIZ WRITTEN

C05-K05-Q-W01

Find the maximum and minimum of $y = \sin^2\theta + \cos\theta + 1$ on $0 \le \theta < 2\pi$.

WORKING AREA

FORMULA BANK

Extrema

$$y = -(t-h)^2 + k$$

Addition

$$\sin(\alpha+\beta) = \sin\alpha\cos\beta + \cos\alpha\sin\beta$$

Double angle

$$\sin 2\alpha = 2\sin\alpha\cos\alpha \qquad \cos 2\alpha = \cos^2\alpha - \sin^2\alpha$$

Harmonic form

$$a\sin\theta + b\cos\theta = r\sin(\theta+\alpha)$$

Angle / range

$$\tan\theta = \frac{m_1 - m_2}{1 + m_1 m_2} \qquad -1 \le \sin\theta, \cos\theta \le 1$$

Concept 5 | Advanced Theorems and Applications

F04 | C05-K05-F04-VQUIZ

Quiz W2 | QUIZ WRITTEN

C05-K05-Q-W02

Find $\sin 2\alpha,\ \cos 2\alpha,\ \tan 2\alpha$ when $\sin\alpha = 1/3$ and α is in QII.

WORKING AREA

FORMULA BANK

Extrema

$$y = -(t-h)^2 + k$$

Addition

$$\sin(\alpha+\beta) = \sin\alpha\cos\beta + \cos\alpha\sin\beta$$

Double angle

$$\sin 2\alpha = 2\sin\alpha\cos\alpha \qquad \cos 2\alpha = \cos^2\alpha - \sin^2\alpha$$

Harmonic form

$$a\sin\theta + b\cos\theta = r\sin(\theta+\alpha)$$

Angle / range

$$\tan\theta = \frac{m_1 - m_2}{1 + m_1 m_2} \qquad -1 \le \sin\theta, \cos\theta \le 1$$

Concept 5 | Advanced Theorems and Applications

F07 | C05-K05-F07-VQUIZ

Quiz W3 | QUIZ WRITTEN

C05-K05-Q-W03

Rewrite $3\sin\theta - \sqrt{3}\cos\theta$ as $r\sin(\theta+\alpha)$.

WORKING AREA

FORMULA BANK

Extrema
$$y = -(t-h)^2 + k$$

Addition
$$\sin(\alpha+\beta) = \sin\alpha\cos\beta + \cos\alpha\sin\beta$$

Double angle
$$\sin 2\alpha = 2\sin\alpha\cos\alpha \quad \cos 2\alpha = \cos^2\alpha - \sin^2\alpha$$

Harmonic form
$$a\sin\theta + b\cos\theta = r\sin(\theta+\alpha)$$

Angle / range
$$\tan\theta = \frac{m_1 - m_2}{1 + m_1 m_2} \quad -1 \le \sin\theta, \cos\theta \le 1$$

Concept 5 | Advanced Theorems and Applications

F08 | C05-K05-F08-VQUIZ

Quiz W4 | QUIZ WRITTEN

C05-K05-Q-W04

Solve $\sin\theta \le \cos\theta$ on $0 \le \theta < 2\pi$.

WORKING AREA

FORMULA BANK

Extrema

$$y = -(t-h)^2 + k$$

Addition

$$\sin(\alpha+\beta) = \sin\alpha\cos\beta + \cos\alpha\sin\beta$$

Double angle

$$\sin 2\alpha = 2\sin\alpha\cos\alpha \qquad \cos 2\alpha = \cos^2\alpha - \sin^2\alpha$$

Harmonic form

$$a\sin\theta + b\cos\theta = r\sin(\theta+\alpha)$$

Angle / range

$$\tan\theta = \frac{m_1 - m_2}{1 + m_1 m_2} \qquad -1 \le \sin\theta, \cos\theta \le 1$$